MIL-C-915B(NAVY)
23 April 1965

SUPERSEDING
MIL-C-915A(SHIPS)
30 July 1952

MILITARY SPECIFICATION

CABLE, CORD AND WIRE, ELECTRICAL (SHIPBOARD USE)

All interested bureaus of the Department of the Navy concurred in the use of this specification.

1. SCOPE

1.1 Scope.- This specification covers electrical cable, cord, and wire for shipboard use (see 6.1 for ratings, characteristics, and applications). As distinguished by the type of insulation used, the general features of construction, and the service involved, electrical cables covered by this specification fall within the following general classifications:

Cables for nonflexing service.
Cables for repeated flexing service.
Wire, miscellaneous.

1.2 Classification.-

1.2.1 Cables for nonflexing service.- Cables for nonflexing service shall be of the following types, as specified (see 6.2):

1.2.1.1 Heat- and flame-resistant types.-

Type SHFR	- Single, heat- and flame-resistant, radio.
Type DHFR	- Double, heat- and flame-resistant, radio.
Type THFR	- Triple, heat- and flame-resistant, radio.
Type MDU	- Multiconductor, degaussing, unarmored.
Type MDY	- Multiconductor, degaussing, steel tape armored.
Type TTHFWA	- Twisted pair, telephone, heat- and flame-resistant, armored.
Type PBTM	- Pyrometer, base, multiple pairs, unarmored.

1.2.1.2 Other types.-

Type DSWS-4	- Double conductor, special purpose, shielded.
Type DSS	- Double conductor, special purpose, shielded.
Type TSS	- Three conductor, special purpose, shielded.
Type FSS	- Four conductor, special purpose, shielded.
Type 7SS	- Multiple conductor, special purpose, shielded.
Type TSP	- Twisted pair, special purpose, plain.
Type TTRSA	- Twisted shielded pair, radio, armored.
Type DBSP	- Double, small boat, shielded, plain.
Type TBSP	- Triple, small boat, shielded plain.
Type FBSP	- Four, small boat, shielded plain.
Type FCSF	- Four, combination, special purpose, flexible.
Type MCSF	- Four, combination, special purpose, flexible.
Type SDU	- Single, degaussing, unarmored.
Type SSF	- Single, special purpose, flexible.

1.2.2 Cables for repeated flexing service.- Cables for repeated flexing service shall be of the following types, as specified (see 6.2):

1.2.2.1 Heat, flame, and oil-resistant types.-

Type DCOP	- Double conductor, oil-resistant, portable.
Type TCOP	- Triple conductor, oil-resistant, portable.

FSC 6145

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Type MHOF - Multiconductor, heat- and flame-resistant, flexible.
Type TTOP - Twisted pair, telephone, oil-resistant, portable.
Type MCOS - Multiple conductor, oil-resistant, shielded.
Type MMOP - Multiconductor, microphone, oil-resistant, portable.
Type SHOF - Single, heat- and oil-resistant, flexible.
Type DHOF - Double, heat- and oil-resistant, flexible.
Type THOF - Triple, heat- and oil-resistant, flexible.
Type FHOF - Four, heat- and oil-resistant, flexible.
Type T3P - Triple-conductor, 3000-volt, portable.

1.2.2.2 Other types.-

Type TRF - Tough rubber jacket, flexible.
Type TRXF - Tough rubber jacket, extra flexible.
Type TTRS - Twisted shielded pair, radio, flexible.
Type CVSF - Combination, aircraft carrier, special purpose, flexible.
Type DLT - Divers' lifeline and telephone cable.
Type JAS - Jet aircraft servicing.

1.2.3 Miscellaneous.- Miscellaneous wire shall be as follows:

Type MRI - Multiconductor resin insulated.

1.3 Cable sizes.- The conductor size in thousands of circular mils, the number of conductors, or the number of twisted pairs in the wire or cable for each type shall be as indicated under detailed requirements by a numeral after the type letter designation.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

J-C-98 - Cable and Wire, Insulated, Methods of Sampling and Testing.
L-P-390 - Plastic, Molding Material, Polyethylene, Low and Medium Density.
QQ-R-175 - Resistance Wire.
TT-P-320 - Pigment, Aluminum; Powder and Paste, for Paint.
UU-P-271 - Paper, Wrapping, Waterproofed Kraft.
PPP-B-576 - Box, Wood, Cleated, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant (For Carton Sealing).

MILITARY

MIL-C-17 - Cables, Radio Frequency: Coaxial, Dual Coaxial, Twin Conductor and Twin Lead.
MIL-G-160 - Grips, Cable (Naval Shipboard Use).
MIL-C-572 - Cords, Yarns, and Monofilaments, Organic Synthetic Fiber.
MIL-I-631 - Insulation, Electrical Synthetic Resin Composition, Non-rigid.
MIL-Y-1140 - Yarn, Cord, Sleeving, Cloth, and Tape — Glass.
MIL-I-3053 - Insulation, Electrical, Asbestos Fiber, Treated and Untreated.
MIL-C-5424 - Cable, Steel (Corrosion-Resistant) Flexible, Preformed (for Aeronautical Use).
MIL-M-20693- Plastic Molding Material, Polyamide (Nylon) Rigid.

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STANDARDS

MILITARY

MIL-STD-104 - Limits for Electrical Insulation Color.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

BUREAU OF SHIPS

815-1197030 - Tubes, Stuffing, Steel, for Passing Electric Cables Through Pressureproof BHDS, Submarines.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Circular NBS 431 - Lubricating Oil; General Information, Requirements and Methods of Test.
Circular 561 - Temperature Relationship of Thermocouple Wire.

(Application for copies should be addressed to National Bureau of Standards, Washington, D.C.)

DEPARTMENT OF COMMERCE

CS 227-59 - Polyethylene Film.

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

AMERICAN SOCIETY FOR TESTING MATERIALS

D353-58 - Specification for Natural Rubber Performance Insulation for Wire and Cable, 60°C Operation.
D470-59T - Methods of Testing Rubber and Thermoplastic Insulated Wire and Cable.
D752-62 - Specification for Heavy Duty Black Neoprene Sheath for Wire and Cable.

(Application for copies should be addressed to American Society For Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

AMERICAN STANDARDS ASSOCIATION

C8.15-1958 - Metallic and Associated Coverings for Insulated Cables, Requirements for.

(Application for copies should be addressed to American Standards Association, 10 East 40th Street, New York, N.Y.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

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3. REQUIREMENTS

3.1 Qualification.-

3.1.1 The types SHFR, DHFR, THFR, TTRSA, TTHFWA, DSS, TSS, FSS, 7SS, TSP, MHOF, TTOP, MCOS, MMOP, TTRS, DHOF, THOF, FHOF, MCSF, FCSF, SDU, PBTM, and T3P electrical cables shall be products which have been tested, and passed the qualification tests specified herein, and have been listed on or approved for listing on the applicable qualified products list.

3.1.2 The types which are subject to qualification (see 3.1.1) are identified with the "approved sample" which was tested and on which the approval was based, and the manufacturer shall not change any of the component materials or the assembly of the components in completed cables offered for inspection under subsequent contracts or orders without first obtaining written approval for such modifications from the Bureau of Ships.

3.2 Preproduction sample, (Types DSWS-4, MDU, MDY, DLT and JAS).- Preproduction samples of the types DSWS-4, MDU, MDY, DLT or JAS cable shall be furnished for the preproduction inspection specified in 4.3.

3.3 Definitions of terms.- The definitions specified hereinafter apply to the technical terms, wherever such terms appear in this specification:

3.3.1 Conductor.- A conductor is a slender rod or filament of drawn metal of circular cross section or a group of such rods or filaments not insulated from one another, suitable for transmitting an electric current.

3.3.2 Stranded conductor.- A stranded conductor is a conductor composed of more than one filament.

3.3.3 Strand.- A strand is one of the filaments, or one of the stranded groups, of a stranded conductor.

3.3.4 Concentric lay.- A concentric lay conductor or cable is composed of a central core surrounded by one or more layers of helically wound strands or insulated conductors.

3.3.5 Bunched lay.- In a bunched lay conductor, the strands are twisted together in the same direction without regard to geometrical arrangement.

3.3.6 Rope lay.- In a rope lay conductor, the stranded members themselves may have either a bunched, concentric, or rope lay.

3.3.7 Direction of lay.- The direction of lay is the lateral direction, either right-hand or left-hand, in which a strand or insulated conductor passes over the top as it recedes from an observer looking along the axis of the conductor or cable. In unilay or unidirectional lay construction, all components are laid in the same direction instead of reversing the direction of alternate layers.

3.3.8 Length of lay.- The lay of the helically wound strand or insulated conductor is the axial length of one turn of the helix, usually expressed in inches or as a multiple of the pitch diameter.

3.3.9 Pitch diameter.- The pitch diameter is the mean diameter of the helix described by the strands or insulated conductors in any layer.

3.3.10 Elongation.- Elongation is the extension or increase in length produced by a tension load in a section of a test specimen between bench marks placed on it, and is either expressed as a percentage of the original length between bench marks or indicated by specifying a minimum distance between bench marks.

3.3.11 Circular mil.- A circular mil is a unit of area equal to 78.54 percent of a square mil. The cross-sectional area of a circle in circular mils is, therefore, equal to the square of its diameter in mils. A circular inch is equal to 1,000,000 circular mils.

3.3.12 Twisted pair.- A twisted pair is composed of two insulated conductors twisted together.

3.3.13 Shielded pair.- A shielded pair is a twisted pair over which a close braid of copper wire has been applied.

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3.3.14 Completed wire or cable.- A completed wire or cable is one on which defective starting and finishing ends have been removed and all manufacturing operations have been completed, and which is ready for inspection.

3.3.15 Insulation resistance.- The insulation resistance of an insulated conductor is the electrical resistance offered by its insulation to an impressed direct current (d.c.) potential tending to produce a leakage of current through the same.

3.3.16 Shipping length.- A shipping length is defined as one continuous length of completed wire or cable, wound in a coil or on a reel, which is ready for packaging, packing, and marking for shipment in accordance with section 5.

3.3.17 Nominal length.- The nominal length for wire or cable on a reel is the unit ordering length, for each type and size, as shown in 6.2.2.1. The nominal length or unit ordering length, for wire or cable in coils (see 5.1.1.1) is 500 feet unless otherwise stated in the detail requirements (see 6.2.1).

3.3.18 Standard length.- A standard length is defined as any length between 110 and 90 percent of the nominal length (see 6.2.2.1).

3.3.19 Random length.- A random length is defined as any length at least 30 and less than 90 percent of the nominal length (see 6.2.2.1).

3.3.20 Remnant length.- A remnant length is defined as any length at least 10 and less than 30 percent of the nominal length (see 6.2.2.1).

3.3.21 Scrap length.- A scrap length is defined as any length less than 10 percent of the nominal length (see 6.2.2.1).

3.4 Material.- All material used in the construction of electrical wires and cables shall be of the quality and form best suited for the purpose intended.

3.5 Minimum standards.- The requirements specified hereinafter for each component material and completed wire or cable constitute the minimum acceptable standards of quality and performance under tests. The detail requirements for cables and wire shall be as specified herein (see 3.13 to 3.40, inclusive); in the event of any conflict between the general requirements of this specification and the detail requirements for the particular type of cable or wire, the latter shall govern.

3.6 Standardized construction.- The necessity for standardization of the fittings, fixtures, and other items, with which the cable is supported, terminated, or served, particularly as regards the watertight integrity of the ship, requires similar uniformity as to the type, size, and construction, as well as electrical performance, of the cable or wire. The general construction details specified hereinafter for each type and size of cable, shall be strictly complied with so as to assure the necessary uniformity and interchangeability without regard to the source of manufacture or date of procurement.

3.7 Watertight construction.- All sizes of cable types required to be of watertight construction (see 3.13, 3.14, 3.15, 3.16, 3.17, 3.18, 3.19, 3.20, 3.23, 3.24, and 3.25) shall contain sealing compounds which have been approved as a result of qualification tests or in writing as a modification of the qualified sample construction (see 3.1.1 and 3.1.2) or as otherwise approved by the Bureau of Ships. Leakage in excess of the value specified in paragraph 50, column 1 of Appendix I is indicative of inadequate processing control, and the cable represented by such samples shall be considered a failure, and shall be sufficient cause for rejection of the entire lot represented by the sample.

3.8 Dimensional tolerances.-

3.8.1 Overall dimensions.- Unless otherwise specified herein, the overall diameter of each completed wire or cable specified hereinafter shall be considered the maximum value allowable. No plus tolerance will be permissible. Where the minimum overall diameter of a completed wire or cable is specified hereinafter, no value below this minimum will be permissible. Where the minus tolerance is not specified definitely, the minimum permissible diameter shall be not less than 92-1/2 percent of the specified overall diameter of the completed wire or cable.

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3.8.2 Nominal dimensions.- Where the diameter over successive component layers of insulation, braid, sheath and armor is specified hereinafter, these diameters shall be considered as nominal values, and the nominal thickness of any layer shall be calculated as one half of the diameter increase of that layer. The average thickness throughout the length shall be within either ±0.003 inch or ±15 percent of the calculated nominal thickness, whichever allows the greater tolerance, provided that the overall diameter of the completed cable is within the specified limits.

3.8.3 Minimum thicknesses.- Where the thicknesses of the conductor insulation or cable sheath are specified as a minimum value, the average thickness throughout the length shall be not less than the specified minimum. The minimum thickness measured at any cross section shall be not less than 90 percent of the specified minimum thickness.

3.8.4 Centering and circularity.-

3.8.4.1 Insulated conductors.- The several types of insulating walls for the various classes of electrical cable shall be applied concentrically about the conductor and so cured, processed, or maintained as to provide for accurate centering of the conductor or core within the finished wall, and retention of a circular cross section. The insulation on the individual conductors shall be uniform in diameter throughout the conductor length, and at any cross section the maximum wall thickness shall not exceed the minimum by more than 25 percent for specified thicknesses greater than 0.025 inch, nor by more than 40 percent for specified thicknesses of 0.025 inch and less.

3.8.4.2 Outer sheaths.- The outer sheath shall be applied concentrically to the cable core in order to maintain circularity in the completed cable. The maximum wall thickness of the sheath at any cross section shall not exceed the minimum by more than 66 percent.

3.9 Manufacturer's identification.-

3.9.1 Except where otherwise specifically required or approved, all wires and cables shall be provided with an easy and indelible means of identifying the manufacturer. The means employed shall be such that it cannot be confused with the requirements for circuit identification specified in 3.10. The use of a printed marker tape 1/8 inch wide or colored marker threads, as applicable to the particular type or types of wire and cable in question, shall be considered standard; any departures therefrom shall be subject to specific approval of the bureau or agency concerned.

3.9.2 Marker tape.- Marker tape shall be printed legibly in 1/16-inch characters at intervals of 1 foot showing the following information:

(a) Name of manufacturer.
(b) Name or location of plant.
(c) Year of manufacture.
(d) Specification number.
(e) Progressive serial number.

The serial number is not necessarily a footage marker. The marker tape shall be placed directly under the sheath, or other location as specifically approved by the bureau or agency concerned.

3.9.3 Marker threads.- Marker threads shall be used only when satisfactory to the bureau or agency concerned, and shall be limited to the miscellaneous wire (see 1.2.3) of 0.300-inch overall diameter and smaller, and to the small flexible cables (see 1.2.2.1) of 0.300-inch overall diameter and smaller, or where it is not practicable to employ marker tape. Marker threads shall be of cotton, rayon, or nylon. The number of marker threads and the color of each thread used by the manufacturer shall be satisfactory to the bureau or agency concerned.

3.9.4 Molded marking.- Except for the specific requirements of 3.25 and 3.38, molded marking will be permitted at the manufacturer's option on cables for repeated flexing service (see 1.2.2) but not for cables which require hydrostatic test; however, when molded marking is used, it shall be in addition to and not a substitute for standard marker tape or threads. The marking shall show the manufacturer's name and his trademark or trade-name and the year of manufacture. This marking shall be molded into the sheath during processing, and it shall be either by indentation of or by raised lettering on the sheath itself, or shown on thin slips, colored if desired, which are themselves molded into the sheath and do not project above the surface of the sheath.

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3.10 Circuit identification.-

3.10.1 Means shall be provided for ready identification of each individual insulated conductor with respect to the electrical distribution system of which it is to form a part. This shall be so clear and distinctive as to permit its connection into a standard arrangement (phase sequence, circuit arrangement, and load tracing) in the system; and also to provide a rapid and accurate means for its identification and reconnection into the system after a serious casualty. The standard means for circuit identification shall be:

(a) Colored insulation.
(b) Colored tape.
(c) Color-coded braid.
(d) Printed code, as specifically required or permitted by 3.13 to 3.40, inclusive.

By any of these methods, each color or legend shall be readily distinguishable from every other color or legend and, in the completed cable the shade of colors or legends, shall conform closely to those specified in 3.10.6.

3.10.2 Colored insulation.- Where circuit identification is obtained through the use of colored insulation, the following sequence of solid colors shall be employed: (1) black, (2) white, (3) red, (4) green, (5) orange, (6) blue, (7) brown, (8) gray, (9) yellow, (10) purple, (11) tan, and (12) pink. Where a wall of saturated asbestos or glass fiber insulation is employed on individual conductors in combination with other insulation, circuit identification may be obtained by coloring this saturated wall.

3.10.3 Colored tape.- In general, colored tape shall be limited to those cable types wherein a tape or separator is specified for the accomplishment of other purposes, and coloring simply permits the added purpose of circuit identification to be served. This paragraph shall not be considered as authority for the addition of a tape not already specified; such additions shall be satisfactory to the bureau or agency concerned. The tape colors shall conform to the sequence of solid colors specified in 3.10.2.

3.10.4 Color-coded braid.- In general, color-coded braid shall be limited to those cable types wherein an inner braid (see 3.12.16) is specified over the individual conductor insulation of a multiple conductor or twisted pair telephone cable. This paragraph should not be considered as authority for the addition of a braid not already specified; such addition shall be satisfactory to the bureau or agency concerned. Where a combination of two or more colors is specified for the braid, the "base" color shall be the background, with the "tracer" color (or colors) providing the particular combination required. The "tracer" colors shall be on three adjacent carriers for each color. Where two tracers are specified, they shall be applied in opposite directions.

3.10.5 Printed code.- Where circuit identification is obtained through the use of printing, each conductor shall be identified by printing the appropriate conductor number followed by the corresponding spelled-out word representing the base color, or spelled-out words for the base and one or two tracer colors (see table I). The legends shall be printed in contrasting color, white on a black background, and repeated along the wire axis at maximum intervals of 3 inches. Alternate legends shall be inverted, for example; 10 - ORANGE-BLACK ꓘCAꓶ-EGNARO - 01. The printing shall remain intact and easily read after the conductors are fanned out at the cable ends and wiped clean of sealing compounds. Also, the printing shall be legible, and shall not smudge or rub off after exposure to hydraulic fluids and handling incident to installation and maintenance.

3.10.6 Color shades.- The shades of color used shall be fast to light and to heat, and shall fall within the light and dark limits of the color chips specified in MIL-STD-104. In the case of synthetic rubber insulation (see 3.12.4 and 3.12.6), or compound-filled tapes (see 3.12.15), for circuit identification, somewhat wider limits will be permitted in color shades, provided all colors in the cable are easily distinguishable from each other.

3.10.7 Color code.-

3.10.7.1 Multiple conductor cable.- Except where otherwise specifically required hereinafter, the color code shown in table I shall apply to all multiple conductor (not twisted pair) cables having from 2 to 127 individually insulated conductors within a common protective sheath. For example, all single conductor cables would be "1 Black"; all 2-conductor cables would consist of "1 Black" and "2 White"; all 3-conductor cables would consist of "1 Black", "2 White" and "3 Red" etc., up to a 127-conductor cable, where all the color

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combinations listed would be included. In the case of cables having more than one layer of conductors, the numbering shown above shall be from the innermost to the outermost that is the number (No.) 1 conductor shall be the center conductor (or one of the center conductors where two or more are used as a center) of the concentric lay.

Table I - Color code of multiple conductor cable.

Conductor No.	Background or base color	First tracer color	Second tracer color
1	Black	---	---
2	White	---	---
3	Red	---	---
4	Green	---	---
5	Orange	---	---
6	Blue	---	---
7	White	Black	---
8	Red	Black	---
9	Green	Black	---
10	Orange	Black	---
11	Blue	Black	---
12	Black	White	---
13	Red	White	---
14	Green	White	---
15	Blue	White	---
16	Black	Red	---
17	White	Red	---
18	Orange	Red	---
19	Blue	Red	---
20	Red	Green	---
21	Orange	Green	---
22	Black	White	Red
23	White	Black	Red
24	Red	Black	White
25	Green	Black	White
26	Orange	Black	White
27	Blue	Black	White
28	Black	Red	Green
29	White	Red	Green
30	Red	Black	Green
31	Green	Black	Orange
32	Orange	Black	Green
33	Blue	White	Orange
34	Black	White	Orange
35	White	Red	Orange
36	Orange	White	Blue
37	White	Red	Blue
38	Brown	---	---
39	Brown	Black	---
40	Brown	White	---
41	Brown	Red	---
42	Brown	Green	---
43	Brown	Orange	---
44	Brown	Blue	---
45	White	Black	Blue
46	Red	White	Blue
47	Green	Orange	Red

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Table I - Color code of multiple conductor cable (Cont'd.).

Conductor No.	Background or base color	First tracer color	Second tracer color
48	Orange	Red	Blue
49	Blue	Red	Orange
50	Black	Orange	Red
51	White	Black	Orange
52	Red	Orange	Black
53	Green	Red	Blue
54	Orange	Black	Blue
55	Blue	Black	Orange
56	Black	Orange	Green
57	White	Orange	Green
58	Red	Orange	Green
59	Green	Black	Blue
60	Orange	Green	Blue
61	Blue	Green	Orange
62	Black	Red	Blue
63	White	Orange	Blue
64	Red	Black	Blue
65	Green	Orange	Blue
66	Orange	White	Red
67	Blue	White	Red
68	Black	Green	Blue
69	White	Green	Blue
70	Red	Green	Blue
71	Green	White	Red
72	Orange	Red	Black
73	Blue	Red	Black
74	Black	Orange	Blue
75	Red	Orange	Blue
76	Green	Red	Black
77	Orange	White	Green
78	Blue	White	Green
79	Red	White	Orange
80	Green	White	Orange
81	Blue	Black	Green
82	Orange	White	---
83	Green	Red	---
84	Black	Green	---
85	White	Green	---
86	Blue	Green	---
87	Black	Orange	---
88	White	Orange	---
89	Red	Orange	---
90	Green	Orange	---
91	Blue	Orange	---
92	Black	Blue	---
93	White	Blue	---
94	Red	Blue	---
95	Green	Blue	---
96	Orange	Blue	---
97	Yellow	---	---
98	Yellow	Black	---
99	Yellow	White	---
100	Yellow	Red	---

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Table I - Color code of multiple conductor cable (Cont'd.).

Conductor No.	Background or base color	First tracer color	Second tracer color
101	Yellow	Green	---
102	Yellow	Orange	---
103	Yellow	Blue	---
104	Black	Yellow	---
105	White	Yellow	---
106	Red	Yellow	---
107	Green	Yellow	---
108	Orange	Yellow	---
109	Blue	Yellow	---
110	Black	Yellow	Red
111	White	Yellow	Red
112	Green	Yellow	Red
113	Orange	Yellow	Red
115	Black	Yellow	White
116	Red	Yellow	White
117	Green	Yellow	White
118	Orange	Yellow	White
119	Blue	Yellow	White
120	Black	Yellow	Green
121	White	Yellow	Green
122	Red	Yellow	Green
123	Orange	Yellow	Green
124	Blue	Yellow	Green
125	Black	Yellow	Blue
126	White	Yellow	Blue
127	Red	Yellow	Blue

3.10.7.2 Twisted pair telephone cable.- Individual conductors and pairs in twisted pair telephone cables shall be color coded by pairing the solid colors in sequence (see 3.10.2) first, 1 (black) with 2 to 12, inclusive, in the order listed, then 2 (white) with 3 to 12, inclusive, in the order listed, then 3 (red) with 4 to 12, inclusive, in the order listed. The color that is common to a series of pairs will be considered the first conductor of a pair and the other colored conductor (its mate) the second conductor. For example, in pairs 1 through 12 the black color identifies the first conductor of each pair.

3.11 Standard copper conductors.-

3.11.1 Conductors shall be of soft or annealed round copper wire having the properties specified in 3.12.1.1.

3.11.2 Area.- The cross sectional area of each copper conductor in a completed wire or cable shall be not less than the minimum area specified in tables II and III. Nominal areas are based on the exact or calculated wire diameters specified in table IV. The nominal strand diameters are specified in tables II, III and IV for convenience.

3.11.3 Coating, metallic.- Except where otherwise specified in the detail requirements, all conductor strands shall be bare, uncoated copper wires. When a metallic coating is specified in the detail requirements the wire shall be thoroughly and evenly coated with a thin adherent layer of commercially pure tin which, prior to stranding or braiding, shall meet the tests specified in 4.8.22. An alloy coating may be employed only when specifically approved by the bureau or agency concerned as an exception limited to a specific contract or order.

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Table II - Conductor data - concentric strand.

Standard copper conductor size	Number of strands	Strand diameter	Diameter over conductor	Cross sectional area of conductor		Maximum conductor resistance (d.c.) per 1000 feet at 25°C.		Weight per 1000 feet
	Minimum	Nominal	Nominal	Nominal	Minimum	Bare	Coated	Approx.
		(Inch)	(Inches)	(Cir. mils)	(Cir. mils)	(Ohms)	(Ohms)	(Pounds)
1(7)	7	0.013	0.038	1,119	1,096	9.89	10.3	3.4
2(7)	7	.016	.048	1,779	1,743	6.25	6.50	5.5
3(7)	7	.020	.060	2,828	2,771	3.92	4.07	8.7
3(19)	19	.013	.063	3,036	2,975	3.64	3.80	9.3
4(7)	7	.025	.076	4,497	4,407	2.46	2.52	14
6(7)	7	.031	.092	6,512	6,382	1.69	1.73	20
9(7)	7	.036	.108	9,016	8,836	1.23	1.25	28
14(7)	7	.045	.136	14,340	14,050	0.770	0.784	44
23(7)	7	.057	.171	22,800	22,340	.486	.493	70
30(19)	19	.040	.202	30,860	30,240	.358	.365	95
40(19)	19	.045	.226	38,910	38,130	.284	.288	120
50(19)	19	.051	.254	49,080	48,100	.225	.228	150
60(37)	37	.040	.282	60,090	58,890	.185	.189	190
75(37)	37	.045	.317	75,780	74,260	.146	.149	230
100(61)	61	.040	.363	99,060	97,080	.112	.115	310
125(61)	61	.045	.407	124,900	122,400	.0888	.0904	390
150(61)	61	.051	.457	157,600	154,400	.0704	.0716	490
200(61)	61	.057	.514	198,700	194,700	.0560	.0570	610
250(61)	61	.064	.577	250,500	245,500	.0444	.0453	770
300(91)	91	.057	.628	296,400	290,500	.0375	.0382	910
350(91)	91	.062	.682	349,800	342,800	.0316	.0321	1,100
400(127)	127	.057	.742	413,600	405,400	.0268	.0273	1,300
500(127)	127	.064	.832	521,600	511,100	.0214	.0217	1,600
650(127)	127	.072	.936	657,600	644,500	.0169	.0172	2,000
800(127)	127	.081	1.050	829,300	812,700	.0134	.0136	2,600
1000(127)	127	.091	1.180	1,046,000	1,025,000	.0106	.0108	3,200
1300(127)	127	.102	1.325	1,318,000	1,292,000	.00843	.00851	4,100
1600(127)	127	.114	1.485	1,662,000	1,629,000	.00668	.00676	5,100
2000(127)	127	.128	1.670	2,097,000	2,055,000	.00530	.00536	6,300

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Table III - Conductor data-bunch and rope strand.

Standard copper conductor size	Number of strands	Strand diameter	Diameter over conductor	Cross sectional area of conductor		Maximum conductor resistance (d.c.) per 1000 feet at 25°C.		Weight per 1000 feet
	Minimum	Nominal	Nominal	Nominal	Minimum	Bare	Coated	Approx.
		(Inch)	(Inches)	(Cir. mils)	(Cir. mils)	(Ohms)	(Ohms)	(Pounds)
Bunch:								
1/2(21)	21	0.005	0.028	525	515	21.1	22.2	1.6
3/5(7)	7	.010	.030	703	689	15.7	16.4	2.2
1(10)	10	.010	.038	1,005	985	11.0	11.5	3.1
1(26)	26	.006	.042	1,034	1,014	10.7	11.2	3.2
1-1/2(16)	16	.010	.049	1,608	1,576	6.88	7.17	4.9
1-1/2(41)	41	.006	.049	1,630	1,597	6.78	7.15	5.0
2-1/2(19)	19	.011	.057	2,426	2,300	4.75	4.90	6.3
2-1/2(26)	26	.010	.061	2,613	2,561	4.23	4.48	8.0
2-1/2(65)	65	.006	.061	2,594	2,542	4.26	4.51	8.0
4(41)	41	.010	.077	4,121	4,038	2.68	2.81	13
6(65)	65	.010	.097	6,533	6,402	1.69	1.78	20
9(90)	90	.010	.120	9,045	8,864	1.22	1.28	28
14(140)	140	.010	.145	14,070	13,790	.786	.823	43
Rope:								
20(49)	7x7	.020	.180	19,800	19,400	.562	.582	61
23(228)	19x12	.010	.190	22,910	22,460	.499	.523	73
26(49)	7x7	.023	.210	26,250	25,730	.428	.440	82
30(304)	19x16	.010	.220	30,550	29,940	.374	.392	97
33(336)	7x48	.010	.235	33,370	33,090	.344	.360	106
42(209)	19x11	.014	.260	42,110	41,280	.272	.284	130
42(49)	7x7	.029	.260	41,740	40,910	.268	.275	130
53(532)	19x28	.010	.304	53,470	52,400	.213	.218	169
60(304)	19x16	.014	.310	61,260	60,040	.187	.196	190
66(133)	19x7	.022	.330	66,370	65,040	.171	.175	210
84(2107)	2107	1/	.410	83,690	82,020	.138	-------	270
83(418)	19x22	.014	.380	84,230	82,560	.136	.142	270
105(2646)	2646	1/	.460	105,500	103,400	.108	-------	340
105(259)	37x7	1/	.410	105,500	103,400	.108	.113	330
133(3325)	3325	1/	.520	133,100	130,400	.0876	-------	430
133(259)	37x7	1/	.460	133,100	130,400	.0856	.0892	420
133(684)	19x36	.014	.480	137,800	135,100	.0830	.0867	440
150(760)	19x40	.014	.510	153,100	150,100	.0747	.0780	490
168(427)	61x7	1/	.520	167,800	164,400	.0681	.0710	530
200(988)	19x52	.014	.580	199,100	195,100	.0575	.0600	630
220(259)	37x7	.029	.610	220,700	216,300	.0494	.0507	718
250(1254)	19x66	.014	.680	252,700	247,700	.0453	.0472	800
250(6384)	19x7x48	.006	.713	250,000	245,000	.0461	-------	821
300(259)	37x7	.034	.714	300,000	294,000	.0378	-------	926
400(2052)	19x108	.014	.820	413,500	405,300	.0277	.0289	1,300
500(259)	37x7	.044	.922	500,000	490,000	.0234	-------	1,585
600(427)	61x7	.038	1.013	600,000	588,000	.0196	-------	1,910
650(427)	61x7	.039	1.053	650,000	637,000	.0154	-------	2,070
800(4033)	37x109	.014	1.150	812,700	796,500	.0141	.0148	2,600

1/ These are conductors for flexible cables, types TRF and TRXF, for which the minimum number of strands and minimum area are specified. The individual strand diameter is determined from the required area and the actual number of strands used.

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Table IV - Standard copper wires.

Standard copper wire sizes				Maximum resistance (d.c.) per 1000 feet at 25°C.		Approx. weight per 1000 feet
A.W.G. No.	Strand diameter		Cross sectional area			
	Nominal	Calculated	Nominal	Bare	Coated	
	(Inch)	(Inch)	(Cir.mils)	(Ohms)	(Ohms)	(Pounds)
36	0.005	0.005000	25.00	449	473	0.0757
34	.006	.006305	39.75	280	296	.120
30	.010	.01003	100.5	108	114	.304
29	.011	.0113	127.69	89.8	97.0	.394
28	.013	.01264	159.8	68.2	71.2	.484
27	.014	.01420	201.5	54.1	56.5	.610
--	.015	.0152	231	47.9	49.7	.730
26	.016	.01594	254.1	42.9	44.7	.769
24	.020	.02010	404.0	27.0	27.5	1.22
--	.022	.02234	499.1	21.9	22.4	1.55
--	.023	.02315	535.9	20.3	20.7	1.70
22	.025	.02535	642.4	16.9	17.2	1.94
--	.029	.02919	852.0	12.8	13.1	2.69
--	.031	.03050	930.3	11.7	11.9	2.82
--	.034	.03403	1,158	9.60	9.79	3.51
19	.036	.03589	1,288	8.46	8.64	3.90
18	.040	.04030	1,624	6.71	6.85	4.92
17	.045	.04526	2,048	5.31	5.42	6.20
16	.051	.05082	2,583	4.21	4.29	7.82
15	.057	.05707	3,257	3.35	3.42	9.86
--	.062	.06200	3,844	2.82	2.88	11.6
14	.064	.06408	4,107	2.66	2.72	12.4
13	.072	.07196	5,178	2.10	2.14	15.7
12	.081	.08081	6,530	1.67	1.69	19.8
11	.091	.09074	8,234	1.32	1.34	24.9
10	.102	.1019	10,380	1.05	1.06	31.4
9	.114	.1144	13,090	0.832	.844	39.6
8	.128	.1285	16,510	.660	.668	50.0

3.11.4 Types of stranding.- Stranded conductors shall be furnished with the types of stranding as specified hereinafter and in tables II and III.

3.11.4.1 Concentric stranding.- Except where otherwise specified herein, all conductors shall be concentrically stranded. One strand shall form the central core, and be surrounded by one or more layers of helically laid wires, the direction of lay being alternated for the successive layers. The pitch or lay of any layer of wires shall not exceed the values specified in table V, except that for conductors composed of 37 wires or more, this requirement shall apply only to the two outer layers. The number of strands shall be as specified in table II. The diameter of all strands shall be substantially equal.

3.11.4.2 Bunch stranding.- Where specifically required under detailed requirements, stranded conductors shall be bunch stranded. A collection of approximately the specified number of strands shall be twisted together in the same direction without regard to geometrical arrangement. The pitch or lay shall not exceed the values specified in table V. The diameter of all strands shall be equal.

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Table V - Lay of wire stranding.

Concentric stranding			Bunch stranding	
Wires or strands per conductor	Number of layer	Lay expressed in diameters of individual wires or strands	Nominal area of bunch	Maximum lay
		Maximum	Cir. mils	Inches
			0 to 1000	0.63
7	First	48	1001 to 2500	.80
19	Second	80	2501 to 4000	1.00
37	Third	112	4001 to 6000	1.30
61	Fourth	144	6001 to 8000	1.60
91	Fifth	176	8001 to 12000	2.10
127	Sixth	208	12001 to 17000	2.75
			17001 to 24000	3.50

3.11.4.3 Rope stranding.- Where specifically required, stranded conductors shall be rope stranded. The required number of stranded members shall be laid up concentrically to form the completed conductor. The lay of wires or strands within each stranded member, whether concentric, bunch or rope lay, and the lay of the two outer layers of stranded members shall not exceed the values specified in table V. In the case of the extra flexible rope stranded cable, types TRXF and JAS, the direction of lay of the component members may all be in the same direction in lieu of the alternately reversed direction specified in 3.11.4.1.

3.11.5 Joints.- Necessary joints in solid wires and rods shall be made prior to final drawing in accordance with the best commercial practice. The joint shall be made in a neat and workmanlike manner so as not to increase the diameter of the strand nor lessen its breaking strength by more than 5 percent. No splice shall be made in a stranded conductor as a whole. Not more than one of the solid wires in a concentric conductor of 19 wires or less, nor more than 5 percent of the solid wires in a concentric conductor of more than 19 wires, shall be spliced in any 1 foot of the conductor. In rope stranded conductors, the component members forming the rope shall be considered the equivalent of a solid wire in a concentric conductor, and may be spliced as a unit under the same limitations.

3.11.6 Electrical resistance.-

3.11.6.1 The resistance of completed single conductor cable, and of insulated single conductors intended for use in multiple conductor cable when measured in the preliminary stage, shall not exceed the maximum value specified in table II or III. The resistance of the conductors in completed multiple conductor cable shall not exceed the maximum value shown in the detail requirements (see 3.13 to 3.40 inclusive).

3.11.6.2 When the resistance measurements are made at a temperature other than 25° Centigrade (C.), suitable correction shall be made in accordance with 3.12.1.2.2.

3.12 Principal components.- The requirements specified hereinafter cover the several principal components used in the manufacture of electrical wires and cables.

3.12.1 Copper wire.-

3.12.1.1 Physical properties.-

3.12.1.1.1 Each solid conductor and each wire previous to stranding shall be so drawn and annealed that it shall conform to the values specified in table VI. Each wire shall be uniform in cross section, and shall be free from flaws, scales, and other imperfections.

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3.12.1.1.2 After stranding, the minimum elongation permitted shall be reduced 5 percent in numerical value (for example, from 30 to 25 percent) from the values specified in table VI.

Table VI - Physical properties of round copper wires.

Diameter of copper wire	Elongation in 10 inches (min.)	
	Bare	Coated
(Inch)	(Percent)	(Percent)
0.0030 to 0.0116	20	10
0117 to .0207	20	15
.0208 to .105	25	20
.106 to .298	30	25

3.12.1.2 Electrical resistance.-

3.12.1.2.1 The resistance of the copper wires prior to stranding or insulating shall not exceed the maximum values specified in table IV, and shall be determined only when specifically required.

3.12.1.2.2 Copper resistance shall be referred to a temperature of 25°C. by multiplying the measured value by a conversion factor corresponding to the temperature of the measurement. These conversion factors are shown in table VII. For temperatures between those tabulated, the conversion factor may be obtained by interpolation or calculated from the formula:

$$\text{Conversion factor} = \frac{259.5}{234.5 + t}$$

Where:

t = Test temperature, °C

Table VII - Copper resistance temperature conversion factor.

Temperature of copper		Conversion factor	Temperature of copper		Conversion factor	Temperature of copper		Conversion factor
(°C.)	(°F.)		(°C.)	(°F.)		(°C.)	(°F.)	
0	32.0	1.107	20	68.0	1.020	40	104.0	0.945
2	35.6	1.097	22	71.6	1.012	42	107.6	.939
4	39.2	1.088	24	75.2	1.004	44	111.2	.932
6	42.8	1.079	26	78.8	0.996	46	114.8	.925
8	46.4	1.070	28	82.4	.989	48	118.4	.919
10	50.0	1.061	30	86.0	.981	50	122.0	.912
12	53.6	1.052	32	89.6	.974	52	125.6	.906
14	57.2	1.044	34	93.2	.967	54	129.2	.900
16	60.8	1.036	36	96.8	.959	56	132.8	.893
18	64.4	1.028	38	100.4	.952	58	136.4	.887

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3.12.2 Insulation, synthetic resin. -

3.12.2.1 Specimens of insulation removed from the insulated conductor of a complete wire or cable shall meet the requirements specified hereinafter when tested as specified in 4.8.1.

3.12.2.1.1 Tensile strength. - The tensile strength shall be not less than 1800 p.s.i.

3.12.2.1.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 4-1/2 inches apart before rupture.

3.12.2.1.3 Air oven aging. - After 96 hours aging at 100°C. in the air oven (see 4.8.1.3), the ultimate elongation shall be not less than 80 percent of the value measured before aging.

3.12.3 Insulation, synthetic resin (105°C.). -

3.12.3.1 Specimens of insulation removed from the insulated conductor of a complete wire or cable shall meet the requirements specified hereinafter when tested as specified in 4.8.1.

3.12.3.1.1 Tensile strength. - The tensile strength shall be not less than 1800 p.s.i.

3.12.3.1.2 Elongation. - The specimen shall stretch until the 2-inch bench marks are not less than 4 inches apart before rupture.

3.12.3.1.3 Air oven aging. - After 96 hours aging at 136°C. in the air oven (see 4.8.1.3) the ultimate elongation shall be not less than 90 percent of the value measured before aging.

3.12.4 Insulation, synthetic rubber. -

3.12.4.1 Specimens of insulation removed from the insulated conductor of a completed wire or cable shall meet the requirements specified hereinafter when tested as specified in 4.8.1.

3.12.4.1.1 Specimens taken from conductor sizes 14 and larger. -

3.12.4.1.1.1 Tensile strength. - The tensile strength shall be not less than 600 p.s.i.

3.12.4.1.1.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 6-1/2 inches apart before rupture.

3.12.4.1.1.3 Set. - The set, after elongating the specimen until the 2-inch bench marks are 6 inches apart, shall not exceed 30/64 inch.

3.12.4.1.1.4 Air pressure heat aging. - After 20 hours aging at 126.7°C. in the air pressure heat bomb (see 4.8.1.3), the tensile strength and elongation shall be not less than 50 percent of the values measured before aging.

3.12.4.1.1.5 Oxygen pressure aging. - After 168 hours aging at 80°C. in the oxygen pressure bomb (see 4.8.1.3), the tensile strength and elongation shall be not less than 50 percent of the value measured before aging.

3.12.4.1.2 Specimens taken from conductor sizes 9 and smaller. -

3.12.4.1.2.1 Tensile strength. - The tensile strength shall be not less than 600 pounds per square inch.

3.12.4.1.2.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 6-1/2 inches apart before rupture.

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3.12.4.1.2.3 Oxygen pressure aging. - After 96 hours aging at 70°C. in the oxygen pressure bomb (see 4.8.1.3), the elongation shall be not less than 70 percent of the value measured before aging.

3.12.5 Insulation, polythene. - Polythene insulation shall be in accordance with type II, grade 7 of L-P-390.

3.12.5.1 Specimens of the insulation removed from the insulated conductor of the completed cable shall conform to the following requirements when tested as specified in 4.8.1.

3.12.5.1.1 Tensile strength. - The tensile strength shall be not less than 1400 pounds per square inch.

3.12.5.1.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 8 inches apart before rupture.

3.12.5.1.3 Set. - The set, after elongating specimens until the 2-inch bench marks are 4 inches apart, shall not exceed 35/64 inch.

3.12.5.1.4 Air oven aging. - After 96 hours aging at 100°C. in the air oven (see 4.8.1.3), the ultimate elongation shall be not less than 60 percent of the value measured before aging.

3.12.6 Insulation, butyl rubber. -

3.12.6.1 Specimens of the insulation removed from the insulated conductor of the completed cable shall conform to the following requirements when tested as specified in 4.8.1.

3.12.6.1.1 Tensile strength. - The tensile strength shall be not less than 600 p.s.i.

3.12.6.1.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 11 inches apart before rupture.

3.12.6.1.3 Set. - The set, after elongating specimens until the 2-inch bench marks are 6 inches apart, shall not exceed 30/64 inch.

3.12.6.1.4 Air pressure heat aging. - After 20 hours aging at 126.7°C. in the air pressure heat bomb (see 4.8.1.3), the tensile strength shall be not less than 70 percent, and the elongation shall be not less than 75 percent of the values measured before aging.

3.12.7 Insulation, varnished-cambric. -

3.12.7.1 Insulation. - The wall of varnished cambric insulation shall consist of multiple layers of high grade varnished fabric as follows, with a lubricating compound conforming to 3.12.8. Samples of tape removed from the completed cable shall pass the tests specified in 4.8.3.

3.12.7.2 Fabric. - The fabric shall be a closely woven cotton cloth having at least 60 and 55 threads per inch for warp and filler, respectively.

3.12.7.3 Varnish coating. - The highest grade of electrical insulating varnish, shall be applied to the fabric, preferably in two or three successive coats, each coat being dried slowly at a low baking temperature. The finished product shall be of uniform thickness, free from wrinkles, blisters, and other inperfections. The varnish coating shall be pliable and shall not crack when the varnished cloth is doubled upon itself.

3.12.7.4 Thickness of varnished cloth. - The finished varnish cloth shall have a uniform average thickness of not less than 0.005 inch nor more than 0.013 inch. The allowable variation of individual tapes shall be ±20 percent but shall not exceed 0.002 inch.

3.12.7.5 Tensile strength. - The tensile strength of the varnished tapes (measured lengthwise) (see 4.8.3.2), at room temperature, shall be not less than 2,400 pounds per square inch cross section, upon samples removed from the completed cable.

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3.12.7.6 Varnished-cambric tape. - Used as a tape, the varnished cambric shall be of a suitable width, and so applied that samples taken from the completed cable shall meet the physical and electrical tests specified in 4.8.3. The nominal thickness of tape used shall be not less than 0.005 inch nor more than 0.013 inch, as may be most suitable for the type and size of cable in which used. The tension used in wrapping successive layers of tape around the conductor shall be such that all layers fit snugly and that there is no crinkling of the inner layers attributable to failures in applying or maintaining correct tension at all times.

3.12.7.7 Lap. - In general, and except where otherwise specifically required, the amount of lap used in applying the varnished cambric tape, and whether positive or negative, will be left to the discretion of the manufacturer.

3.12.7.8 Thin varnished cambric walls. - For thin walls of varnished cambric, particularly those of 0.048 inch thickness or less, or where used in combination with asbestos, the thickness of varnished cambric tape used shall be given special consideration. For such walls it is required that there be a multiplicity of tape layers, with the best possible distribution of joints, and hence improved structure as a dielectric. Unless otherwise specified or approved for a particular type and size of cable (see 3.13 to 3.40 inclusive), the tape thickness shall be such that at no point in the insulation shall there be less than three thicknesses of varnished cambric.

3.12.7.9 Alternates for varnished cambric. - Where varnished cambric is specified as a component of electrical cables, the use of the suitable alternate material will be permitted when satisfactory to the bureau or agency concerned. The suitability of such alternates shall be determined on the basis of tests specified in 4.2.

3.12.7.10 Effect of hot petrolatum. - After aging 15 minutes at 150°C. in hot petrolatum (see 4.8.3.3) and cooling to room temperature, the varnish coating on the tapes shall not be materially softer or more tacky than before being placed therein.

3.12.7.11 Dielectric strength. - The average dielectric strength of the varnished-cambric tapes (see 4.8.3.4), as removed from the completed cable, in terms of volts per mil of thickness at room temperature, shall be not less than 750 for single conductor cables and not less than 600 for multiple conductor cables.

3.12.7.12 Aging. - After 150 hours aging at 125°C. in an air oven (see 4.8.3.5) the varnish film on the tape shall show no evidence of cracking or peeling visible to the unaided eye, when bent around a mandrel 1/8 inch in diameter.

3.12.8 Lubricating compound for varnished-cambric insulation. -

3.12.8.1 Use. - A suitable compound shall be used for excluding air and moisture and for the lubrication of the varnished cambric insulation. It shall assure freedom from tearing or rupturing of the insulation when the completed cables are subject to the bending test (see 4.8.9).

3.12.8.2 General properties. - The compound shall be a mineral base, acid-free compound, which will not dry out, oxidize, or combine with the varnish on the cambric. The compound shall have no deleterious effect upon any other component material in the cable construction.

3.12.8.3 Amount used. - The amount of compound used shall be such that the completed cable will pass the drip test specified in 4.8.8. The outer layers of varnished cambric, in the larger size conductors especially, may be applied without added compound, provided the tape itself is sufficiently lubricated to allow free motion between the layers in bending of the cable.

3.12.9 Asbestos. -

3.12.9.1 All asbestos used in the construction of electrical cables shall be of a form, type, and class suitable for the particular purpose, and shall conform to the applicable requirements of MIL-I-3053. The composition and structure shall be such as will enable the completed cable to meet all specified test requirements.

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3.12.9.2 Asbestos electrical insulating walls. - Asbestos electrical insulating walls shall consist of plain or reinforced asbestos applied either in treated or untreated form. The asbestos shall be suitably applied and compressed about the conductor so as to provide a dense felted wall of circular cross sections, in which the conductor is well centered. Such insulating walls are normally to be employed in combination with walls of insulating material of high dielectric strength such as synthetic resin or varnished cambric. Asbestos for this application shall conform to class 2 or 4 of MIL-I-3053.

3.12.9.3 Asbestos heat and flame-resistant walls. - The asbestos for the wall or beltings applied about two or more individually insulated conductors, which have been cabled together, shall consist of plain or reinforced asbestos, applied either as treated or untreated. The asbestos shall be suitably applied and compressed about the cable core so as to provide a dense felted wall of circular cross section in which the cable core is well centered. Asbestos for this application shall conform to class 4 or 6 of MIL-I-3053. The thickness of this felted wall shall be such as to permit strict adherence of the completed cable with the specified overall dimensions.

3.12.9.4 Impregnation. - The asbestos over the individually insulated conductor (or conductors) and over the cable core shall be thoroughly impregnated with a moisture and flame-retardant insulating compound in such a manner as to fill all interstices and make the insulation one homogeneous structure. The compound used to saturate the asbestos shall consist of suitable high grade ingredient and shall retain its initial qualities during the normal service life of the cable. The compound shall develop no injurious chemical action within itself or with any other of the component parts of the completed cable. The compound shall not drip from the completed cable when tested as specified in 4.8.8.

3.12.9.5 Asbestos fillers. - Fillers shall meet the requirements of 3.12.11.

3.12.10 Glass fiber. -

3.12.10.1 All glass fibers used in the construction of electrical cables shall be of a class and form suitable for the particular purpose, and shall conform to the applicable requirements of MIL-Y-1140.

3.12.10.2 Admixtures. - Where glass fiber insulation is specified as such, it shall not be used in a direct combination or mixture with any organic fabric or textile material, that is, glass fiber rovings, yarn, tape, cloth, braid, cord, bonded mat, or other form of glass fibrous material, and shall not be part cotton, part silk, part linen, and so forth, unless otherwise specifically permitted or required in 3.13 to 3.40 inclusive.

3.12.10.3 Glass fiber servings. - Servings shall be of multiple wound continuous fiber yarn, size 900-2/2 or the approximate equivalent, except where otherwise specified or approved by the bureau or agency concerned.

3.12.10.4 Glass fiber braids. - The yarn for glass fiber braids shall be the appropriate size of either staple or continuous fiber yarn conforming to MIL-Y-1140. The braid angle, weave and thickness shall be as specified in 3.12.16.

3.12.10.5 Glass fiber electrical insulating walls. - Glass fiber electrical insulating walls shall consist of staple, fiber roving or condensed sliver suitably applied and compressed about the conductor so as to provide a dense wall of circular cross section in which the conductor is well centered. Such electrical insulating walls are normally to be employed in combination with walls of insulating materials of high dielectric strength, such as synthetic resin or varnished cambric.

3.12.10.6 Glass fiber heat and flame-resistant walls. - The glass fiber for walls or beltings applied about two or more individually insulated conductors, which have been cabled together, shall consist of staple fiber roving or a condensed sliver. The glass fiber shall be suitably applied and compressed about the cable core so as to provide a dense wall of circular cross section in which the cable core is well centered. The thickness of the felted wall shall be such as to permit strict adherence of the completed cable with the specified overall dimensions.

3.12.10.7 Impregnation. - The glass fiber over the individually insulated conductor (or conductors), including braids, and over the cable core shall be thoroughly impregnated with a moisture and flame retardant insulating compound in such a manner as to fill all interstices, and make the insulation one homogeneous

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structure. The compound used to saturate the glass fibers shall consist of suitable high grade ingredients and shall retain its initial qualities during the normal service life of the cable. The compound shall develop no injurious chemical action within itself or with any other of the component parts of the completed cable. The compound shall not drip from the completed cable under the test conditions specified in 4.8.8.

3.12.10.8 Glass fiber fillers. - Fillers shall meet the requirements of 3.12.11.

3.12.10.9 Glass fiber impervious sheath reinforcement. - Where glass fiber is used as a heat and flame-resistant textile reinforcement in the impervious sheath construction for cables (see 3.12.18), it shall be of either staple or continuous fiber yarn.

3.12.11 Fillers. -

3.12.11.1 Except where otherwise specified, filler material shall be cotton, jute, synthetic fibers, glass fibers, or asbestos. Filler material shall be of long fiber and shall be free from defects, dirt, knots, and lumps. Dry or untreated fillers shall not be used.

3.12.11.2 For heat and flame-resistant cable types. -

3.12.11.2.1 Asbestos or glass fibers, pre-impregnated with suitable moisture and flame-retardant insulating compounds, shall be used in conjunction with the approved watertight core sealing compounds as fillers for heat and flame-resistant cables (see 1.2.1.1). Fibrous fillers may be omitted at the manufacturer's option, only from cables having three or more conductors of standard size 3(7) or smaller, provided the required firmness of core is retained with the sealing compound alone. For the larger size cables, the asbestos or glass fibers shall occupy not less than 50 percent of the total valley space so as to provide a firm, well-rounded assembly as specified in the detail requirements.

3.12.11.2.2 Asbestos. - Where asbestos is employed as heat-and flame-resistant fillers, it shall conform to the applicable requirements for class 4 or class 6 of MIL-I-3053.

3.12.11.2.3 Glass fiber. - Where glass fiber is employed as heat and flame-resistant fillers it shall be of staple rovings and condensed sliver, and shall conform to the applicable requirements of MIL-Y-1140.

3.12.11.2.4 Filler impregnation. - The moisture and flame-retardant impregnating compound shall consist of high grade insulating ingredients which retain their initial qualities during the normal service life of the cable. The compound shall develop no injurious chemical action within itself, or with any other of the component parts of the completed cable. Except in flexible cables, the fillers shall be compacted and thoroughly impregnated in such a manner as to completely fill all interstices of the core and make the fillers one homogeneous structure. The amount of compound shall be not less than 10 percent of the weight of the saturated filler. In no case shall the compound from the fillers drip from the completed cable under the test conditions specified in 4.8.8.

3.12.11.3 For other cable types. - Fillers used in all multiple conductor cables (see 1.2.1.2 and 1.2.2), except heat-and flame-resistant types (see 1.2.1.1), shall be of cotton, jute, glass, or asbestos treated with a moisture-resistant compound. The amount of compound shall be not less than 10 percent of the weight of the saturated fillers.

3.12.12 Cotton. -

3.12.12.1 All cotton used shall be clean, dry, and free from all foreign matter deleterious to the insulating qualities.

3.12.12.2 Separator. - Where a serving of cotton is used as a separator around a copper conductor, a minimum No. 60 single-ply cotton yarn shall be applied with full coverage.

3.12.12.3 Reinforcement. - Cotton used for reinforcing the sheath of cables intended for repeated flexing service shall be multiple ends of seine twine, hawser cord, or heavy cabled cotton yarn of a size most suitable for the particular construction in which used. Alternate reinforcement may be used only when satisfactory to the bureau or agency concerned.

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3.12.12.4 Tie cord. - Cotton tie cord shall be of a braided construction, having a breaking strength of not less than 30 pounds (see 4.8.28.2), and impregnated with a waterproofing flexible compound to a diameter of not more than 0.065 inch.

3.12.12.5 Fillers and braids. - When used as fillers and braids, the requirements of 3.12.11 and 3.12.16 shall also be met.

3.12.13 Rayon and other synthetic fiber. - Rayon and other synthetic fiber used in the construction of electrical cable and wire shall be of the best quality and in the best form suited for the purpose intended. It shall be clean, dry, and free from foreign matter deleterious to its insulating qualities, and shall conform to MIL-C-572.

3.12.14 Separators. -

3.12.14.1 Where a separator is specified or permitted, the materials specified hereinafter, when properly employed, will be considered suitable. Other materials shall be used only when satisfactory to the Bureau of Ships.

3.12.14.1.1 For heat-and flame-resistant types. - A wind of glass fibers, or other suitable, heat-resistant synthetic fiber or a thin tape of moisture-resistant polyester, cellophane, cellulose acetate, or flame-resistant synthetic resin will be suitable. The wall thickness of the applied separator shall not exceed 0.004 inch.

3.12.14.1.2 For other types. - A wind or tape of cotton, paper, rayon, cellulose acetate or polyester will be suitable. This type of separator shall not be used for circuit identification.

3.12.15 Tape, compound-filled. -

3.12.15.1 The tape shall be made of cotton sheeting, evenly and firmly woven from good cotton, free from unsightly defects, dirt, knots, lumps and irregularities of twist. It shall be thoroughly impregnated and evenly coated on both sides or one side when specifically approved by the bureau or agency concerned with an insulating compound. The compound shall adhere firmly to the fabric. The compound shall develop no injurious chemical action within itself, or with any other of the component parts of the complete cable.

3.12.15.2 Thickness. - The thickness of the tape shall be not less than 0.008 inch.

3.12.15.3 Application. - The tape shall be applied in a manner to insure a smooth surface and a circular section. The tape shall be applied with not less than 1/8-inch lap for diameters under 0.2-inch, and not less than 1/4-inch lap for larger diameters.

3.12.16 Braids, textile. -

3.12.16.1 Braids shall be of cotton (see 3.12.12), rayon (see 3.12.13), or glass fiber (see 3.12.10), as specified for the particular type and size of cable.

3.12.16.2 Inner (marker) braids. -

3.12.16.2.1 Except where otherwise specified or approved by the bureau or agency concerned, inner braids shall be closely woven of 2-ply cotton, or weave of rayon or glass fibers at the manufacturer's option. These braids shall be color-coded for circuit identification, as specified in 3.10. The braids shall be suitably treated so that they will not fray or loosen at the cut ends after reasonable handling incident to use.

3.12.16.2.2 Thread size. - Cotton marker braids shall be woven from yarn not smaller than that specified in table VIII.

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Table VIII - Thread size.

Diameter under braid	Minimum thread size
Inches	size/ply
0.050 to 0.400	40/2
.401 to .800	30/2
.801 to 3.000	20/2

3.12.16.2.3 Braid angle. - The tangent of the angle between the cotton thread and the axis of the wire shall be not less than that specified in table IX.

Table IX - Textile braid angle.

Diameter under braid	Value of tangent	Corresponding braid angle
(Inches)		Degrees
0.050 to 1.000	0.700	35
1.001 to 1.500	.839	40
1.501 to 3.000	1.000	45

3.12.16.3 Outer cotton braids. -

3.12.16.3.1 Outer cotton braids shall be closely woven from cotton thread having not less than 2 plies. The braid shall be thoroughly saturated with a flexible, hard, black (except where otherwise specified) moisture-and flame-retardant, insulating compound, which shall not ooze or drip off at a temperature of 95°C.

3.12.16.3.2 Braid angle. - The tangent of the angle between the cotton thread and the axis of the wire shall be not less than that specified in table IX.

3.12.16.3.3 Braid thickness. - The thickness of outer cotton braids shall be not less than that specified in table X.

Table X - Braid thickness.

Diameter under braid	Thickness of braid minimum	Corresponding size and ply of cotton yarn minimum
(Inches)	(Inch)	(size/ply)
0.050 to 0.100	0.014	40/2
.101 to .200	.016	30/2
.201 to .350	.017	26/2
.351 to .800	.020	20/2
.801 to 1.500	.026	12/2
1.501 to 3.000	.032	8/2

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3.12.16.4 Glass braids. - Glass braids shall conform to 3.12.10 and 3.12.16.1 to 3.12.16.2, inclusive, as far as practicable.

3.12.16.5 Rayon braids. - Rayon braids shall conform to 3.12.13, and 3.12.16.1 to 3.12.16.2, inclusive, as far as practicable.

3.12.16.6 Copper shielding braid. - Shielding braids shall be of bare, uncoated copper if applied over synthetic resin insulation (see 3.12.2 or 3.12.3) or polythene insulation (see 3.12.5). Tinned copper shall be employed over rubber insulation (see 3.12.4 and 3.12.6). The percent of coverage shall conform to 3.12.20.4. The braid angle shall not be less than 30 degrees.

3.12.17 Sheath, polychloroprene. - Polychloroprene sheath shall fit closely to the cable core and, except in types TRF and SHOF, shall be readily removable without damage to the conductor insulation. Samples of the sheath, removed from completed cable, shall meet the requirements specified hereinafter, when tested as specified in 4.8.1.

3.12.17.1 Tensile strength. - The tensile strength shall be not less than 1800 p.s.i.

3.12.17.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 8 inches apart before rupture.

3.12.17.3 Set. - The set, after elongating specimens until the 2-inch bench marks are 6 inches apart, shall not exceed 3/8 inch.

3.12.17.4 Air pressure heat aging. - After 20 hours aging at a temperature of 126.7°C. in the air pressure heat bomb (see 4.8.1.3), the tensile strength and elongation shall be not less than 65 percent of the values measured before aging.

3.12.17.5 Hot oil immersion. - After subjection to the hot oil test (see 4.8.17) the tensile strength and elongation shall be not less than 65 percent of the values measured before aging.

3.12.18 Sheath, impervious. -

3.12.18.1 Where specified in 3.13 to 3.40, inclusive, cable shall be provided with an impervious sheath as approved on the basis of qualification tests (see 4.2). The sheath shall be applied to the cable core as a close-fitting, extruded tube to provide a concentric wall of uniform texture throughout. For cables with braided metal armor, application of the sheath in laminated form, or the inclusion of textile reinforcement within the sheath wall or to the outer surface of the sheath, with the object of subsequently molding such laminations or several component materials together to form an impervious sheath, will not be acceptable. Cables for repeated flexing service may have a suitable reinforcement under or within the body of the impervious sheath, when so employed in the approved sample (see 3.1.2). The sheath compound shall be black, unless a colored sheath is specified. Samples of the sheath removed from completed cable shall meet the requirements specified hereinafter, when tested as specified in 4.8.1.

3.12.18.2 Tensile strength. - The tensile strength shall be not less than 1800 p.s.i.

3.12.18.3 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 6 1/2 inches apart before rupture.

3.12.18.4 Set. - The set, after elongating specimens until the 2-inch bench marks are 4 inches apart, shall not exceed 30/64 inch.

3.12.18.5 Hot oil immersion. - After subjection to the hot oil test (see 4.8.17), the tensile strength shall be not less than 1800 p.s.i., and the elongation shall be not less than 50 percent.

3.12.18.6 Imperviousness. - The sheath shall be of such composition and construction as to render the completed cable impervious to water and air under pressure, when tested as specified in 4.8.13.

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3.12.18.7 Specific characteristics. -

3.12.18.7.1 The composition and construction of the impervious sheath shall provide a combination of the following basic properties that is properly proportioned for the particular purpose intended:

(a) Resistance to flame.
(b) Stability under long continued exposure to elevated temperatures.
(c) Absence of noticeable odor at cable operating temperatures.
(d) Deformability at low temperature (that is, cold bending qualities).
(e) Stability under exposure to oil, sulfuric acid, and sea water.
(f) Stability under exposure to sunlight.

3.12.18.7.2 Nonflexing service. - Particular emphasis shall be placed on the flame resistance and the stability under heating. The remaining characteristics shall be maintained at as high a level as can be made compatible with this objective.

3.12.18.7.3 Repeated flexing service. - Particular emphasis shall be placed on the flexing endurance characteristic. The remaining characteristics shall be maintained at as high a level as can be made compatible with this objective.

3.12.19 Sheath, impervious special purpose, unarmored. -

3.12.19.1 Where specified in the detail requirements or purchase order, cable shall be provided with a special impervious sheath approved by the Bureau of Ships. When used on cable that is normally armored, the cable designation shall be modified by changing the A to U; for example Unarmored TTHFWA would be designated TTHFWU. In such cases the maximum diameter of the unarmored cable shall be 0.050 inch less than the maximum specified diameter of the armored counterpart.

3.12.19.1.1 Tensile. - The tensile strength shall be not less than 2500 p.s.i.

3.12.19.1.2 Elongation. - Specimens shall stretch until the 2-inch bench marks are not less than 8 inches apart before rupture.

3.12.19.1.3 Set. - The set, after elongating specimens until the 2-inch bench marks are 6 inches apart, shall not exceed 1.0 inch.

3.12.19.1.4 Air pressure heat aging. - After 20 hours aging at a temperature of 126.7°C. in the air pressure heat bomb (see 4.8.1.3), the tensile strength and elongation shall be not less than 80 percent of the values measured before aging.

3.12.19.1.5 Hot oil immersion. - After subjection to the hot oil test (see 4.8.17) the tensile strength and elongation shall not be less than 80 percent of the values measured before aging.

3.12.19.1.6 Heat shock. - Completed cable shall meet the following requirements when tested as specified in method 5011 of J-C-98:

0 to 3/4-inch o.d.	3 turns on 3 x Mandrel	No cracks
over 3/4-inch to 1-1/2 inch o.d.	1 turn on 4 x Mandrel	No cracks
over 1-1/2 inch o.d.	180° turn on 5 x Mandrel	No cracks

3.12.20 Armor, braided metal. -

3.12.20.1 Armoring wire shall be of 0.0126 ± 0.0005 inch diameter aluminum alloy, except where otherwise specified herein. The wire shall be of uniform diameter and shall be free from cracks, splits, or other flaws. Splices in the wire shall be infrequent, staggered, and inconspicuous, so as not to increase the normal diameter of the cable or result in rough spots. The weave shall be such that the wires of a strand are laid closely together, flat and parallel, tightly and firmly binding the sheathed cable. The braid shall be applied with sufficient tension to prevent loosening and creeping, but without so much tension that excessive broken ends occur. It shall remain snug to the cable when cut and not spring away from the sheath. It shall withstand the tests specified in 4.8.7. It shall provide reasonable mechanical protection to the sheath during installation and handling, and shall retard deformation of the cable under compression from stuffing tubes and cable hangers.

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3.12.20.2 Aluminum armor. -

3.12.20.2.1 Aluminum armor shall be Alclad 5056 aluminum wire. The wire shall be 0.0126 ± 0.0005 inch in diameter.

3.12.20.2.2 Aluminum wire shall have a tensile strength of not less than 50,000 p.s.i., and the elongation in 10 inches shall be not less than 2 percent, as tested before application to the cable.

3.12.20.3 Weave. - The weave shall be of either the "one over-one under" or the "two over-two under" type. The selection of the number of ends per carrier and the number of carriers per braider shall be such as to produce a basket weave with a braid angle within the limits shown in table XI. The number of ends per carrier shall be not more than the values shown in table XII.

Table XI - Armor braid angle.

Diameter over sheath	Braid angle	
	Minimum	Maximum
(Inches)	(Degrees)	(Degrees)
0.100 to 0.600	30	60
.601 to 1.000	35	60
1.001 to 1.500	40	70
1.501 to 2.000	45	70
2.001 to 3.000	50	80

Table XII - Ends per carrier.

Cable diameter under braid	Maximum number of ends per carrier	
	One over-one under	Two over-two under
(Inches)		
0.100 to 0.400	8	5
.401 to .800	12	10
.801 to 3.000	16	10

3.12.20.4 Coverage. - The coverage for all types and sizes of cable shall be not less than 88 percent, as determined by the following formula:

$K = (2F - F^2) \times 100.$

Where: K = Percent coverage.
a = Angle of braid with axis of cable.
Tan a = $2\pi DP/C$.
d = Diameter of individual braid wire in inches.
C = Number of carriers.
D = Diameter of cable under armor in inches.
F = NPd/Sin a.
N = Number of wires per carrier.
P = Picks per inch of cable length.

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3.12.21 Paint. -

3.12.21.1 Where specified (see 3.13 to 3.40, inclusive), braided metal armored cables shall be suitably coated with aluminum paint.

3.12.21.2 Aluminum paint. - Aluminum paint shall consist of aluminum pigment (paste form) conforming to TT-P-320, in a suitable vehicle of the synthetic resinuous (phenol-formaldehyde or glyceride) type designed for good penetration, high bonding power, resistance to water and to oil, and with flexibility at temperatures up to 100°C.

3.12.21.3 Application. - Painting shall be done as soon as practicable after completion of prior manufacturing operations so as to prevent accumulation of dirt, moisture, and foreign matter on the braid or in the interstices. The paint shall be properly dried and shall not be tacky. The painting shall be done preferably by the immersion method so as to fill in all interstices in the outer cable surface. The paint finish shall provide a relatively smooth overall coating that will facilitate making watertight glands with Navy standard stuffing tubes and approved packing. It shall be flexible and adhere firmly so that it will not crack, peel, or flake off under the cold bend (see 4.8.9) and the heat endurance (see 4.8.12) tests.

3.12.22 End seals. -

3.12.22.1 All types and sizes of cable specified herein, except miscellaneous wires (see 1.2.4), shall have both ends of every shipping length provided with a seal to prevent the entrance of moisture. The end seals shall be constructed as shown on figure 1, unless specific approval is given to the manufacturer for the use of an alternate method. Approval of alternate methods shall be granted only after qualification tests (see 4.2), and when the design is considered to be such that uniformly satisfactory sealing will be obtained in routine factory applications. Having once fixed upon a given design of end seal and having received qualification, the manufacturer shall adhere strictly under subsequent deliveries of cable to the construction so tested and approved.

3.12.22.2 Watertightness. - The end seal shall be of such composition and shall be applied in such manner as to prevent the entrance of water into the cable under an externally applied pressure. A sample of the completed cable except watertight cables with end seal in place shall withstand, without leakage, the water and air leakage test specified in 4.8.13.

3.12.23 Natural rubber. - Unless specifically authorized by the Bureau of Ships or as specified herein, the use of natural rubber either as a component of electric cables or as a component of end seals will not be permitted.

3.12.24 Metal splices. - In process, metal to metal splices are not permitted.

3.13 Type SHFR, DHFR and THFR cable (for 3,000 volt service). -

Construction

First - A standard conductor of bare, uncoated copper (see 3.11).
Second - A wall of synthetic resin insulation (see 3.12.2).
Third - With or without separator (see 3.12.14) at manufacturer's option.
Fourth - A wall of varnished-cambric insulation (see 3.12.7).
Fifth - A separator (see 3.12.14).
Sixth - A wall of felted asbestos (see 3.12.9) or glass fiber (see 3.12.10) with or without separator (see 3.12.14) on SHFR at manufacturer's option.
Seventh - Conductors of DHFR and THFR shall be twisted together with a lay not greater than 24 times the pitch diameter. Fillers (see 3.12.11) shall be employed to obtain a firm, well-rounded assembly.
Eighth - A wall of felted asbestos (see 3.12.9) or glass fiber (see 3.12.10).
Ninth - With or without separator (see 3.12.14) at manufacturer's option.
Tenth - An impervious sheath (see 3.12.18).
Eleventh - Braided metal armor (see 3.12.20).
Twelfth - Paint (see 3.12.21).

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Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XIII
Conductor resistance (see 4.8.24)	See table XIII
Insulation resistance	See table XIII
Watertightness (see Appendix)	
Cubic inches in 6 hours	1

Group B	Requirements
Cold bend (see 4.8.9 and table XIII)	
Insulation (see 3.12.2)	
Tensile - p.s.i., minimum	1800
Elongation - Percent, minimum	125
Sheath (see 3.12.18)	
Tensile - p.s.i., minimum	1800
Elongation - Percent, minimum	225
Set - Inches, maximum	30/64

Group C	Requirements
Flammability (see 4.8.10)	(See 4.8.10.3)
Insulation (see 3.12.2.1.4)	
Elongation - Percent of original value, minimum	80
Sheath (see 3.12.18.5)	
Tensile p.s.i., minimum	1800
Elongation - Percent, minimum	50
Conductors (see 4.8.21)	

Table XIII - Types SHFR, DHFR, and THFR cable.

Type	Standard copper conductor	Number of conductors	Diameters				Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C (see 4.8.24)	Insulation resistance 1000 feet (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9)
			Over synthetic resin	Over varnished cambric	Over first asbestos or glass	Overall				
			Minimum				Minimum	Bare / Maximum	Minimum	
			Inch	Inch	Inch	Inch	Volts	Ohms	Megohms	Inches
SHFR-4	4(7)	1	0.115	0.185	0.247	0.540	8000	2.46	300	7
DHFR-4	4(7)	2	.115	.185	.247	.844	8000	2.50	400	10
THFR-4	4(7)	3	.115	.185	.247	.883	8000	2.50	400	11

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3.14 Type MDU cable. -

Construction

First - A standard conductor of bare, uncoated copper (see 3.11)

Second - A separator (see 3.12.14) at manufacturer's option.

Third - A wall of black synthetic resin insulation (see 3.12.3).

Fourth - Nineteen conductors, each marked with printed code (see 3.10.6) cabled together with a lay not greater than 24 times the pitch diameter of the outer layer. The color of print shall be white. Fillers may be employed at manufacturer's option in sizes 6 and 14. Fillers are required in sizes 23 and larger.

Fifth - A binder over conductor assembly.

Sixth - A thermoplastic sheath (see 3.12.19) applied tightly over and preferably adhering to the binder type.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XIV
Conductor resistance (see 4.8.24)	See table XIV
Insulation resistance (see 4.8.25)	See table XIV
Watertightness (see Appendix)	
Cubic inches in 6 hours maximum	1

Group B	Requirements
Cold bend (see 4.8.9 and table XIV)	No material shall drip or exude from end of cable.
Drip (see 4.8.8)	
Insulation (see 3.12.3)	
Tensile - p.s.i., minimum	1800
Elongation - Percent minimum	100
Sheath (see 3.12.19)	
Tensile - p.s.i., minimum	2500
Elongation - Percent, minimum	300
Set - Inches, maximum	1

Group C	Requirements
Flammability (see 4.8.10)	(See 4.8.10.3)
Insulation (see 3.12.3.1.3)	
Elongation - Percent of original value, minimum	90
Sheath (see 3.12.19.1.4 and 3.12.19.1.5)	
Tensile - Percent of original value, minimum	80
Elongation - Percent of original value, minimum	80
Heat shock (see 3.12.19.1.6)	No cracks
Conductor (see 4.8.21)	

Note: Type MDU replaces type MDGA of MIL-C-915.

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Table XIV - Type MDU cable.

Type	Standard copper conductor	Diameter over Insulation minimum	Sheath thickness minimum	Overall diameter maximum	Dielectric strength full reel minimum (see 4.8.19)	Conductor resistance per 1000 feet at 25°C. maximum (see 4.8.24)	Insulation resistance per 1000 feet minimum (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9)
		(Inch)	(Inch)	(Inch)	(Volts)	(Ohms)	(Megohms)	(Inches)
MDU-6	6(7)	0.156	0.060	1.000	2500	1.72	100	13
MDU-14	14(7)	.230	.060	1.395	2500	.785	100	19
MDU-23	23(7)	.296	.075	1.765	2500	.495	100	23
MDU-40	40(19)	.351	.075	2.040	2500	.290	90	27
MDU-60	60(37)	.407	.075	2.330	2500	.189	75	30

3.15 Type MDY cable. -

CONSTRUCTION

First - A stranded conductor of bare uncoated copper (see 3.11).
Second - A separator (see 3.12.14) at manufacturer's option.
Third - A wall of black synthetic resin insulation (see 3.12.3).
Fourth - Nineteen conductors, each marked with printed code (see 3.10.6) cabled together with a lay not greater than 24 times the pitch diameter of the layer. The color of the printed code shall be white. Fillers may be employed at manufacturer's option in sizes 6 and 14. Fillers are required in sizes 23 and larger.
Fifth - A binder over conductor assembly.
Sixth - Thermoplastic sheath (see 3.2.18) applied tightly over and preferable adhering to the binder tape.
Seventh - Braided metal armor and paint (see 3.12.20).
Eighth - The impervious sheath (see 3.12.19).

Test requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	
Volts, r.m.s., minimum	
Conductor to conductor	2500
Conductor to armor	2500
Armor to water	500
Insulation resistance (see 4.8.25)	See table XV
Conductor resistance (see 4.8.25)	See table XV
Watertightness (see Appendix I) with outer sheath removed, cubic inches in 6 hours maximum	1

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Test Requirements (Cont'd)

Group B	Requirements
Cold bend (see 4.8.9 and table XV)	
Drip (see 4.8.8)	0
Insulation (see 3.12.3)	
Tensile - p.s.i., minimum	1800
Elongation - Percent, minimum	100
Inner sheath (see 3.12.18)	
Tensile - p.s.i. minimum	1800
Elongation - Percent, minimum	225
Set - Inches, maximum	3/8
Outer sheath (see 3.12.19)	
Tensile - p.s.i., minimum	2500
Elongation - Percent, minimum	300
Set - Inches, maximum	1

Group C	Requirements
Flammability (see 3.8.10)	See 4.8.10.3
Insulation (see 3.12.3.1.3)	
Elongation - Percent original value, minimum	90
Inner sheath (see 3.12.18.5)	
Tensile - p.s.i. minimum	1800
Elongation - Percent, minimum	50
Outer sheath (see 3.12.19.1.4 and 3.12.19.1.5)	
Tensile - Percent of original value, minimum	80
Elongation - Percent of original value, minimum	80
Heat shock (see 3.12.19.1.6)	No cracks
Conductor (see 4.8.21)	

Note: Type MDY replaces type MDGL of MIL-C-915 and MDGT and MDGY of MIL-C-2681.

Table XV - Type MDY cable.

Type	Standard copper conductor	Diameter over Insulation minimum	Inner sheath thickness minimum	Diameter over Armor Nominal	Conductor resistance per 1000 feet at 25°C. (see 4.8.24) maximum	Insulation resistance per 1000 feet (see 4.8.25 maximum	Outer sheath thickness minimum	Overall diameter minimum	Overall diameter maximum	Cold bend mandrel diameter (see 4.8.9)
		(Inch)	(Inch)	(Inch)	(Ohms)	(Megohms)	(inch)	(inch)	(inch)	(inch)
MDY-6	6(7)	0.156	0.050	1.000	1.72	100	0.060	1.120	1.190	15
MDY-14	14(7)	.230	.050	1.380	.785	100	.060	1.500	1.570	21
MDY-13	23(7)	.296	.065	1.710	.495	100	.075	1.880	1.960	25
MDY-40	40(19)	.351	.065	1.960	.290	90	.075	2.150	2.240	29
MDY-60	60(37)	.407	.065	2.250	.189	75	.075	2.425	2.525	33

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3.16 Type TTHFWA cable. -

Construction

First - The standard conductor size 3/5(7) of bare, uncoated copper (see 3.11).
Second - A wall of synthetic resin insulation (see 3.12.2).
Third - Two (see note) conductors shall be twisted together with a lay not more than 2-1/4 inches.
Fourth - The specified number of twisted pairs shall be cabled together with a lay not greater than 10 inches. Fibrous fillers (see 3.12.11) shall not be employed. Color coded (see 3.10.7.2)
Fifth - An approved binder over the assembled pairs.
Sixth - With or without separator (see 3.12.14) at manufacturer's option.
Seventh - An impervious sheath (see 3.12.18).
Eighth - Braided metal armor (see 3.12.20).
Ninth - Paint (see 3.12.21).

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Watertightness (see Appendix)	
Cubic inches in 6 hours maximum	1
Dielectric strength (see 4.8.19)	(see table XVI)
Conductor resistance (see 4.8.24)	(see table XVI)
Insulation resistance (see 4.8.25)	(see table XVI)

Group B	Requirements
Cold bend (see 4.8.9 and table XVI)	
Insulation (see 3.12.2)	
Tensile - p.s.i., minimum	1800
Elongation - Percent, minimum	125
Sheath (see 3.12.18)	
Tensile - p.s.i., minimum	1800
Elongation - Percent, minimum	225
Set - Inch, maximum	30/64

Group C	Requirements
Flammability (see 4.8.10)	(see 4.8.10.3)
Insulation (see 3.12.2.1.4)	
Elongation - Percent of original value, minimum	80
Sheath (see 3.12.18.5)	
Tensile - p.s.i., minimum	1800
Elongation - Percent, minimum	50
Conductors (see 4.8.21)	

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Table XVI - Type TTHFWA cable.

Type	Number of pairs	Diameters		Dielectric strength (see 4.8.19)		Conductor resistance (d.c.) per 1,000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1,000 feet (see 4.8.25	Cold bend mandrel diameter (see 4.8.9)
		Over insulated conductor	Overall	Between conductors	Conductors to ground	Bare		
		Minimum		Minimum	Minimum	Maximum	Minimum	
		Inch	Inches	Volts	Volts	Ohms	Megohms	Inches
1/ TTHFWA-1-1/2	1-1/2	0.062	0.380	1,000	2,000		100	5
TTHFWA-3	3	.062	.500	1,000	2,000		100	7
TTHFWA-5	5	.062	.590	1,000	2,000		100	8
TTHFWA-10	10	.062	.690	1,000	2,000		100	9
TTHFWA-15	15	.062	.800	1,000	2,000	16.20	100	10
TTHFWA-20	20	.062	.880	1,000	2,000		100	11
TTHFWA-30	30	.062	1.030	1,000	2,000		100	13
TTHFWA-40	40	.062	1.130	1,000	2,000		100	15
TTHFWA-50	50	.062	1.265	1,000	2,000		100	16
TTHFWA-60	60	.062	1.350	1,000	2,000		100	17

1/ Type TTHFWA-1-1/2 comprises 1-1/2 pairs, or three, instead of two, conductors twisted.

3.17 Type PBTM cable. -

Construction

First - One conductor of bare copper and one conductor of constantan (see 3.7 and 3.11). Temperature electromotive force relationships shall follow National Bureau of Standards Circular No. 561. Any necessary joints shall be silver soldered in copper wire and welded in constantan wire.
Second - A wall of synthetic resin insulation (see 3.12.2).
Third - One copper and one constantan conductor twisted with a maximum lay of 3 inches to form a pair.
Fourth - The specified number of twisted pairs cabled with a maximum lay of 10 inches in the outer layer. Nonfibrous sealing-compound fillers in all valley spaces (see 3.7). Telephone circuit identification (see 3.10.7.2).
Fifth - An approved binder over the assembled pairs to form a round core.
Sixth - An impervious sheath (see 3.12.18).
Seventh - Braided metal armor (see 3.12.20) and paint (see 3.12.21).

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XVII
Conductor resistance (see 4.8.24)	See table XVII
Insulation resistance (see 4.8.25)	See table XVII
Watertightness (see Appendix)	
Cubic inches in 6 hours at 25 p.s.i., maximum	

Group B	Requirements
Cold bend (see 4.10.9 and table XVII)	(4.8.9)
Insulation (see 3.12.2)	
Tensile - P.s.i., minimum	1800
Elongation - percent, minimum	125
Sheath (see 3.12.18)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	225
Set - Inches, maximum	30/64

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Test Requirements (Cont'd.)

Group C	Requirements
Flammability (see 4.8.10)	(see 4.8.10.3)
Insulation (see 3.12.2.1.4)	
Elongation - Percent of original value, minimum	80
Sheath (see 3.12.18.5)	
Tensile - p.s.i., minimum	1800
Elongation - Percent, minimum	50

Table XVII - Type PBTM cable.

Type	Number of pairs	Conductors		Diameter over insulation	Diameter, overall	Dielectric strength	Conductor resistance (d.c.) per 1000 ft. at 70°F.(4.8.19)	Insulation resistance per 1000 ft. (4.8.25)	Cold bend mandrel diameter (4.8.9)
		Number of strands	Size of strands	Minimum	Maximum	Minimum	Maximum	Minimum	Nominal
			AWG	Inch	Inches	Volts	Loop, ohms	Meghoms	Inches
PBTM-5	5	7	30	0.062	0.590	1000	460.0	100	8
PBTM-15	15	7	30	.062	.800	1000	460.0	100	10
PBTM-30	30	7	30	.062	1.030	1000	460.0	100	13

3.18 Type DSWS-4 cable. -

Construction

First - A standard copper conductor size 4(7) with metallic coating (see 3.11).

Second - A wall of ozone-resistant synthetic insulation (see 3.12.4 or 3.12.6) or approved equivalent. Color coding (see 3.10.7) is required and may be obtained at manufacturer's option by colored insulation (see 3.10.2) or by applying over the insulation a flexible colored adhering film of lacquer, varnish, or paint which will not crack or peel, and which will exhibit its original color coding in the completed cable. An additional covering of clear polyamide may be employed at the manufacturer's option.

Third - Two conductors shall be twisted together with a lay not exceeding 5 inches. Fillers of synthetic rubber-like compound shall be used to attain a firm, well-rounded cross section. Fillers may be laid in separately with a minimum amount of fibrous material, or may be integral with or vulcanized to the sheath compound so as to meet the hydrostatic test (see 4.8.31). Fillers in the outer valleys may be extruded into the valleys with an integral belt approximately 1/32-inch thick over the conductor assembly. Fillers and belt shall be readily separable from, and without damage to, the conductor insulation and color coding.

Fourth - A binder tape over the conductor assembly is permitted but not required, and generally will not be used when the 1/32-inch belted construction is supplied.

Fifth - A shielding braid of 0.0063 inch diameter coated copper wire over the taped or belted assembly providing not less than 88 percent coverage. The braiding angle shall be approximately 30 and not more than 40 degrees.

Sixth - An overall sheath of total thickness 0.060 inch minimum average. If needed to meet electrical tests, the sheath may comprise an inner layer of insulating compound, such as synthetic rubber (see 3.12.4 or 3.12.6), and an outer layer of polychloroprene (see 3.12.17), or other compound approved as a component of completed qualified product. The sheath shall be readily separable from the underlying shield, leaving the shield clean for soldering. The overall diameter shall not exceed 0.800 inch nor be less than 0.770 inch.

Length - Nominal length (see 6.2.2.1): 500-foot coils.

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Test Requirements

Preproduction	Requirements
Insulation resistance per 1000 feet at room temperature from shield to water, 50 megohms, minimum after 90 days continuous submergence in water; values recorded at 1-week intervals.	
Flexing endurance (see 4.8.14) - 1,000 cycles of bending on a 5-foot specimen, 4-inch diameter mandrel at 40°C. Upon successful completion of this test, specimen shall meet the dielectric test and the open end hydrostatic test.	There shall be no visible derangement.

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (4.8.19)	
Volts, root mean square (r.m.s.), minimum	
5 minutes conductor to conductor	10,000
5 minutes conductor to shield	10,000
5 minutes shield to water	500
Volts, d.c., (minimum)	
15 minutes conductor to conductor	25,000
15 minutes conductor to shield	25,000
Insulation resistance (see 4.8.25)	
Megohms per 1000 feet (minimum)	
Shield to water	100
Conductor and conductor to shield	200
Capacitance (see 4.8.27.3)	
Picofarads per foot at 25°C. at 1 kilocycle (white insulated conductor with black conductor connected to shield at one end).	40 ± 10
Hydrostatic (see Appendix I and 4.8.31.1)	
Cubic inches in 2 hours at 1000 p.s.i.	0
Slippage of cable or component, inch, maximum	0.25

Group B	Requirements
Insulation (see 3.12.4 or 3.12.6)	
Tensile - p.s.i., minimum	600
Elongation - Percent, minimum	225
Tensile - p.s.i., minimum	600
Elongation - Percent minimum	450
Sheath (see 3.12.17)	
Tensile - p.s.i., minimum	1800
Elongation - Percent, minimum	300
Set - Inches, maximum	3/8

Group C	Requirements
Insulation (see 3.12.4.1.1.4 and 3.12.4.1.1.5)	
Tensile - Percent of original value, minimum	50
Elongation - Percent of original value, minimum	50
Insulation (see 3.12.6.1.4)	
Tensile - Percent of original value, minimum	70
Elongation - Percent of original value, minimum	75
Sheath (see 3.12.17.4 and 3.12.17.5)	
Tensile - Percent of original value, minimum	65
Elongation - percent of original value, minimum	65
Conductors (see 4.8.21 and 4.8.22)	

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3.19 Types DSS, TSS, FSS, and 7SS cable. -

Construction

First - A standard copper conductor (see 3.11) with metallic coating (see 3.11.3)
Second - A wall of synthetic insulation (see 3.12.4). Color coding (see 3.10.7) is required and may be obtained at manufacturer's option by colored insulation (see 3.10.2), except brown, for the seventh in 7SS or by applying over the insulation a flexible adhering film of lacquer, varnish, or paint which will not crack or peel, and which will exhibit its original color coding in the completed cable.
Third - Types DSS, TSS, FSS and 7SS shall have 2, 3, 4 and 7 conductors, respectively. The required number of conductors shall be twisted together with a lay not exceeding that specified in table XVIII. Fillers of synthetic rubber-like compound shall be used to obtain a firm, well-rounded cross-section so as to meet the hydrostatic test (see 4.8.31). Fillers in the outer valleys may be extended into the valleys with an integral belt approximately 1/32 inch thick over the conductor assembly. Fillers, belt, and sheath (see "Fifth" herein) shall be readily separable from and without damage to the conductor, the conductor insulation, and color coding.
Fourth - A shielding braid of 0.0063 inch diameter tin coated copper wire over the belted assembly, providing not less than 88 percent coverage. The braiding angle shall be 50 to 55 degrees.
Fifth - An overall sheath of total thickness specified in the table XVIII. If needed to meet electrical tests, the sheath may comprise an inner layer of insulating compound (see 3.12.4 or 3.12.6) and an outer layer of polychloroprene (see 3.12.17) or chlorosulfonated polyethylene.

Length Requirement

Nominal length (see 5.1.1.1) 500-foot coils for all types and sizes.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	
Volts, r.m.s., minimum	
5 minutes, conductor to conductor	3000
5 minutes, conductor to shield	1000
5 minutes, shield to water	500
Conductor resistance (see 4.8.24)	(see table XVIII)
Insulation resistance (see 4.8.25)	
Megohms per 1000 feet at 25°C.	
Between conductors and conductors to shield, minimum	500
Between shield and water, minimum	250
Capacitance, (see 4.8.27)	
DSS only - Between black and white insulated with black conductor connected to shield at one end.	
Picofarads per foot at 1 kilocycle	40±2.5
Hydrostatic (see Appendix I and 4.8.31.1)	
Cubic inches in 2 hours at 1000 p.s.i.	0
Slippage of cable or component, inch, maximum	0.25
Pressure cycling (see 4.8.31.2)	
P.s.i. pressure minimum	1000
All electrical tests must be met before, during, and after pressure cycling.	
Capacitance of DSS-3 cable shall not vary more than 10 percent during pressure cycling.	

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Test Requirements (Cont'd)

Group B	Requirements
Insulation (see 3.12.4)	
Tensile - P.s.i., minimum	600
Elongation - Percent, minimum	225
Sheath (see 3.12.17)	
Tensile - p.s.i., minimum	1800
Elongation - percent minimum	300
Set - Inch, maximum	3/8

Group C	Requirements
Flammability (see 4.8.10)	(see 4.8.10.3)
Insulation (see 3.12.4.1.2.3)	
Elongation - percent of original value, minimum	70
Sheath (see 3.12.17.4 and 3.12.17.5)	
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Conductors (see 4.8.21 and 4.8.22)	
Flexing endurance (see 4.8.14)	
1000 cycles of bending on a 5-foot specimen, 4-inch diameter mandrel at - 40°C. Upon successful completion of this test, specimens shall meet the dielectric test (a), (b), (c), of group A tests and also the open end hydrostatic test (see 4.8.31.1).	No visible derangement

Table XVIII - Types DDS, TSS, FSS and 7SS[1/] cable.

Type	Standard copper conductor	Number of conductors	Insulation thickness [2/]	Lay	Sheath thickness [2/]	Overall diameter		Conductor resistance (d.c.) per 1000 feet at 25°C.
			Minimum	Maximum	Minimum	Minimum	Maximum	Maximum
			Inch	Inch	Inch	Inch	Inch	Ohms
DSS-2	2(7)	2	0.020	2.0	0.050	0.370	0.390	6.64
DSS-3	3(7)	2	.025	2.0	.080	.480	.500	4.15
DSS-4	4(7)	2	.025	2.5	.065	.480	.500	2.57
TSS-2	2(7)	3	.020	2.0	.050	.385	.400	6.64
TSS-3	3(7)	3	.025	2.5	.070	.480	.500	4.15
TSS-4	4(7)	3	.025	2.5	.055	.480	.500	2.57
FSS-2	2(7)	4	.020	2.5	.080	.480	.500	6.64
FSS-3	3(7)	4	.025	2.5	.060	.480	.500	4.15
FSS-4	4(7)	4	.025	3.0	.085	.600	.625	2.57
7SS-2	2(7)	7	.025	3.0	.085	.600	.625	6.64

[1/] Type 7SS replaces MSS-6.

[2/] Because of the compacted construction necessary to meet the hydrostatic test required on these cables, the centering and and circularity of the insulation and sheath (see 3.8.4) is waived provided the specified minimum wall (see 3.8.3) is met, and the least thickness at any point (see 3.8.3) is not less than 80 percent of the specified value.

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3.20 Type TSP cable. -

Construction

First - A standard copper conductor with metallic coating, size 3/5(7) (see 3.11).
Second - A wall of synthetic resin insulation (see 3.12.2).
Third - Two conductors twisted together with a maximum lay of 2-1/2 inches to form a twisted pair with color-coded insulation (see 3.10.7.2).
Fourth - The specified number of twisted pairs cabled together with a maximum lay of 10 inches in the outer layer. No fibrous fillers shall be employed.
Fifth - A suitable binder or separator or both may be applied over the assembled pairs.
Sixth - An impervious sheath (see 3.12.18) of the thermoplastic type.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XIX
Conductor resistance (see 4.8.24)	See table XIX
Insulation resistance (see 4.8.25)	See table XIX
Watertightness (see Appendix)	
Cubic inches in 6 hours at 25 p.s.i., maximum	1

Group B	Requirements
Insulation (see 3.12.2)	
Tensile - p.s.i., minimum	1800
Elongation - percent, minimum	125
Sheath (see 3.12.18)	
Tensile - p.s.i., minimum	1800
Elongation - percent, minimum	225
Set - Inch, maximum	30/64

Group C	Requirements
Insulation (see 3.12.2.1.4)	
Elongation - percent of original value, minimum	80
Sheath (see 3.12.18.5)	
Tensile - p.s.i., minimum	1800
Elongation - percent, minimum	50
Conductors (see 4.8.21 and 4.8.22)	
Flammability (see 4.8.10)	(See 4.8.10.3)

Table XIX - Type TSP cable.

Type	Number of		Thickness of		Overall diameter	Dielectric strength conductor to ground (see 4.8.19)	Insulation resistance per 1000 ft. (see 4.8.25)	Conductor resistance per 1000 ft. at 25°C. (see 4.8.24)
	Conductors	Pairs	Insulation	Sheath				
			Minimum	Minimum	Maximum	Minimum	Minimum	Maximum
			Inch	Inch	Inches	Volts	Megohms	Ohms
TSP-11	22	11	0.015	0.080	0.735	2,500	100	16.6
TSP-31	62	31	.015	.080	1.062	2,500	100	16.6

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3.21 Type TTRSA cable. -

Construction

First - A standard conductor of bare or tinned copper (see 3.11) size 1(7).
Second - A wall of polythene (see 3.12.5) to the diameter and within the dimensional tolerances necessary to comply with all performance test requirements and overall diameter of the completed cable. A thin jacket of suitable clear polyamide insulating material conforming to type III, grade E of MIL-M-20693 shall be applied over the polythene.
Third - Two such insulated conductors twisted together with maximum lay of 2-1/2 inches to form a balanced pair. One conductor in each pair shall be white or natural, and its mate shall be a dark contrasting color such as blue or black.
Fourth - A braided shield of 0.005 or 0.0063-inch diameter bare or tinned copper wire (see 3.12.16.6) on each pair. The braid angle shall be approximately 30 degrees.
Fifth - Shield insulation of one white polyester tape (conforming to type G, form T of MIL-I-631) applied helically with overlap and covered with a nominal 3 mil thickness of extruded clear polyamide conforming to type III, grade E of MIL-M-20693. Serial printing on the tape under the polyamide jacket at 3-inch maximum intervals shall be black or red on a colored white or gray background. Alternate legends shall be inverted; for example:

10 ORANGE BLACK BLACK 10 ORANGE

The printing shall be legible without removal of the polyamide jacket. Color coding (see 3.10.7) shall be provided in sequence for each shielded pair.
Sixth - The specified number of shielded pairs cabled together with a lay not exceeding 14 inches. Fillers (see 3.12.11) shall be employed, where necessary, to obtain a firm, well rounded assembly.
Seventh - A wall of felted asbestos (see 3.12.9) or glass fiber (see 3.12.10) over the assembly of pairs.
Eighth - An impervious sheath (see 3.12.18).
Ninth - Braided metal armor (see 3.12.20).
Tenth - Paint (see 3.12.21).

Test Requirement

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XX
Conductor resistance (see 4.8.24)	See table XX
Insulation resistance (see 4.8.25)	
Megohms per 1000 feet minimum	1000
Capacitance (see 4.8.27.3)	
Picofarads, maximum, per foot at 25°C. and 1 megacycle per second (mc./s)	28
Impedance (4.8.27) Ohms, maximum at 25°C. and 1 mc/s.	75 ± 5
Capacitance unbalance (see 4.8.32) percent, maximum	10

Group B	Requirements
Cold bend (see 4.8.9 and table XX)	
Insulation (see 3.12.5)	
Tensile - P.s.i., minimum	1400
Elongation - Percent, minimum	300
Sheath (see 3.12.18)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	225
Set - Inch, maximum	30/64

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Test Requirements (Cont'd)

Group C	Requirements
Insulation (see 3.12.5.1.4) - Percent of original value	60
Sheath (see 3.12.18.5)	
Tensile P.s.i., minimum	1800
Elongation - Percent, minimum	50
Flammability (see 4.8.10)	(see 4.8.10.3)
Thermoplastic flow (see 4.8.11) no electrical failure (hours)	6
Conductors (see 4.8.21)	

Table XX - Type TTRSA cable.

Type	Number of shielded pairs	Overall diameter (maximum)	Dielectric strength, (min.)			Conductor resistance (d.c.) per 1000 feet at 25°C. (maximum)	Cold bend mandrel diameter
			Conductor to mate	Conductor to shield	Shield to shield		
		Inches	Volts	Volts	Volts	Ohms	Inches
TTRSA-2	2	0.740	2000	1000	200	10.3	9
TTRSA-4	4	.800	2000	1000	200	10.3	10
TTRSA-6	6	.940	2000	1000	200	10.3	12
TTRSA-8	8	1.050	2000	1000	200	10.3	13
TTRSA-10	10	1.140	2000	1000	200	10.3	15
TTRSA-12	12	1.160	2000	1000	200	10.3	15
TTRSA-16	16	1.250	2000	1000	200	10.3	16

3.22 Types DBSP, TBSP, and FBSP cable.-

Construction

First - A standard conductor of bare, uncoated copper (see 3.11). A separator (see 3.12.14) shall be used when synthetic insulation (see 3.12.4 and 3.16.6) is employed.

Second - A wall of extruded synthetic insulation.

Third - The specified number of conductors shall be twisted together with a lay not greater than 20 times the pitch diameter. Fillers may be employed only where necessary to obtain a smooth, uniform overall diameter on the completed cable.

Fourth - A thin copper tape not more than 0.003 inch thick, applied helically over the conductor assembly with a positive lap.

Fifth - A gray impervious sheath of polyvinylchloride.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	See table XXI
Dielectric strength (see 4.8.19)	See table XXI
Conductor resistance (see 4.8.24)	See table XXI
Insulation resistance (see 4.8.25)	

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Test Requirements (Cont'd)

Group B	Requirements
Cold bend (see 4.8.9) - Bend over a mandrel having a diameter twice that of the cable specimen.	No cracking of sheath.
Insulation (see 3.12.2, 3.12.3, 3.12.4 or 3.12.6)	Applicable paragraph

Group C	Requirements
Hot oil (see 4.8.17)	
Sheath compound - Elongation, percent minimum	200
Heat shock (see 4.8.30)	
Dissection of specimen after test	No damage to insulation, sheath or tape.
Insulation (see 3.12.2.1.4; 3.12.3.1.3; 3.12.4.1.1.4; 3.12.4.1.1.5; 3.12.4.1.2.3; 3.12.6.1.4)	Applicable paragraph
Flammability (see 4.8.10)	See 4.8.10.3

Table XXI - Types DBSP, TBSP, and FBSP cable.

Size	Standard copper conductor	Insulation thickness	Sheath thickness	Maximum overall diameter			Dielectric strength (see 4.8.19)		Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)		Insulation resistance per 1000 ft. (see 4.8.25)
				Type DBSP	Type TBSP	Type FBSP	Shield to ground	Conductors to ground	Bare	Tinned	
		Minimum	Minimum				Minimum	Minimum	Maximum	Maximum	Minimum
		Inch	Inch	Inch	Inch	Inch	Volts	Volts	Ohms	Ohms	Megohms
3/5	3/5(7)	0.015	0.030	0.206	0.216	0.232	500	1,000	16.0	16.7	100
1	1(7)	.015	.030	.224	.234	.253	500	1,000	10.1	10.5	100
2	2(7)	.015	.030	.245	.257	.278	500	1,000	6.37	6.64	100
3	3(7)	.020	.035	.303	.319	.346	500	1,500	4.00	4.15	100
4	4(7)	.025	.035	.359	.378	.413	500	1,500	2.51	2.57	100
6	6(7)	.030	.040	.426	.450	-----	500	1,500	1.72	1.76	100
9	9(7)	.030	.045	.471	.498	-----	500	1,500	1.25	1.27	100
14	14(7)	.035	.045	.552	.586	-----	500	2,000	.785	.800	100
23	23(7)	.035	.050	.639	.677	-----	500	2,000	.495	.503	100

3.23 Type FCSF cable.-

Construction

First - Standard conductors of bare, uncoated copper (see 3.7, 3.11 and table XXII). At manufacturer's option, a concentric strand (see 3.11.4.1) of 37 wires in lieu of 20(49) and 91 wires in lieu of 66(133) may be furnished, provided the specified conductor resistance is not exceeded.
Second - A separator (see 3.12.14) if bare copper conductor is employed.
Third - A wall of butyl rubber (see 3.12.6).

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Construction (Cont'd)

Fourth - A compound filled tape (see 3.12.15) or an inner braid (see 3.12.16.2) on each conductor is permitted but not required. Color coding, one black and one white on each conductor size is required.

Fifth - The four conductors cabled together with fillers of synthetic rubber or polychloroprene to obtain a firm, well-rounded assembly, with a minimum amount of fibrous material. Outer fillers may be integral with the sheath compound or may be laid in separately.

Sixth - A suitable binder over the conductor assembly is permitted but not required.

Seventh - A polychloroprene sheath (see 3.12.17). Fillers and sheath material in the completed cable shall be readily separable from, and without damage to, the conductor insulation. Reinforcement (see 3.12.12) shall be applied under the sheath, or under at least half the sheath thickness.

Length Requirement

Shipping lengths (feet) - FCSF-66 - 1590 minimum, 1650 maximum.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXII
Conductor resistance (see 4.8.24)	See table XXII
Insulation resistance (see 4.8.25)	See table XXII
Watertightness (see Appendix)	
Cubic inches in 6 hours maximum	20

Group B	Requirements
Cold bend (4.8.9). When bent two turns around mandrel with diameter twice that of specimen at -30°C.	No cracking of sheath.
Insulation (see 3.12.6)	
Tensile - P.s.i., minimum	600
Elongation - Percent, minimum	450
Sheath (see 3.12.17)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.6.1.4)	
Tensile - Percent of original value, minimum	70
Elongation - Percent of original value, minimum	75
Sheath (see 3.12.17.4 and 3.12.17.5)	
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Conductors (see 4.8.21)	

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Table XXII - Type FCSF cable.

Type	Standard copper conductor	Number of conductors	Insulation thickness	Lay of conductors	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet (see 4.8.25)
		total	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
			Inch	Inches	Inches	Inches	Volts	Ohm	Megohms
FCSF-66	2 of 20(49) and 2 of 66(133)	4	0.062 .062	12	1.300	1.370	2500	0.573 .174	300

3.24 Type MCSF cable.-

Construction

First - Two bare or coated bronze conductors (approximately No. 6 AWG) each consisting of 49 by 0.023 inches bunched strand, and two bare or coated bronze conductors (approximately No. 1 AWG) each consisting of 7 x 19 by 0.025 inch strands.
Second - A compound filled tape (see 3.12.15) over each conductor, at option of the contractor.
Third - A wall of SBR on each conductor.
Fourth - Four conductors cabled together. Color coding; one black and one white of each conductor size. Fillers shall be integral with sheath; or, if laid in separately, a binder tape or reinforcement shall be applied under the sheath.
Fifth - A polychloroprene sheath (see ASTM-D752-62). Fillers and sheath material in the completed cable shall be separable from, and without damage to, the conductor insulation or color coding.

Length Requirement

Shipping lengths (feet) - 1590 minimum, 1650 maximum.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXIII
Conductor resistance (see 4.8.24)	See table XXIII
Insulation resistance (see 4.8.25)	See table XXIII
Watertightness (see Appendix I)	
Cubic inches in 6 hours maximum	10

Group B	Requirements
Cold bend (see 4.8.9) 2 turns at -30°C.	
Mandrel twice diameter of specimen	No damage
Tension test (see 4.8.20)	No physical or
12,000 pounds load minimum	electrical damage.
Insulation (ASTM D470-59T)	
Tensile - P.s.i., minimum	700
Elongation - Percent, minimum	300
Sheath (see ASTM D752-62)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	300

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Test Requirements (Cont'd)

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (after accelerated aging)	
Air pressure test (20) hours at 80 p.s.i. and 127°C.)	
Tensile - Percent of original value, minimum	50
Elongation - Percent of original value, minimum	50
Oxygen - pressure test (168 hours at 300 p.s.i. and 80°C.)	
Tensile - Percent of original value, minimum	50
Elongation - percent of original value minimum	50
Sheath (see ASTM D752-62)	
Oxygen pressure test (168 hours at 80 ± 1°C.)	
Tensile - Percent of original value, minimum	50
Elongation - Percent of original value, minimum	50
Oil immersion test (18 hours at 121 ± 1°C.)	
Tensile and elongation - Percent of original value, minimum	60

Table XXIII - Type MCSF cable.

Type	Number of conductors	Insulation thickness	Lay of conductors	Sheath thickness	Overall diameter	Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet (see 4.8.25)	Weight per 1000 feet
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
		Inches	Inches	Inch	Inches	Volts	Ohms	Megohms	Pounds
MCSF-4	4	0.070	20	0.125	1.50	2500	2 conductors 0.573 2 conductors 0.175	1000	1625

3.25 Type SDU cable.-

Construction

First - A standard conductor of bare, uncoated copper (see 3.11), except that all of the 36 strands comprising the outer layer of the stranded conductor shall be metallically coated (see 3.11.3).

Second - A wall of butyl rubber insulation (see 3.12.6).

Third - With or without a nonmetallic reinforcement at manufacturer's option. When reinforcement is employed, it may be applied under or over the insulation but not over the sheath.

Fourth - A polychloroprene sheath (see 3.12.17). Manufacturer's identification shall consist of a molded marking (see 3.9.4) at intervals not greater than 25 feet, showing the cable type letter designation, the manufacturer's name or trade-mark, and the year of manufacture.

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Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXIV
Conductor resistance (see 4.8.24)	See table XXIV
Insulation resistance (see 4.8.25)	See table XXIV
Hydrostatic (see 4.8.31 and Appendix)	
Cubic inches in 2 hours at 500 p.s.i.	1

Group B	Requirements
Insulation (see 3.12.6)	
Tensile - P.s.i., minimum	600
Elongation - Percent, minimum	450
Sheath (see 3.12.17)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.6.1.4)	
Tensile - Percent of original value, minimum	70
Elongation - Percent of original value, minimum	75
Sheath (see 3.12.17.4 and 3.12.17.5)	
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Conductor (see 4.8.21 and 4.8.22)	

Table XXIV - Type SDU cable.

Type and size	Standard copper conductor	Insulation thickness	Sheath thickness	Overall diameter	Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet at 15.6°C. (see 4.8.25)
		Minimum	Minimum	Maximum	Minimum	Maximum	Minimum
		Inch	Inch	Inches	Volts	Ohms	Megohms
SDU-500	500(127)	0.050	0.062	1.210	2000	0.0216	200
SDU-800	800(127)	.066	.062	1.465	2000	.0136	200

3.26 Type SSF cable.-

Construction

First - A standard conductor of bare, uncoated copper (see note 1).
Second - A separator (see 3.12.14).
Third - A wall of synthetic rubber (see 3.12.4) or butyl rubber (see 3.12.6) having a minimum wall thickness (see 3.8.3) of 0.075 inch.
Fourth - A reinforcement (see 3.12.12) at manufacturer's option.
Fifth - A polychloroprene sheath (see 3.12.17) bonded to the underlying insulation, having a minimum wall thickness (see 3.8.3) of 0.075 inch.

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Length Requirement

Shipping lengths (feet) - 1045 minimum, 1155 maximum

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXV
Conductor resistance (see 4.8.24)	See table XXV
Insulation resistance (see 4.8.25)	See table XXV

Group B	Requirements
Cold bend - Two turns around mandrel twice the diameter of the specimen at -30°C.	(See 4.8.9)
Insulation (3.12.4 or 3.16.6)	
Tensile - P.s.i., minimum	600
Elongation - Percent minimum (see 3.12.4)	225
Elongation - Percent minimum (see 3.12.6)	450
Sheath (see 3.12.17)	
Tensile - P.s.i., minimum	1800
Elongation - Percent minimum	300

Group C	Requirements
Accelerated aging (see 4.8.1.3)	
Insulation (see 3.12.4.1.1.4; 3.12.4.1.1.5)	
Tensile and elongation percent of original value minimum	50
Insulation (see 3.12.5.1.4)	
Tensile-Percent of original value, minimum	70
Elongation-Percent of original value, minimum	75
Sheath (see 3.12.17.4 - 3.12.17.5)	
Tensile and elongation-percent of original value minimum	65
Flammability (see 4.8.10)	see 4.8.10.3
Conductors (see 4.8.21)	

Table XXV - Type SSF cable.

Type	Standard copper conductor	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. 1/	Insulation resistance per 1000 feet (see 4.8.25)
		Minimum	Maximum	Minimum	Maximum	Minimum
		Inches	Inches	Volts	Ohm	Megohms
SSF-300	300 (259)	1.020	1.100	2500	0.0388	100

1/ The copper shall be hard drawn with a minimum tensile strength of 67,200 p.s.i., and a minimum elongation of 0.85 before stranding. The resistance specified is based on 102.7 percent of the values required for annealed copper.

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3.27 Types DCOP and TCOP cable.-

Construction

First - A standard conductor of bare, uncoated copper (see 3.11).
Second - A separator (see 3.12.14) when synthetic rubber insulation (see 3.12.4) is used, or with or without separator (see 3.12.14), at manufacturer's option when synthetic resin insulation (see 3.12.2) is used.
Third - A wall of synthetic insulation (see 3.12.2 or 3.12.4).
Fourth - Plain colored insulation is required on size DCOP-1 and DCOP-1-1/2 1/. An inner braid (see 3.12.16) will be permitted on sizes DCOP-2 and TCOP-2.
Fifth - The specified number of conductors shall be twisted together with either a right-hand or left-hand lay, with the lay not greater than that specified in table XXVI. Cotton tie cord (see 3.12.12) simulating a third or dummy conductor shall be laid straight as a central core is size DCOP-1-1/2. Fillers (see 3.12.11) shall be employed, where necessary, to obtain a firm, well-rounded assembly.
Sixth - A braid of minimum No. 60/2 ply cotton or approved alternate shall be used over the conductor assembly of size DCOP-1 and is permitted but not required on size DCOP-1-1/2. An approved binder shall be used over the conductor assembly of sizes DCOP-2 and TCOP-2.
Seventh - A sheath of polychloroprene (see 3.12.17) or an impervious sheath (see 3.12.18).

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXVI
Conductor resistance (see 4.8.24)	See table XXVI
Insulation resistance (see 4.8.25)	See table XXVI

Group B	Requirements
Flexing endurance (see 4.8.14)	
Twisting and bending cycles at 75°C. minimum	1000
Twisting and bending cycles at -30°C. minimum	500
Insulation (see 3.12.4)	When applicable
Tensile - P.s.i., minimum	600
Elongation - Percent, minimum	225
Insulation (see 3.12.2)	When applicable
Tensile - P.s.i. minimum	1800
Elongation - Percent minimum	125
Sheath (see 3.12.17 or 3.12.18)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	225
Set - inch, maximum	30/64

Group C	Requirements
Insulation (see 3.12.4.1.2.3)	When applicable
Elongation - Percent of original value, minimum	70
Insulation (see 3.12.2.1.4)	When applicable
Elongation - Percent minimum	80

1/ Because of peculiar lay-up of type DCOP-1-1/2 cable, the centering and circularity requirements (see 3.8.4) shall not apply, provided the minimum thickness requirements (see 3.8.3) are met.

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Test Requirements (Cont'd)

Group C	Requirements
Sheath (see 3.12.17.4 and 3.12.17.5 or 3.12.18.5)	
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	50
Conductor (see 4.8.21)	

Table XXVI - Type DCOP and TCOP cable.

Type	Standard copper conductor	Insulation thickness	Lay of conductors	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet (see 4.8.25)
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
		Inch	Inches	Inch	Inch	Volts	Ohms	Megohms
DCOP-1	1(10)	0.015	1	0.235	0.250	1000	11.60	100
DCOP-1-1/2	1-1/2(16)	.015	3/8	.300	.315	1500	13.10	100
DCOP-2	1-1/2(16)	.015	2	.310	.330	1500	7.22	100
TCOP-2	1-1/2(16)	.015	2	.325	.345	1500	7.22	100

3.28 Type MHOF cable.-

Construction

First - The standard conductor size 2-1/2 (19) of bare, uncoated copper (see 3.11).
Second - A wall of synthetic resin insulation (see 3.12.2) to a diameter of 0.092 inch, minimum.
Third - Printed circuit identification (see 3.10.5).
Fourth - The specified number of conductors shall be cabled together in one or more layers, all in a left-hand direction, with a minimum amount of asbestos or glass fiber fillers (see 3.12.11) in the interstices between conductors. The length of lay of the first and successive layers shall be a suitable function of the pitch diameter of that layer, designed to meet the flexing endurance test on the completed cable. The conductors may be laid around a central core of asbestos or glass fiber (see 3.12.11). A single conductor shall not be used as a center for the cable core.
Fifth - An approved binder over the conductor assembly.
Sixth - An impervious sheath (see 3.12.18) of the thermoplastic type. The minimum sheath thickness shall be not less than 0.060 inch.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (4.8.19)	See table XXVII
Conductor resistance (4.8.24)	See table XXVII
Insulation resistance (4.8.25)	See table XXVII

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Test Requirements (Cont'd)

Group B	Requirements
Flexing endurance (4.8.14)	
Twisting and bending cycles at 75°C., minimum	1000
Twisting and bending cycles at -30°C., minimum	500
Cold bend (4.8.9 and table XXVII)	
Insulation (see 3.12.2)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	125
Sheath (see 3.12.18)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	225
Set - Inch, maximum	30/64

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.5.1.4)	
Elongation - Percent of original value, minimum	80
Sheath (see 3.12.18.5)	
Tensile - P.s.i., minimum	1400
Elongation - Percent, minimum	50
Conductors (see 4.8.21)	

Table XXVII - Type MHOF cable.

Type	Number of conductors	Overall diameter	Dielectric strength (see 4.8.19)		Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet (see 4.8.25)	Cold bend mandrel diameter (see 4.8.9)	
			After flame test	Full reel	Bare		Individual conductors	Completed cable
		Inches Min. Max.	Minimum	Minimum	Maximum	Minimum		
			Volts	Volts	Ohms	Megohms	Inch	Inches
MHOF-7	7	0.465 0.500	750	2,000	4.75	500	0.50	6.0
MHOF-10	10	.540 .585	750	2,000		500	0.50	7.0
MHOF-14	14	.585 .635	750	2,000		500	.50	7.5
MHOF-19	19	.650 .705	750	2,000		500	.50	8.5
MHOF-24	24	.735 .795	750	2,000		500	.50	10.0
MHOF-30	30	.775 .835	750	2,000		500	.50	10.5
MHOF-37	37	.855 .925	750	2,000		500	.50	11.5
MHOF-44	44	.925 1.000	750	2,000		500	.50	12.0

3.29 Type TTOP cable.-

Construction

First - A standard conductor of bare, uncoated copper (see 3.11), size 1(10).
Second - A wall of synthetic resin insulation (see 3.12.2). Color coding (see 3.10.2) unless accomplished under the Third step.

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Construction (Cont'd)

Third - With or without an inner braid (see 3.10.4), at manufacturer's option suitably treated to prevent fraying and to provide adequate resistance to abrasion due to flexing of the cable. Color coding of this covering is required unless accomplished under the Second step.
Fourth - Two conductors shall be twisted together with a suitable lay to form a twisted pair with color-coded insulation (see 3.10.7.2)
Fifth - The specified number of twisted pairs shall be cabled together in one or more layers all in a left-hand direction. The length of lay of the first and successive layers shall be suitable function of the pitch diameter of that layer, and shall be designed to meet the flexing endurance test on completed cable. Circularity to cross section shall be obtained by the use of a minimum amount of asbestos or glass fiber fillers (see 3.12.11) lightly impregnated. A single pair shall not be used as a center for the cable core.
Sixth - An approved binder over the assembled pairs.
Seventh - An impervious sheath (see 3.12.18) of the thermoplastic type. The minimum sheath thickness shall be not less than 0.060 inch.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXVIII
Conductor resistance (see 4.8.24)	See table XXVIII
Insulation resistance (see 4.8.25)	See table XXVIII

Group B	Requirements
Flexing endurance (see 4.8.14)	
Twisting and bending cycles at 75°C. minimum	1000
Twisting and bending cycles at -30°C. minimum	500
Insulation (see 3.12.2)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	125
Sheath (see 3.12.18)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	225
Set - Inch, maximum	30/64

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.2.1.4)	
Elongation - Percent of original value, minimum	80
Sheath (see 3.12.18.5)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	50
Conductors (see 4.8.21)	

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Table XXVIII - Type TTOP cable.

Type	Number of pairs	Overall diameter	Dielectric strength (see 4.8.19)		Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet (see 4.8.25)
			Between conductors	Conductors to ground	Bare	
			Minimum	Minimum	Maximum	Minimum
		Inch	Volts	Volts		Megohms
TTOP-3	3	0.480	1,000	2,000	11.4	100
TTOP-5	5	.590	1,000	2,000		100
TTOP-10	10	.700	1,000	2,000		100
TTOP-15	15	.830	1,000	2,000		100

3.30 Type MCOS cable.-

Construction.

First - A standard conductor of bare, uncoated copper (see 3.11).
Second - A wall of synthetic resin insulation (see 3.12.2).
Third - Sizes 2, 4 and 7 shall have 2, 4 and 7 conductors, respectively, with plain colored insulation (see 3.10.2).
Size 5 shall have five insulated conductors comprising one shielded pair and three singles. The shielded pair shall comprise two insulated conductors twisted together with a braided shield of either 0.005 or 0.0063 inch diameter copper wire (see 3.12.16.6) over the pair. The colors shall be black and white in the pair; singles shall be one red, one green, and one blue and shall be obtained by colored insulation (see 3.10.2).
Size 6 shall have six insulated conductors comprising two shielded pairs and two singles. Each shielded pair shall comprise two insulated conductors twisted together with a braided shield of either 0.005 or 0.0063 inch diameter copper wire (see 3.12.16.6) over the pair. The colors shall be black and white in one pair; blue and red in the other pair; one yellow single, and one green single and shall be obtained by colored insulation (see 3.10.2) or color-coded braid (see 3.10.4) at the manufacturer's option.
Fourth - The required single conductors and shielded pairs shall be cabled together all in a left-hand direction with a minimum amount of filler (see 3.12.11) in the interstices. The length of lay shall be a suitable function of the pitch diameter, designed to meet the flexing endurance (see 4.8.14) on completed cable and, in addition for sizes 5 and 6 only, the electrical requirements of 4.8.29. The conductors of size 7 shall be laid around a central core of cotton, asbestos, or glass fiber (see 3.12.11). A single conductor shall not be used as a center for the cable core.
Fifth - A braided shield of either 0.005-or 0.0063-inch diameter copper wire (see 3.12.16.6) plus a separator (see 3.12.14) over the assembly of sizes 2, 4 and 7. At the manufacturer's option a separator (see 3.12.14) or a tape (see 3.12.15) may be applied under the copper braid.
An approved binder over the assembly of sizes 5 and 6.
Sixth - For sizes 2, 4, 6 and 7, an impervious sheath (see 3.12.18) of the thermoplastic type, having a minimum thickness of 0.060 inch. For size 5, a polychloroprene sheath (see 3.12.17) or an impervious sheath (see 3.12.18), having a minimum thickness of 0.030 inch.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXIX
Conductor resistance (see 4.8.24)	See table XXIX
Insulation resistance (see 4.8.25)	See table XXIX

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Test Requirements (Cont'd)

Group B	Requirements
Flexing endurance (see 4.8.14)	
Twisting and bending cycles at 75°C. minimum	1000
Twisting and bending cycles at -30°C. minimum	500
Induced voltage (4.8.29) - (MCOS-5;MCOS-6 only)	
Microvolts, maximum	5
Insulation (see 3.12.2)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	125
Sheath (see 3.12.17)	When applicable
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8
Sheath (see 3.12.18)	When applicable
Tensile P.s.i., minimum	1800
Elongation - Percent, minimum	225
Set - Inch, maximum	30/64

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.2.1.4)	
Elongation - Percent of original value, minimum	80
Sheath (3.12.17.4; 3.12.17.5)	When applicable
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Sheath (3.12.18.5)	When applicable
Tensile - P.s.i., per square inch minimum	1800
Elongation - Percent, minimum	50
Conductors (4.8.21)	

Table XXIX - MCOS cable.

Type	Number of conductors	Standard copper conductors	Insulation thickness	Overall diameter		Dielectric strength (see 4.8.19)		Insulation resistance per 1,000 feet (see 4.8.25)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)
						Between conductors	Shield to ground		Bare
			Minimum	Minimum	Maximum	Minimum	Minimum	Minimum	Maximum
			Inch	Inch	Inch	Volts	Volts	Megohms	Ohms
MCOS-2	2	1-1/2(16)	0.025	0.440	0.460	2,000	1,500	200	7.08
MCOS-4	4	1-1/2(16)	.025	.490	.510	2,000	1,500	200	7.08
MCOS-5	5	1(26)	.015	.375	.390	1,000	500	100	10.90
MCOS-6	6	1(10)	.015	.450	.465	1,000	1,500	100	11.40
MCOS-7	7	1-1/2(16)	.025	.575	.595	2,000	1,500	200	7.08

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3.31 Type MMOP cable.-

Construction

First: A standard conductor of bare, uncoated copper (see 3.11).
Second: A wall of synthetic resin insulation (see 3.12.2). Color coding (see 3.10.2).
Third: Five such color-coded conductors shall be twisted together with a maximum lay not greater than 2 inches. Glass fibers tie cord (see 3.12.10) shall be cabled with the five conductors laid straight as a central core. Fillers (see 3.12.11) shall be employed, where necessary, to obtain a firm, well-rounded assembly.
Fourth: A braid of a minimum No. 60/2 ply cotton or approved alternate over the conductor assembly.
Fifth: An impervious sheath (see 3.12.18).

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXX
Conductor resistance (see 4.8.24)	See table XXX
Insulation resistance (see 4.8.25)	See table XXX

Group B	Requirements
Flexing endurance (see 4.8.14)	
Twisting and bending cycles at 75°C. min.	1000
Twisting and bending cycles at -30°C. min.	500
Insulation (see 3.12.2)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	125
Sheath (see 3.12.18)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	225
Set - Inch, maximum	30/64

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.2.1.4)	
Elongation - Percent of original value, minimum	80
Sheath (see 3.12.18.5)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	50
Conductors (see 4.8.21)	

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Table XXX - Type MMOP cable.

Type	Standard copper conductor	Number of conductors	Diameters		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet (see 4.8.25)
			Over insulated conductor	Overall		Bare	
			Minimum	Maximum	Minimum	Maximum	Minimum
			Inch	Inch	Volts	Ohms	Megohms
MMOP-5	1/2(21)	5	0.065	0.305	1000	21.7	100

3.32 Types SHOF, DHOF, THOF and FHOF cable.-

Construction

First: A standard conductor of bare, uncoated copper (see 3.11).
Second: A separator (see 3.12.14).
Third: A wall of butyl rubber insulation (see 3.12.6).
Fourth: An inner braid (see 3.12.16) will be permitted on sizes 3 and larger. A tape (see 3.12.15) or an inner braid (see 3.12.16) will be permitted on sizes 23 and larger.
Fifth: Reinforcement (see 3.12.12) on type SHOF 23 and larger only. The required number of conductors twisted together with a left-hand lay not greater than specified in table XXXI. Fillers (see 3.12.11) shall be employed to obtain a firm, well rounded assembly. An approved binder over the conductor assembly.
Sixth: A polychloroprene sheath (see 3.12.17) or impervious sheath (see 3.12.18) on sizes 42 and smaller except THOF-14 which shall be polychloroprene. A polychloroprene sheath (see 3.12.17) on sizes larger than 42.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXXI
Conductor resistance (see 4.8.24)	See table XXXI
Insulation resistance (see 4.8.25)	See table XXXI

Group B	Requirements
Cold bend (see 4.8.9) - Bend two turns around mandrel with diameter twice that of specimen at -30°C.	No visual or electrical damage
Flexing endurance (see 4.8.14) (Types DHOF, THOF and FHOF only)	
Twisting and bending cycles at +75°C.	1000
Twisting and bending cycles at -30°C.	500

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Test Requirements (Cont'd)

Group B	Requirements
Insulation (see 3.12.6)	
Tensile - P.s.i., minimum	600
Elongation - Percent, minimum	450
Sheath (see 3.12.17)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.6.1.4)	
Tensile - Percent of original value, minimum	70
Elongation - Percent of original value, minimum	75
Sheath (see 3.12.17.4 and 3.12.17.5)	
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Conductors (see 4.8.21)	

Table XXXI - Types SHOF, DHOF, THOF and FHOF cables.

Type	Standard copper conductor	Insulation thickness	Lay of conductors	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet (see 4.8.25)
							Bare	
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
		Inch	Inches	Inches	Inches	Volts	Ohms	Megohms
SHOF-3	2-1/2(65)	0.031	----	0.195	0.210	2,000	4.26	300
SHOF-23	23(228)	.040	----	.440	.460	2,500	.499	300
SHOF-60	60(304)	.050	----	.570	.600	2,500	.187	250
SHOF-150	150(760)	.070	----	.830	.870	2,500	.0747	200
SHOF-200	200(988)	.070	----	.940	.980	2,500	.0575	200
SHOF-250	250(1254)	.070	----	1.035	1.085	2,500	.0141	150
SHOF-500	259(37x7)	.090	----	1.380	1.450	2,500	0.0234	100
SHOF-650	427(61x7)	.100	----	1.540	1.610	2,500	0.0154	100
SHOF-800	800(4033)	.100	----	1.600	1.670	2,500	.0141	100
DHOF-3	2-1/2(26)	.031	2	.405	.425	2,000	4.43	300
DHOF-4	4(41)	.031	2	.440	.460	2,000	2.81	250
DHOF-6	6(65)	.031	2-1/2	.490	.510	2,000	1.76	200
DHOF-9	9(90)	.031	3	.540	.570	2,000	1.27	200
DHOF-14	14(140)	.040	4	.675	.705	2,500	0.810	200
DHOF-23	23(228)	.040	5-1/2	.820	.860	2,500	.513	150
DHOF-30	30(304)	.050	7-1/2	.920	.960	2,500	.382	150
DHOF-83	83(418)	.070	16	1.390	1.450	2,500	.139	150
DHOF-250	250(1254)	.070	24	2.000	2.100	2,500	.0462	100
DHOF-400	400(2052)	.090	30	2.400	2.500	2,500	.0283	100

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Table XXXI - Types SHOF, DHOF, THOF and FHOF cables (Cont'd)

Type	Standard copper conductor	Insulation thickness	Lay of conductors	Overall diameter		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet (see 4.8.25)
							Bare	
		Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
		Inch	Inches	Inches	Inches	Volts	Ohms	Megohms
THOF-3	2-1/2(26)	.031	2	.430	.450	2,000	4.43	300
THOF-4	4(41)	.031	2-1/2	.460	.480	2,000	2.81	250
THOF-6	6(65)	.031	3	.520	.550	2,000	1.76	200
THOF-9	9(90)	.031	3-1/2	.570	.600	2,000	1.27	200
THOF-14	14(140)	.040	4-1/2	.720	.750	2,500	.810	200
THOF-23	23(228)	.040	6	.860	.900	2,500	.513	150
THOF-42	42(209)	.070	12	1.200	1.250	2,500	.278	150
THOF-150	150(760)	.070	18	1.740	1.820	2,500	.0762	125
THOF-250	250(1254)	.070	24	2.140	2.240	2,500	.0462	100
THOF-400	400(2052)	.090	30	2.680	2.800	2,500	.0283	100
THOF-600	427(61x7)	.100	36	2.980	3.150	2,500	0.0196	100
FHOF-3	2-1/2(26)	.031	2-1/2	.460	.480	2,000	4.43	300
FHOF-4	4(41)	.031	3	.520	.550	2,000	2.81	250
FHOF-9	9(90)	.031	4	.630	.660	2,000	1.27	200
FHOF-42	42(209)	.070	15	1.300	1.380	2,500	0.278	150
FHOF-60	60(304)	.070	18	1.430	1.510	2,500	0.187	150
FHOF-133	133(684)	.070	20	1.920	2.000	2,500	0.0846	125

3.33 Type T3P cable.-

CONSTRUCTION

First: Conductors and ground wires of coated copper wires (see 3.11) and table XXXII.
Second: A wall of butyl rubber (see 3.12.6) on insulated conductors.
Third: (a) A color-coded tape (see 3.12.11) or an inner braid (see 3.12.16) on each insulated conductor.
(b) A green inner braid (see 3.12.16) on each uninsulated ground wire.
Fourth: Three insulated conductors twisted together with a left hand lay not greater than twenty times the pitch diameter. One ground wire shall be placed in each of the three valleys with sufficient fillers (see 3.12.11) to obtain a well-rounded assembly.
A compound filled tape (See 3.12.15) over the conductor assembly.
Fifth: A polychloroprene sheath (see 3.12.17). Reinforcement (see 3.12.12.3) shall be applied under the sheath or approximately near the middle of the sheath thickness so as not to show as an overall surface marking.

TEST REQUIREMENTS

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXXII
Conductor resistance (see 4.8.24)	See table XXXII
Insulation resistance (see 4.8.25)	See table XXXII

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Test Requirements (Cont'd)

Group B	Requirements
Cold bend (see 4.8.9) 2 turns at -30°C.; mandrel twice diameter of specimen	No damage
Insulation (see 3.12.6)	
Tensile-P.s.i., minimum	600
Elongation-Percent, minimum	450
Sheath (see 3.12.17)	
Tensile-P.s.i., minimum	1200
Elongation-Percent, minimum	250

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.6.1.4)	
Tensile - P.s.i., minimum	70
Elongation - Percent, minimum	75
Sheath (see 3.12.17.4 and 3.12.17.5)	
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Conductors (see 4.8.21)	

Table XXXII - Type T3P cable.

Type	Conductors		Insulation thickness	Sheath thickness	Overall diameter	Dielectric strength	Conductor resistance (d.c.) per 1000 feet at 25°C.	Insulation resistance per 1000 feet
	Number	Stranding	Minimum	Minimum	Maximum	Minimum	Maximum	Minimum
			Inch	Inch	Inches	Volts	Ohm	Megohms
T3P-42								
Insulated	3	7x7/ 0.0292	0.125	0.172	1.67	11,500	0.273	500
Uninsulated	3	7x7/ 0.0164	0	-----	----	0	.874	0
T3P-400								
Insulated	3	37x7/ 0.0393	0.141	0.266	3.20	13,000	0.0284	250
Uninsulated	3	19x7/ 0.0316	0	-----	----	0	.0863	0

3.34 Type TRF cable.-

Construction

First: A standard conductor of bare, uncoated copper (see 3.11).
Second: A tape separator (see 3.12.14).
Third: A wall of synthetic insulation (see 3.12.4 or 3.12.6) with or without tape (see 3.12.15) at the manufacturer's option.
Fourth: Cotton reinforcement (see 3.12.12).
Fifth: A sheath of polychloroprene (see 3.12.17) bonded to the underlying insulation.

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Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength(see 4.8.19)	See table XXXIII
Conductor resistance (see 4.8.24)	See table XXXIII

Group B	Requirements
Cold bend (see 4.8.9) - Bend specimen two turns around mandrel with diameter twice that of specimen at -30°C.	No cracks
Insulation (see 3.12.4 and 3.12.6), respectively.	
Tensile - P.s.i. minimum	600
Elongation - Percent minimum	225
Tensile - P.s.i., minimum	600
Elongation - Percent minimum	450
Sheath (see 3.12.17)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.4.1.1.4 and 3.12.4.1.1.5)	
Tensile - Percent of original value, minimum	50
Elongation - Percent of original value, minimum	50
Insulation (see 3.12.6.1.4)	
Tensile - Percent of original value, minimum	70
Elongation - Percent of original value, minimum	75
Sheath (see 3.12.17.4 and 3.12.17.5)	
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Conductors (see 4.8.21)	

Table XXXIII - Type TRF cable.

Type	Standard copper conductor	Diameters		Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)
		Over insulation	Overall		Bare
				Minimum	Maximum
		Inch	Inch	Volts	Ohms
TRF-105	105(259)	0.56	0.76	1,200	0.108
TRF-133	133(259)	.61	.81	1,200	.0856
TRF-168	168(427)	.67	.86	1,200	.0681

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3.35 Type TRXF cable.-

Construction

First: A standard conductor of bare, uncoated copper (see 3.11).
Second: A tape separator (see 3.12.14).
Third: A sheath of polychloroprene (see 3.12.17).

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXXIV
Conductor resistance (see 4.8.24)	See table XXXIV

Group B	Requirements
Cold bend (see 4.8.9) - Two turns around mandrel with diameter twice that of specimen at -30°C.	No damage
Sheath (see 3.12.17)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Sheath (see 3.12.17.4 and 3.12.17.5)	
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Conductors (see 4.8.21)	

Table XXXIV - Type TRXF cable.

Type	Standard copper conductor	Overall diameter	Dielectric strength (see 4.8.19)	Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)
				Bare
			Minimum	Maximum
		Inch	Volts	Ohms
TRXF-84	84(2107)	0.60	1,200	0.138
TRXF-105	105(2646)	.68	1,200	.108
TRXF-133	133(3325)	.75	1,200	.0876

3.36 Type TTRS cable.-

Construction

First: A standard conductor of bare, or tinned copper (see 3.11), size 1(7).
Second: A wall of polythene (see 3.12.5) to the diameter, and within the dimensional tolerances required to obtain the specified degree of performance as limited by the maximum overall

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Test Requirements (Cont'd.)

Group B	Requirements
Flexing endurance (see 4.8.14)	
Bending cycles only at -30°C., minimum	500
Insulation (see 3.12.4)	
Tensile - P.s.i., minimum	600
Elongation - Percent, minimum	225
Sheath (see 3.12.17)	
Tensile - P.s.i., minimum	1200
Elongation - Percent, minimum	250

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.4.1.1.4 and 3.12.4.1.1.5)	
Tensile - Percent of original value, minimum	50
Elongation - Percent of original value, minimum	50
Sheath (see 3.12.17.4 and 3.12.17.5)	
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Conductor (see 4.8.21)	

3.38 Type DLT cable.-

Construction

First: A stranded steel strength member conforming to MIL-C-5424, 7 by 19 strand, 0.156 inch in diameter, not preformed, insulated with 0.046 inch minimum wall (see 3.8.3) of 60 percent natural rubber in accordance with the applicable requirements of Publication ASTM 353-58 to a nominal diameter of 0.262 inch. The insulation shall fill all outer interstices of the steel strand. This wall may be applied in one or two layers and shall be properly vulcanized so that the completed cable meets all test requirements.

Second: A stranded conductor of coated copper, size 1-1/2(41) (see 3.11) covered with a cotton separator (see 3.12.12.2) and insulated with 0.031 inch minimum wall (see 3.8.3 and prior to cabling) of 30 percent natural rubber in accordance with Publication ASTM 353-58. Circuit identification (see 3.10.7).

Third: Four insulated conductors shall be cabled in a close helix about the insulated strength member with a lay of approximately 0.625 inch. The insulated conductors shall be partially embedded in a relatively soft partially vulcanized insulation of the strength member.

Fourth: A minimum 0.031 inch belt (see 3.8.3) of butyl rubber (see 3.12.6) filling the outer interstices over the insulated conductor assembly.

Fifth: Cotton reinforcement (see 3.12.12.3).

Sixth: A polychloroprene sheath (see 3.12.17) having a minimum thickness (see 3.8.3) of 0.078 inch bonded to the underlying butyl belt. The assembled components shall be so molded and vulcanized that the completed cable shall be one compact mass from which the individual conductors shall be readily separable without damage to the insulation. The overall diameter shall not exceed 0.710 inch. Manufacturer's identification (see 3.9) shall include molded marking (see 3.9.4) at intervals not greater than 25 feet, showing the cable type letter designation (type DLT), the manufacturer's name or trade mark, and the year of manufacture.

Length Requirement

Type DLT cable shall be delivered in nominal lengths of not less than 600 feet and not more than 630 feet. Except where authorized by the contracting agency, not more than 30 percent of the total footage of each lot offered for inspection may be accepted in random lengths of not less than 400 feet, and not more than 420 feet or in random lengths of not less than 200 feet and not more than 210 feet. Combinations thereof of random lengths will be permitted. Other lengths are not suitable for the purpose.

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Test Requirements

Preproduction	Requirements
Endurance (see 4.8.33) 500 cycles	No conductor breakage or electrical failures between conductor or conductor to strength member.

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	
Volts, r.m.s., minimum	
Conductor to conductor	2000
Conductor to ground (strength member and water bath)	2000
Insulation resistance (see 4.8.25)	
Megohms per 1000 feet, minimum	200
Conductor resistance (see 4.8.24)	
Ohms per 1000 feet at 25°C., maximum	17.0

Group B	Requirements
Flexing endurance (see 4.8.14)	
Bending cycles only, 2-inch mandrel at room temperature	1000
Breaking strength (see 4.8.28) pounds, minimum	2500
Insulation (see ASTM D353-58)	
Tensile - P.s.i., minimum	900
Elongation - Percent, minimum	300
Sheath (see 3.12.17)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	300

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see ASTM D353-58)	
Air oven test at 70° ± 1°C. for 96 hours	
Tensile and elongation - Percent original value, minimum	75
Oxygen bomb pressure test at 70° ± 1°C. for 96 hours	
Tensile and elongation - Percent of original value, minimum	65
Sheath (see 3.12.17.4 and 3.12.17)	
Tensile - Percent of original value, minimum	65
Elongation percent of original value, minimum	65
Conductor (see 4.8.21)	
Steel strength member	MIL-C-5424

3.39 Type JAS cable.-

Construction

First: Standard conductors of bare uncoated copper (see 3.11 and table XXXVI) with separator (see 3.12.14).
Second: A wall of butyl rubber (see 3.12.6), or synthetic rubber (see 3.12.4)
Third: An inner braid (see 3.12.16.2) on size 6(65). A tape (see 3.12.15) or inner braid (see 3.12.16.2) on size 250(6384) is not required but will be permitted at manufacturer's option. Color coding, one conductor white and one conductor black of each size.

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Construction (Cont'd)

diameter of the finished cable. At the manufacturer's option a thin wall of polyamide (type III, grade E of MIL-M-20693) may be applied over the polythene.

Third: Two such insulated conductors shall be combined to form a pair. They shall be twisted together with any lay necessary to obtain the specified mechanical and electrical characteristics. One conductor in each pair shall be white or natural and its mate shall be a dark contrasting color such as blue or black.

Fourth: A braided shield of 0.005 or 0.0063 inch diameter bare or tinned copper wire (see 3.12.16.6) on each pair. The braid angle shall be approximately 30 degrees.

Fifth: Shield insulation on each pair comprised of one or more polyester tapes (type G of MIL-I-631), applied helically with adhesive under the overlaps to prevent any loosening or unwinding at the ends where pairs are separated incident to terminating the cable. Color-coding (see 3.10.7) shall be provided for each shielded pair by printing on the insulating tape at 3-inch, maximum, intervals; the printing shall be black or red on a colored white background for legibility.

Sixth: The specified number of shielded pairs (see table XXXV) shall be cabled together with a left-hand lay. The length of lay shall be a suitable function of the pitch diameter of that layer, designed to meet the flexing endurance test on completed cable. Wherever practicable, the component pairs shall be cabled in a single layer, using, if necessary, a soft center core. A single pair shall not be used as a core for the cabled construction. Fillers (see 3.12.11.3) shall be used, if necessary, to obtain circularity of cross section.

Seventh: The cabled pairs shall be bound by a loose braid or fabric binder in such a manner as to permit sliding between the pairs themselves and between the pairs and the overall braid or binder when the cable is sharply and repeatedly bent and twisted.

Eighth: An impervious sheath (see 3.12.18) of the thermoplastic type.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	
Volts, r.m.s., minimum	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	200
Conductor resistance (see 4.8.24)	
Ohms per 1000 feet at 25°C., maximum	10.3
Insulation resistance (see 4.8.25)	
Megohms per 1000 feet at 25°C., minimum	1000
Capacitance (see 4.8.27.3)	
Picofarads per foot at 25°C., and 1 mc/s maximum	26
Impedance (see 4.8.27)	75 minimum
Ohms at 25°C. and 1 mc/s.	90 maximum
Capacitance unbalance (see 4.8.32) -	
Percent, maximum	10

Group B	Requirements
Flexing endurance (see 4.8.14)	
Twisting and bending cycles at 75°C., minimum	1000
Twisting and bending cycles at -30°C., minimum	500
Insulation (see 3.12.5)	
Tensile - P.s.i., minimum	1400
Elongation - Percent, minimum	300
Sheath (see 3.12.18)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	225
Set - Inch, maximum	30/64

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Test Requirements (Cont'd.)

Group C	Requirements
Insulation (see 3.12.5.1.4)	
Elongation - Percent of original value, minimum	60
Sheath (see 3.12.18)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	50
Flammability (see 4.8.10)	See 4.8.10.3
Conductors (see 4.8.21; 4.8.22)	

Table XXXV - Type TTRS cable.

Type	Number of shielded pairs	Diameter over sheath (maximum)
		Inches
TTRS-2	2	0.680
TTRS-4	4	.740
TTRS-6	6	.880
TTRS-8	8	.990
TTRS-10	10	1.080
TTRS-12	12	1.100
TTRS-16	16	1.190

3.37 Type CVSF cable.-

Construction

First: Three uncoated copper conductors (see 3.11) size 53(532) (see table III), with separator (see 3.12.14) insulated with synthetic insulation (see 3.12.4 or 3.12.6) having a minimum wall thickness (see 3.8.3) of 0.070 inch. Color-coding (see 3.10.2 or 3.10.4) is required.

Second: One uncoated copper conductor (see 3.11), size 33(336) (see table III) with no covering.

Third: The three large insulated conductors cabled with a lay of not more than 14 inches, placing the uninsulated conductor in one outer valley employing fillers (see 3.12.11) to obtain a well-rounded flexible core, and bound with a compound-filled tape (see 3.12.15) or braid (see 3.12.16).

Fourth: A polychloroprene sheath (see 3.12.17) having a minimum thickness (see 3.8.3) of 0.125 inch to an overall diameter not exceeding 1.45 inches for the completed cable.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	
Volts, r.m.s., minimum	2500
Conductor resistance (see 4.8.24)	
Ohms per 1000 feet at 25°C., maximum	
Size 53(532)	0.217
Size 33(336)	0.351
Size 2-1/2(26)	4.35
Insulation resistance (see 4.8.25)	
Megohms per 1000 feet, minimum	150

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Construction (Cont'd.)

Fourth: Two conductors of size 6(65) twisted with a maximum lay of 4 inches to form a pair with fillers (see 3.12.11.3). Two conductors of size 250(6384) laid parallel with the twisted pair in between in the geometric center. Fillers of synthetic elastomer or polychloroprene shall be employed to fill all valley spaces, and make a firm oval cross section. At manufacturer's option, fillers may be an integral part of the sheath.

Fifth: An open mesh reinforcement of type P, form C in accordance with MIL-C-572 over the conductor assembly.

Sixth: A polychloroprene sheath (see 3.12.17). Fillers and sheath material shall be readily removed from the completed cable without damage to the conductor insulation.

Test Requirements

Preproduction	Requirements
Bending endurance (see 4.8.14)	
14-inch diameter mandrel, room temperature, 100,000 cycles, flatwise minimum	No physical or electrical damage

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	
Volts, r.m.s., minimum	2000
Conductor resistance (see 4.8.24)	
Size 6(65) ohms per 1000 feet at 25°C., maximum	176
Size 250(6384) ohms per 1000 feet at 25°C., maximum	0.0461
Insulation resistance (see 4.8.25)	
Megohms per 1000 feet at 25°C., minimum	
Size 6(65)	200
Size 250(6384)	100

Group B	Requirements
Bending endurance (see 4.8.14)	
14-inch mandrel room temperature, 1000 cycles flatwise, minimum	No physical or electrical damage
Insulation (see 3.12.6)	
Tensile - P.s.i., minimum	600
Elongation - Percent, minimum	250
Insulation (see 3.12.4)	
Tensile - P.s.i., minimum	600
Elongation - Percent, minimum	225
Sheath (see 3.12.17)	
Tensile - P.s.i., minimum	1200
Elongation - Percent, minimum	250

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.6.1.4)	
Tensile - Percent of original value, minimum	70
Elongation - Percent or original value, minimum	75
Sheath (see 3.12.17.4 and 3.12.17.5)	
Tensile - Percent of original value, minimum	65
Elongation - Percent of original value, minimum	65
Conductors (see 4.8.21)	

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Table XXXVI - Type JAS cable.

Type	Conductors				Insulation wall 1/	Sheath wall	Overall diameter (oval shape)
	Number of conductors	Strands per conductor	Size of Strand	Conductor diameter			
		Minimum	Nominal	Nominal	Minimum	Minimum	Maximum
			Inch	Inch	Inch	Inch	Inches
JAS-250	2 of 6(65) and 2 of 250(6384)	65 6384	0.010 .006	0.097 .680	0.031 .070	0.156	1.260 minor and 2.480 major

1/ When the insulation is appled as a "D" shaped section to fit halves of the oval lay-up, the eccentricity requirement (see 3.8.4.1) will not apply.

3.40 Type MRI wire.-

Construction

First: A standard conductor of bare, uncoated copper (see 3.11).
Second: A wall of synthetic resin insulation (see 3.12.2); circuit identification (see 3.10.7).
Third: The specified number of conductors shall be twisted together with a lay not greater than that specified in table XXXVII.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	See table XXXVII
Conductor resistance (see 4.8.24)	See table XXXVII
Insulation resistance (see 4.8.25)	See table XXXVII

Group B	Requirements
Insulation (see 3.12.2)	
Tensile - P.s.i., minimum	1800
Elongation - Percent, minimum	125

Group C	Requirements
Flammability (see 4.8.10)	See 4.8.10.3
Insulation (see 3.12.2.1.4)	
Elongation - Percent of original value, minimum	80
Conductors (see 4.8.21)	

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Table XXXVII - Type MRI wire.

Type	Number of conductors	Standard copper conductor	Diameter over synthetic resin		Lay of conductors	Dielectric strength (see 4.8.19)			Conductor resistance (d.c.) per 1000 feet at 25°C. (see 4.8.24)	Insulation resistance per 1000 feet (see 4.8.25)
						Spark	Full reel	Short length	Bare	Minimum
			Minimum	Maximum	Maximum	Minimum	Minimum	Minimum	Maximum	
			Inch	Inch	Inches	Volts	Volts	Volts	Ohms	Megohms
1/ MRI-D-1	2	1(10)	0.068	0.075	2	2,500	1,000	6,000	11.7	100
MRI-D-2-1/2	2	2-1/2(26)	.092	.102	3	5,000	1,500	6,000	4.52	100
MRI-T-2-1/2	3	2-1/2(26)	.092	.102	3	5,000	1,500	6,000	4.52	100

1/ The nominal length or unit ordering length for type MRI-D-1 is a 600-foot coil.

3.41 Workmanship.- The workmanship shall be first class in every respect and uniform throughout each lot of cable, cord or wire offered for delivery.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The supplier shall provide and maintain a Quality Control System which meets the requirements specified in Appendix II.

4.2 Qualification tests[1/].- (Types SHFR, DHFR, THFR, TTRSA, TTHFWA, DSS, TSS, FSS, 7SS, TSP, MHOF, TTOP, MCOS, MMOP, TTRS, DHOF, THOF, FHOF, MCSF, FCSF, SDU, PBTM and T3P.) Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.2.1.

4.2.1 List of qualification tests.- The following tests, as applicable to the particular types of cable, shall be conducted:

Visual and dimensional examination (see 4.7).
Synthetic resin, synthetic rubber, butyl rubber and polythene, and polychloroprene (see 4.8.1).
Chemical analysis (see 4.8.2).
Varnished cambric (see 4.8.3).
Asbestos (see 4.8.4).
Glass (see 4.8.5).
Impervious sheath (see 4.8.6).
Metal armor (see 4.8.7).
Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Heat endurance (see 4.8.12).
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14).
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Pressure cycling (see 4.8.31.2).
Hot oil test (see 4.8.17).
Oil, water and chemical resistance (see 4.8.18).
Dielectric tests (see 4.8.19).
Cable tension test (see 4.8.20).
Tensile strength and elongation, conductors (see 4.8.21).
Coating metallic (see 4.8.22).
Tensile strength, elongation set and aging, insulation and sheath (see 4.8.23).
Conductor resistance (see 4.8.24).
Insulation resistance (see 4.8.25).[2/]
Flame retardance (see 4.8.26).
Capacitance and characteristic impedance (see 4.8.27).
Capacitance unbalance (see 4.8.32).
Breaking strength (see 4.8.28).
Induced voltage (see 4.8.29).
Heat shock (see 4.8.30).
Hydrostatic test (see 4.8.31).
Thermoplastic flow (see 4.8.11).
Watertightness (see Appendix I).

1/ Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.4.1).

2/ To qualify DSS, FSS, TSS, and 7SS cables the contractor shall conduct and submit test data via the Government inspector for approval of Bureau of Ships showing that the proposed cable design retains the insulation resistance values during 90 days submergence in tap water.

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4.3 Preproduction inspection (Types MDU, MDY, DLT, JAS, DSWS-4).- These cable types shall be subjected to and pass the preproduction tests of 4.3.1. Cables subject to preproduction approval, when specified in the detail requirements, shall be products for which reports of preproduction inspection have been submitted to the bureau or agency concerned via the Government Inspector's Office and found acceptable. These cables are identified with the "approval construction" which was inspected and on which the approval was based. The manufacturer shall not change any of the component materials or the assembly of the components in completed cables offered for delivery without first obtaining written approval for such modification from the bureau or agency concerned.

4.3.1 Preproduction tests.- The MDU, MDY, DLT, JAS and DSWS-4 cables, shall be subjected to the following examination and tests as applicable:

Visual and dimensional (see 4.7).
Synthetic resin, synthetic rubber, butyl rubber and polythene (see 4.8.1).
Chemical analysis (see 4.8.2).
Varnished cambric as removed from completed cable (see 4.8.3).
Asbestos (see 4.8.4).
Glass (see 4.8.5).
Impervious sheath (see 4.8.6).
Metal armor (see 4.8.7).
Drip (see 4.8.8).
Cold bend (see 4.8.9).
Flammability (see 4.8.10).
Thermoplastic flow (see 4.8.11).
Heat endurance (see 4.8.12).
Water and air leakage (see 4.8.13).
Flexing endurance (see 4.8.14).
Impact resistance (see 4.8.15).
Abrasion resistance (see 4.8.16).
Hot oil test (see 4.8.17).
Oil, water, and chemical resistance (see 4.8.18).
Dielectric tests (see 4.8.19).
Cable tension (see 4.8.20).
Tensile strength and elongation, conductors (see 4.8.21).
Coating, metallic (see 4.8.22).
Tensile strength, elongation set and aging insulation and sheath (see 4.8.23).
Conductor resistance (see 4.8.24).
Insulation resistance (see 4.8.25).
Flame retardance (see 4.8.26).
Capacitance and characteristic impedance (see 4.8.27).
Breaking strength (see 4.8.28).
Induced voltage (see 4.8.29).
Heat shock (see 4.8.30).
Hydrostatic test (see 4.8.31).
Capacitance, unbalance (see 4.8.32).
Watertightness (see Appendix I).
Endurance (see 4.8.33).

4.4 Sampling for quality conformance inspection.-

4.4.1 Lot.- A lot of completed wire, cord, and cable shall consist of the total number of units of product (see 4.4.2) of completed wire and cable of any one type, size, and composition manufactured under essentially the same conditions and offered for delivery at one time.

4.4.2 Unit of product.- The unit of product is defined as the standard lengths of cable (see 6.2.1 except as noted herein) having the same type designation and size, whether that length be part of a coil, or reel, or consists of one or more coils, or reels containing standard, remnant or random lengths.

NOTE 1 - The unit of product of cable put up in coils of 500 feet will be 5000 feet or less.
NOTE 2 - The unit of product of cable size 200 or over will be 1000 feet or less.

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4.4.3 Sample.- A sample is one or more units of product drawn from a lot.

4.4.4 Specimen.- A specimen is an individual piece of cable or a component of cable taken from a sample.

4.4.5 Defective unit.- Failure in any test shall constitute a defect. If, however, the sample fails more than one test in any test group, it shall be counted as only one defective unit.

4.4.6 Selection of samples.- The number of samples required shall be chosen without regard to quality by selecting coils or reels.

4.4.6.1 Completed wire or cable.- The contractor shall remove and dispose of nonconforming starting and finishing ends to insure that samples are representative of the completed product to be offered for delivery.

4.4.6.2 Copper wires or conductors.- Sample sections of individual wires or stranded conductor shall be selected in such a manner as to be representative of the material to be used in the completed wire, cord or cable.

4.4.6.3 Number of samples.-

4.4.6.3.1 Group A.- Samples shall be selected for group A examination and tests in accordance with table XXXVIII.

Table XXXVII - Sampling for group A examination and tests.

Number of units of product in lot	Number of samples	Acceptance number defectives	Rejection number defectives
10 or under	2	0	1
11 to 30	3	0	1
31 to 90	7	0	1
91 to 210	15	1	2
211 or over	25	2	3

4.4.6.3.2 Group B.- Samples for group B tests shall be selected in accordance with table XXXIX.

Table XXXIX - Sampling for group B tests.

Units of product	Number of samples	Acceptance number defectives	Rejection number defectives
30 or under	2	0	1
31 to 90	3	0	1
91 to 210	4	0	1
211 and over	5	0	1

4.4.6.3.3 Group C.- Sample units shall be selected from the first lot of each type, size, and composition offered for delivery. Thenceforth, sample units shall be selected from each 2-months production of one type, size, and composition in accordance with table XL. No failures will be permitted for group C tests.

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Table XL - Sampling for group C tests.

Two months production (units of product)	Number of samples
8 or under	1
9 to 30	2
31 to 80	4
81 to 130	5
181 to 240	6

4.4.6.3.4 If the number of defective units in a lot is found to exceed the acceptance number defective in tables XXXVIII and XXXIX, the entire lot shall be rejected. The manufacturer shall have the option of reworking the rejected lot, and after removing all nonconforming lengths which cannot be made to conform, reinspect it under tightened sampling shown in table XLI.

Table XLI - Tightened sampling.

Number of units of product in lot	Number of samples	Acceptance number defectives	Rejection number defectives
10 or under	3	0	1
11 to 30	5	0	1
31 to 90	9	0	1
91 to 210	15	0	1

4.4.6.3.5 Only cables intended for Government end use shall employ a marker tape showing the Navy specification number. Such cable which cannot be made to conform to the specification and is not accepted by the Government shall have the marker tape removed or be painted red along its entire periphery and length, or cut into 10-foot lengths, maximum. The contractor shall furnish evidence to the Government inspector of having taken this action.

4.4.6.4 Sampling for comparison inspection.- (Types SHFR, DHFR, THFR, TTHFWA, DSS, TSS, FSS, 7SS, TSP, TTRSA, MHOF, TTOP, MCOS, MMOP, DHOF, THOF, FHOF, TTRS, MCSF, FCSF, SDU, PBTM, T3P, DSW-4, MDU, MDY and DLT).

4.4.6.4.1 One or more lots.- A full sample shall be selected from the first lot of each type, size, and composition offered for delivery. Thenceforth, comparison inspection samples shall be selected once every 2 months or from one of every five successive lots of the same type, size, and composition offered for delivery. The method of selection which provides the more frequent sampling shall be used.

4.4.6.4.2 Less than one lot.- The provisions of 4.4.6.4.1 shall not apply to any item when the total quantity of that item is 2500 feet, or less, of any one type, size, or composition.

4.4.6.4.3 Action in case of failure.- If any of the samples representing a lot is found to fail one of the specified tests, the entire lot shall be rejected. The manufacturer shall have the option of subjecting each reel length of rejected cable to the test or tests under which failure occurred. After removing all lengths which cannot be made to conform, the lot may be resubmitted for inspection. Acceptance of the first full or fractional lot of completed cable of each type and size shall be withheld pending receipt of a satisfactory report on comparison inspection when required (see 4.4.6.4). After satisfactory completion of comparison inspection on the first lot of any item, action on the next four successive lots of the same item shall be based on the results of quality conformance inspection (see 4.5.1, 4.5.2, and 4.5.3). In case of an unsatisfactory report on comparison inspection, additional samples shall be selected from at least two successive lots of the same item and subjected to comparison tests wherein failure was observed. The additional testing shall be discontinued and normal testing resumed to quality conformance inspection, only after two successive lots have passed comparison inspection (see 4.5.4). Rejected cable shall be handled as specified in 4.4.6.3.5.

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4.4.6.5 Preparation of samples.-

4.4.6.5.1 Length of samples.- The length of cable constituting a sample shall be as follows:

(a) Cables for nonflexing service	24 feet
(except type TTHFWA)	20 feet
(b) Cables for repeated flexing	24 feet
except: SHOF - 150, 250 and 400	5 feet
THOF - 250 and 400	5 feet
SDU	10 feet

4.4.6.5.2 Identification of samples.- All test samples shall be marked with the following information, as applicable:

(a) Name of manufacturer.
(b) Cable designation (type and size).
(c) Manufacturing lot number or order number either of the bureau or agency concerned or the shipbuilder (if submitted for delivery) and footage represented by the sample.
(d) The coil or reel number.
(e) Qualification designation (see 3.1).

4.4.6.5.3 End seals.- In order to obtain representative end seals, the sample for test shall be selected after end seals have been prepared on shipping lengths of cable. The length of such sample shall be 2-1/2 feet. The cut end of the shipping length shall be resealed immediately by the manufacturer in accordance with 3.12.22.1.

4.4.6.5.4 Cables may be shipped in wrapped coils or boxes. When necessary, the total sample length may be cut.

4.5 Quality conformance inspection.-

4.5.1 Group A examination and tests.- Group A examination and tests shall be performed on samples selected in accordance with table XXXVIII, except that the electrical tests shall be performed on all conductors of each coil, or reel length of completed wire, cord or cable offered for delivery. The following examination and tests, shall be performed, as applicable:

Test	Paragraph reference
Visual and dimensional examination	4.7
Watertightness	Appendix I
Hydrostatic	Appendix I and 4.8.3.1
Electrical	
Dielectric	4.8.19
Conductor resistance	4.8.24
Insulation resistance	4.8.25
Dielectric between shield and shield to water	- - -
Capacitance	4.8.27
Capacitance unbalance	4.8.32
Impedance	4.8.27
Pressure cycling	4.8.31.2

4.5.2 Group B tests.- Group B tests shall be conducted on samples selected in accordance with table XXXIX, and shall consist of the following tests, as applicable:

Test	Paragraph reference
Drip	4.8.8
Breaking strength	4.8.28
Induced voltage	4.8.29
Tension	4.8.20

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Test	Paragraph reference
Insulation[1, 2] Unaged	
Tensile	4.8.1
Elongation	4.8.1
Set	4.8.1
Sheath[2]	
Tensile	4.8.1
Elongation	4.8.1
Set	4.8.1
Cold bend	4.8.9

4.5.3 Group C tests.- Group C tests shall be performed on samples selected in accordance with table XL and shall consist of the following tests, as applicable:

Test	Paragraph reference
Hot oil	4.8.17
Accelerated aging	4.8.1.3
Insulation[1, 2]	
Tensile	4.8.1.3
Elongation	4.8.1.3
Sheath[2]	
Tensile	4.8.1.3
Elongation	4.8.1.3
Set	4.8.1.3
Flammability	4.8.10
Thermoplastic flow	4.8.32
Heat shock	4.8.30
Conductors[3]	
Tensile and elongation	4.8.21
Coating, metallic	4.8.22
Flexing endurance	4.8.14

[1]Cables with four or less conductors test all conductors. For cables containing more than four conductors, the square root of the total number of the conductors in the cable shall be tested; however, the figure shall [2]never be less than four. Inspect to the nearest whole number.
[2]Five specimens shall be tested for tensile strength, elongation and set from each sample unit, and the physical property of the inspection shall be taken as the median of the results obtained from the five speci-[3]mens.
[3]The contractor shall establish and maintain records and test data on this test for conductors in the manufacturing process. Tests may be performed before or after stranding, but before insulating.

4.5.4 Comparison inspection.- Comparison inspection shall be conducted on samples selected in accordance with 4.4.6.4 and shall consist of the following tests, as applicable, to determine that the samples are equal to the samples upon which qualification approval was based. When delivery requirements are urgent, comparison inspection may be conducted concurrently with quality conformance inspection upon approval of the Government.

Test	Paragraph reference
Insulation and sheath	
Synthetic resin, synthetic rubber, butyl rubber, polythene, and polychloroprene	4.8.1
Chemical analysis	4.8.2
Varnished cambric as removed from completed cable	4.8.3
Asbestos	4.8.4
Glass	4.8.5
Impervious sheath	4.8.6
Metal armor	4.8.7

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Test	Paragraph reference
Drip	4.8.8
Cold bend	4.8.9
Flammability	4.8.10
Water and air leakage	4.8.13
Flexing endurance	4.8.14
Impact resistance	4.8.15
Abrasion resistance	4.8.16
Hot oil test	4.8.17
Oil, water and chemical resistance	4.8.18
Dielectric	4.8.19
Flame retardance	4.8.26
Capacitance and characteristic impedance	4.8.27
Breaking strength	4.8.28
Heat shock	4.8.30
Hydrostatic	4.8.31
Thermoplastic flow	4.8.11
Watertightness	Appendix

4.6 Reports.- Laboratory reports covering the comparison inspection shall be forwarded directly to the cognizant Government inspector. Two copies of each test report shall be addressed to the Bureau of Ships.

4.7 Visual and dimensional examination.-

4.7.1 Material and workmanship.- Each of the samples of the finished product selected in accordance with 4.4 shall be visually and dimensionally examined, the material inspected and manufacturing processes evaluated to determine that the completed wire, cord, or cable conforms to this specification. The number of random, remnant, and scrap lengths and the total footage of such lengths compared to the number of standard lengths and the total footage in each lot of wire or cable shall be recorded. An excessive number of random, remnant, and scrap lengths may have resulted from the cutting out of obvious defects or from breakage during manufacture, and may be taken as an indication of the need for tighter control of materials, workmanship, and manufacturing process. In order that the number of lengths be kept to a minimum consistent with good manufacturing practices, for each type and size of cable on a contract or order, not less than 70 percent of the total footage to be shipped shall be in standard lengths, and not more than 30 percent may be in any combination of random and remnant lengths. When the total quantity of any one item is six nominal lengths or less, a footage approximating two nominal lengths (in lieu of 30 percent) may be in any combination of random and remnant lengths. When the total quantity of any one item is two nominal lengths or less, the total footage may be in random and remnant lengths. In particular, such of the examination specified hereinafter as applicable to the types and sizes of the completed wire, cord, or cable involved shall be made.

4.7.2 Conductors.- Examination shall be made for the number and size of copper wires (see 3.12.1) composing the standard conductors (see 3.11), of specific stranding total cross sectional area and so forth.

4.7.3 Material examination.- Examination for types of insulation and component materials (see 3.7 and 3.12.2 to 3.12.22, inclusive), and limiting dimensions of each layer or stage, when applicable to the cable under consideration (see 3.13 through 3.40).

4.7.4 Identification and color coding.- Examination shall be made for circuit identification (see 3.10) and color coding, when applicable to the cable under consideration (see 3.13 through 3.40).

4.7.5 Outer coverings.- When applicable, examination shall be made to determine that braids (see 3.12.16) polychloroprene sheath (see 3.12.17), impervious sheath (see 3.12.18 and 3.12.19), metal armor (see 3.12.20) and paint (see 3.12.21) conform to specification requirements.

4.7.6 Dimensions.- The cable, cord, or wire shall be measured to determine that the centering, circularity, and dimensional tolerances (see 3.8 and 3.13 through 3.40, inclusive, as applicable) have not been exceeded. For insulation nominally less than 1/32 inch in thickness, the minimum thickness shall be

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determined with a pingage dial micrometer by placing the specimen on the pin and taking the reading immediately after the movable member comes to rest on the specimen. Overall diameters of cables up to 1 inch diameter and all unarmored cable which are required to pass the hydrostatic test (see 4.8.31) shall be checked by employing a hand micrometer.

4.7.7 Patching.- The manufacturer's methods and practices with regard to making patches in the conductor insulation and cable sheath shall be subject to approval by the Government inspector. The materials and methods employed shall be such that the finished product complies with all specification requirements. The frequency of patching shall be held to a minimum consistent with specification lengths and quantity production, and in no case shall any shipping length (see 5.1.1) contain more than one sheath patch. No build-up or patch that cannot be contained in a single mold shall be made. No sheath build-up or patches are permitted on unarmored cables which are required to pass the hydrostatic test (see 4.8.31).

4.7.8 End seals.- End seals shall be examined at both ends of every shipping length to insure that end seals of the design shown on figure 1 have been applied.

4.8 Methods of tests.-

4.8.1 Synthetic resin, synthetic rubber, butyl rubber, polythene, and polychloroprene (insulation).-

4.8.1.1 Samples of the insulation (see 3.12.2 to 3.12.7, incl.) and sheath (see 3.12.17) taken from the completed wire or cable shall be tested to determine the physical properties in the "as received" condition and after accelerated aging.

4.8.1.2 Physical properties.- The tensile strength, elongation and set shall be determined on specimens prepared and tested as specified hereinafter.

4.8.1.2.1 Specimens.- The test specimen shall be the entire section of the insulation in the case of small conductors or, in the case of a large conductor or in obtaining a specimen of the cable sheath, either a segment of a section cut with a sharp knife held tangentially to the conductor or cable core, or a shaped specimen cut out with a standard die. When the whole cross section is used, the specimen shall not be cut longitudinally. The test specimen shall be as free as possible from surface incisions and imperfections.

4.8.1.2.1.1 Size.- The size of specimen shall be of an area not less than 0.010 square inch and not more than 0.030 square inch. When not possible to use specimens of this size, they shall be as large as practicable.

4.8.1.2.1.2 Condition and age.- Specimens shall not be heated, immersed in water, or subjected to any mechanical or chemical treatment not specifically specified herein. No tests shall be made within 48 hours after vulcanization unless agreed to by the manufacturer.

4.8.1.2.1.3 Buffing.- In event of any irregularities on the surface of a fractional-section specimen, it shall be made smooth and of uniform thickness within 5 percent of the original minimum thickness by buffing, except where large strands are used, in which case the samples shall be buffed sufficiently to remove all inner corrugations.

4.8.1.2.1.4 Calculation of area.- Calculation of the area of the test specimen shall be made as follows:

(a) Where the total cross section of the insulation is used, the area shall be taken as the difference between the area of the circle whose diameter is the minimum average outside diameter of the insulation and the area of the conductor. The area of a stranded conductor shall be figured from its maximum diameter. The following formula may be used to advantage:

$$A = 0.7854 (D + d) (D - d)$$

Where: A = Area of cross section in square inches.
D = Outside diameter of insulation in inches.
d = Diameter over copper in inches.

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(b) Where a slice cut from the insulation by a knife held tangentially to the wire is used, and the slice so cut has the cross section of a segment of a circle, the area shall be calculated as that of the segment of a circle whose diameter is that of the insulation. The height of the segment is the wall of insulation on the side from which the slice is taken. The values may be obtained from a table giving the areas of segments of a unit circle for the ratio of the height of the segment to the diameter of the circle.

(c) Where the cross section of the slice is not a segment of a circle, the area shall be calculated from a direct measurement of the volume or from the specific gravity and the weight of a known length of the specimen having a uniform cross section.

(d) Where a portion of sector of a circle has to be taken as where the conductor is large and the insulation thin, the area shall be calculated as the thickness times the width. This applies either to a straight test piece or one stamped out with a die, and assumes that corrugations have been removed by buffing.

(e) Where a portion of a sector of a circle has to be taken, as where the conductor is large and the insulation thick, the area shall be calculated as the proportional part of the area of the total cross section.

4.8.1.2.2 Apparatus.- The testing machine shall be power-driven and preferably be the pendulum type. A spring balanced type of apparatus may be used if provided with a device which will indicate the actual maximum load at which rupture takes place and provision is made to prevent recoil of the spring. The testing machine shall be accurate within 1 percent of the breaking load.

4.8.1.2.3 Procedure.-

4.8.1.2.3.1 Temperature.- Physical tests shall be conducted with the specimens at a temperature between 20° and 28°C.

4.8.1.2.3.2 Rate.- The specimens shall be stretched at a rate of 20 inches per minute in a testing machine having jaws 4 inches apart.

4.8.1.2.3.3 Tensile strength and elongation.- A test specimen, not less than 6 inches in length, having bench marks 2 inches apart, shall be placed in a testing machine and stretched until the specimen breaks. A scale shall be used to measure the elongation at instant of rupture. The break shall be between the bench marks, and the tensile strength calculated from the area of the original sample.

4.8.1.2.3.4 Set.- The test specimen, not less than 6 inches in length and having bench marks 2 inches apart, shall be placed in the jaws of a testing machine and stretched until the bench marks are separated by the distance specified in 3.12.2 to 3.12.6, inclusive, and 3.12.16, as applicable. The test specimen shall then be released within 5 seconds and the set determined 1 minute, unless otherwise specified, (see 3.12.2) after the beginning of release.

4.8.1.3 Accelerated aging.-

4.8.1.3.1 Specimens.- Specimens for accelerated aging tests shall be duplicates of those prepared for determination of initial physical properties.

4.8.1.3.2 Procedure.- Specimens shall be suspended vertically in a suitable test chamber without touching each other or the walls of the container. The specified temperature shall be maintained within ±2°C. for the specified period of time. At the conclusion of the aging period, the pressure within the test chamber shall be released slowly over a period of not less than 5 minutes. The tension tests shall be conducted not less than 16 nor more than 48 hours after removal from the test chamber. The tensile strength shall be based on the breaking strength after aging and the cross sectional area measured before aging. Aged and unaged specimens shall be tested under the same conditions, insofar as practical.

4.8.1.3.2.1 Oxygen pressure test.- The internal pressure of oxygen in the bomb shall be not less than 290 nor more than 310 pounds per square inch.

4.8.1.3.2.2 Air pressure heat test.- The internal pressure of air in the bomb shall be not less than 78 nor more than 82 pounds per square inch.

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4.8.1.3.2.3 Air oven test.- Heated air at atmospheric pressure shall be circulated so as to maintain a uniform temperature in the oven, using baffles to avoid local overheating or dead spots. The source of heat shall be located in the air supply outside the aging chamber proper, and the supply shall be controlled with upper and lower vents to provide a continuous change of air during test.

4.8.2 Chemical analysis.- Specimens of the conductor insulation and sheath compounds taken from the completed wire or cable shall be subjected to a careful chemical analysis for purposes of establishing the identity of the compounds, and ascertaining the composition of compounds which have been approved on the basis of compliance with the performance requirements of this specification.

4.8.3 Varnished cambric.-

4.8.3.1 Samples of the varnished cambric tape (see 3.12.7) removed from the completed cable shall be tested for tensile strength, effect of hot petrolatum, dielectric strength and aging.

4.8.3.2 Tensile strength.- The tests shall be made in the same general manner as specified in 4.8.1.2, except that the rate of separation of the jaws of the testing machine shall be approximately 12 inches per minute.

4.8.3.3 Effect of hot petrolatum.- Samples of the varnished cambric shall be placed in hot petrolatum at a temperature of 150°C. for a period of 15 minutes, then removed from the petrolatum and allowed to cool to room temperature.

4.8.3.4 Dielectric strength.-

4.8.3.4.1 The apparatus used and the methods of measuring voltage shall be as specified in 4.8.19.

4.8.3.4.2 Specimens.- Not less than 3 (unless the conductor has less than 3 tapes, in which case all tapes shall be taken) nor more than 5 tapes shall be taken from each conductor of a completed cable. Specimens of tape shall be selected to be as representative as possible of all of the tapes used.

4.8.3.4.3 Electrodes.- The electrodes shall be brass or copper, 1/4 inch in diameter, rounded to a radius of 1/32 inch, with flat polished contact surfaces. The electrodes shall be self-aligning, and shall be placed exactly opposite one another, with the specimen in a horizontal plane between them. The electrodes shall be kept clean and polished. The electrodes and the specimen shall be immersed in suitable oil to prevent the occurrence of flashover.

4.8.3.4.4 Procedure.- Starting at zero the voltage shall be increased uniformly to breakdown at a rate of approximately 500 volts per second. Three tests shall be made on each tape at room temperature. The tape thickness shall be measured near the point of puncture with a machinist's micrometer graduated at 0.001 inch, and having anvil and spindle surfaces not less than 0.24 inch nor more than 0.26 inch in diameter. The micrometer shall be tested and calibrated periodically for conformance with accepted commercial standards. The number of thickness readings on the specimen shall be equal to the number of punctures made; these readings shall be distributed along the centerline of the specimen in such manner as to represent closely the thickness at the points of the puncture.

4.8.3.4.5 Test average.- The average of all tests on all tested tapes from any one conductor shall be considered the average breakdown stress, and shall be not less than the appropriate value specified in 3.12.6.11.

4.8.3.5 Aging.- Ten specimens shall be heated in an air oven to a temperature of not less than 124°C. nor more than 126°C. for a period of 150 hours, and then allowed to cool to normal testing room temperature. ~~The specimens shall then be tested by bending them through not less than 180 degrees around a 1/8-~~ inch diameter mandrel. Five specimens shall be bent in the length direction and five in the width direction of the tape.

4.8.4 Asbestos.- Specimens of asbestos taken from the completed cable shall be tested to determine compliance with MIL-I-3053 and 3.12.9 and 3.12.11.

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4.8.5 Glass.- Specimens of glass fiber taken from the completed cable shall be tested to determine compliance with MIL-Y-1140 and 3.12.10 and 3.12.11.

4.8.6 Impervious sheath.- The materials used shall be in conformance with those employed in the approved sample or otherwise specifically approved by the bureau or agency concerned.

4.8.7 Metal armor.-

4.8.7.1 Chemical analysis.- Specimens of armoring wire before application of the completed cable shall be analyzed to determine conformance with 3.12.20.

4.8.7.2 Physical properties.- Specimens of the armoring wire before application to the cable shall be subjected to tests of tensile strength, elongation, springiness and toughness.

4.8.7.3 Springiness.- A length of wire shall be coiled snugly around a mandrel 1/4 inch in diameter at a tension of approximately but not less than 100 gram (gm.) and then released. The outside diameter of the coil shall be measured in several places at or near its middle. The coil so produced shall not exceed 3/8 inch in outside diameter.

4.8.7.4 Toughness.- The toughness of the wire shall be determined by bending a specimen back and forth, in a specially designed fixture, over a 0.030-inch radius support. Each 90-degree movement in either direction shall be taken as one bend. The bends shall be made under a uniform tension of 100 gm., and at a uniform rate not to exceed 50 bends per minute. The wire shall withstand not less than ten 90-degree bends before a rupture.

4.8.8 Drip.-

4.8.8.1 Specimen.- The specimen shall be an 18-inch section of completed cable.

4.8.8.2 Conditioning.- One specimen of each type of cable size 14 and larger shall be suspended in a vertical position in an oven maintained for a period of not less than 18 hours at 95°C. For conductor sizes smaller than 14, no drip test will be required.

4.8.8.3 Observation.- No material shall fall away from the end of the cable during the test.

4.8.9 Cold bend.-

4.8.9.1 Specimens.- Where a cold bend test is specified in section 3, the specimen of cables for nonflexing service (see 1.2.1) shall consist of a 5-foot section of completed cable, and the specimens from cables for repeated flexing service (see 1.2.2) shall consist of 1-foot sections of not less than 20 percent of the insulated and braided conductors removed from completed cable.

4.8.9.2 Apparatus.- Straight wooden bars 5 feet in length, wooden mandrels of specified radii of curvature, and a refrigerator sufficiently large to hold cable specimens and mandrels shall be provided.

4.8.9.3 Procedure.- The specimen shall first be straightened, if necessary, and then kept straight by mounting on a wooden bar. The mounted specimen shall be placed in the refrigerator and conditioned at 0°C. for not less than 6 hours, then removed from the refrigerator, quickly detached from the mounting, and bent around the appropriate wooden mandrel at a rate of approximately 20 degrees per second. The specimen shall be suitably clamped to maintain its bent position, then it shall be dissected and a visual examination made of all its component parts. When defects or disarrangements are discovered in any component, examination shall be made of specimens in the "as received" condition to determine whether the defect existed prior to the cold bend test. Any changes or derangements, such as torn tapes or broken or cracked components resulting from the cold bend test of a nature or to an extent as to affect the electrical properties or performance of the cable in service, shall be reported to the Government inspector.

4.8.9.3.1 Cables for nonflexing service.- Cables for nonflexing service (see 1.2.1) shall be bent 180 degrees over a mandrel having a diameter as indicated in the detail requirements.

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4.8.9.3.2 Braided conductors.- The insulated and braided conductors removed from type MHOF (see 3.28), shall be wrapped three turns around a mandrel having a diameter as specified in the detail requirements, and shall then be subjected to a 60-cycle potential of 2,000 volts r.m.s. for 1 minute.

4.8.10 Flammability.-

4.8.10.1 Specimens.- Specimens shall consist of straight sections of completed wire or cable approximately 18 inches in length.

4.8.10.2 Apparatus.- An enclosure of sufficient size to contain the specimen, supports, heater coil, spark plugs, flame travel gage and their associated accessories shall be arranged to eliminate air drafts and permit clear view of the interior through shatterproof glass windows. Vent holes, distributed around the sides adjacent to the base, shall be provided to admit fresh air when an exhaust fan connected to the top of the enclosure is operated at a minimum suction just sufficient to carry off smoke and gases.

4.8.10.2.1 Supports shall be suitable for holding the specimens in a vertical position with an unsupported span not less than 14 inches in length. The lower end of the specimen shall be wrapped with varnished cambric in such a manner that gases released through this end are diverted toward the spark plug.

4.8.10.2.2 Heater coils shall consist of 7 turns of 0.102 inch diameter resistance wire conforming to composition E of QQ-R-175, space wound to 0.25 inch between centers. The inside diameter of the coil shall exceed the overall diameter of the specimen by not less than 0.5 inch nor more than 0.6 inch. The lower end of the heater coil shall be located 1.5 inches above the top of the lower specimen support.

4.8.10.2.3 Two spark plugs, with extended electrodes spaced 1/8 inch from the surface of the specimen and the plugs located on diametrically opposite sides of the specimen, shall be placed with their longitudinal centerlines in a horizontal plane 1/2 inch above the top of the heater coil, to ignite the gases emitted from the heated specimen. A suitable electric circuit shall be provided to maintain continuous sparking at the electrodes during the specified time. The spark plugs shall be mounted in such a manner that they may be moved away from the specimen after ignition takes place so as not to impede the travel of the flame and to prevent their electrodes from becoming fouled by soot.

4.8.10.2.4 A suitable flame travel gage shall be positioned near the specimen to judge the distance of flame travel without appreciable impeding progress of the flame.

4.8.10.3 Procedure.- Flame tests are specified for the various types of wire and cable in section 3, and the required performance is shown in table XLII. The tests shall be so conducted as to measure, as accurately as practicable, the following factors:

Time of ignition.
Time of heating.
Distance of flame travel.
Time required for self-extinction.

4.8.10.3.1 Time of ignition.- With the enclosure closed and ventilated, the specimen centered in the heater coil, and the spark plugs, and flame gage properly located, a stop watch shall be started simultaneously with the energizing of the heater coil and spark plugs. A constant current as specified in table XLIII shall be supplied from a suitable transformer source to the heater coil. Ignition shall be considered as occurring when the flame transfers from the escaping gases to the surface of the specimen and continues there, disregarding the flashes which may occur in the gassing space prior to the sustained flame. The time required to ignite the specimen shall be recorded.

4.8.10.3.2 Time of heating.- For all armored cable, the heating current shall be turned off 30 seconds after ignition occurs. When armored cable does not ignite in 250 seconds or less, the specimen is considered satisfactory and the test shall be discontinued. For all unarmored cables, where the ignition time is less than 60 seconds, the heating current shall be turned off 60 seconds after being turned on; where the ignition time is greater than 60 seconds, the heating current shall be turned off when ignition occurs.

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Table XLII - Flammability limits.

Cable types	Ignition time	Burning time	Flame travel
	Minimum	Maximum	Maximum
	Seconds	Seconds	Inches
Cables with braided armor---	60	60	2
MHOF, TTOP, MCOS, TTRS, DBSP, TBSP, FBSP, MCSF, SDU------- TRF, TRXF, MMOP, FSCF, MDU, MDY Sizes 14 and larger of CVSF, SHOF, FHOF, THOF, FHOE, SSF, T3P, TSP, JAS----------	30	120	2
Sizes 9 and smaller of DHOF, THOF, FHOF---------- DLT, SHOF	No test required		
MRI----------	25	60	2

Table XLIII - Current in heating coil.

Nominal diameter of cable	Current amperes a.c.	Nominal diameter of cable	Current amperes a.c.
Inch	R.m.s.	Inches	R.m.s.
0.050 to 0.099	45	0.80 to 0.89	53
.100 to .19	46	.90 to .99	54
.20 to .29	47	1.00 to 1.09	55
.30 to .39	48	1.10 to 1.39	56
.40 to .49	49	1.40 to 1.79	57
.50 to .59	50	1.80 to 2.29	58
.60 to .69	51	2.30 to 3.00	59
.70 to .79	52		

4.8.10.3.3 Distance of flame travel.- Immediately after ignition occurs, as defined in 4.8.10.3.1, the electrical supply to the spark plugs shall be cut off, and the plugs shifted away from the flame. The maximum distance to which the flame travels along the surface of the specimen, measured from the top of the heater coil, before extinction shall be noted and recorded.

4.8.10.3.4 Time required for self-extinction.- The number of seconds that the specimen continues to burn, until the cessation of all flaming, after the current in the heater coil has been cut off, shall be recorded as the time required for self-extinction.

4.8.10.3.5 Observations.- The amount and color of the smoke and the amount and size of the soot particles shall be observed and recorded.

4.8.10.4 Number of tests.- If a sample tested fails to comply with any of the limitations specified in table XLII as to flammability upon the first trial, two additional specimens shall be tested, and the results of all 3 trials averaged together.

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4.8.11 Thermoplastic flow test.- A specimen of the completed cable with the ends prepared for electrical testing shall be wound for two turns around a mandrel having a diameter of 10 to 12 times the overall cable diameter. The coiled specimens shall be placed in an air oven at 150°C. with a 60 cycle potential of 120 volts between each conductor and shield and 120 volts between each conductor and its mate for cables having two conductors within a shield. The conductors shall be divided into two approximately equal groups for cables having three or more conductors within a shield, and 120 volts applied between each group and shield, and 120 volts between groups.

4.8.12 Heat endurance.-

4.8.12.1 A heat endurance test shall be made on completed cables, for qualification tests only (see 4.4) when specified in section 3, to determine the effects of prolonged current loading. For purposes of this test, all specimens are designated as power cables, except type MHFF (see 3.28), which is designated as control cable.

4.8.12.2 Specimens.-

4.8.12.2.1 Power cables.- For power cables, there shall be one specimen of completed cable approximately 40 feet in length.

4.8.12.2.2 Control cables.- For control cables, there shall be one specimen of completed cable approximately 40 feet in length.

4.8.12.3 Apparatus.- The testing apparatus shall consist of a test chamber and the necessary temperature-measuring and control equipment. The test chamber shall be essentially a low temperature oven equipped with racks for installing the cable specimens. It may be of any convenient size and constructed of wood and asbestos board or other suitable material. An ambient temperature of 50°C. shall be maintained in the interior of the chamber by utilizing the heat given off by the cable specimens, by additional heat supplied by resistance type heaters installed on the chamber floor, and by the admission of cool air to the chamber through adjustable vents in the top and bottom of the chamber. Convenient windows shall be provided in the chamber walls through which observations of mercury thermometers indicating the interior temperature may be made. Suitable control equipment shall be provided to permit the copper temperature of the cables under test to be held to specified values. The temperature of the copper conductors shall be determined by means of thermocouples placed within the strands of one conductor through a small knife puncture 15 feet from one end. Additional thermocouples for determining the armor temperature shall be provided.

4.8.12.4 Procedure.-

4.8.12.4.1 Power cables.- Specimens shall be racked with thermocouples attached and the conductors of each specimen connected in series to an alternating current power supply adjusted to give a copper temperature of 125°C. This test shall continue for 400 hours.

4.8.12.4.2 Control cables.- The specimen shall be racked within the chamber with thermocouples attached in the manner specified for power cables. The thermocouple to be used for measuring copper temperature shall be in contact with one of the conductors in the inner layer. The conductors shall be connected in series to a power supply adjusted to give a copper temperature of 105°C. in the inner layer of conductors. The test shall continue for 400 hours.

4.8.12.5 Observations.- Readings of copper temperature, armor temperature, current per conductor, chamber temperature, and time shall be taken throughout the duration of the test. Gas or fumes developed from the samples and the oozing of compound from the cable ends or sheath material through the armor during the tests shall be observed and recorded for both power and control cables. At the conclusion of the heat run, the dielectric strength and insulation resistance of the specimens shall be determined, followed by examination and tests (see 4.8.1 to 4.8.3, incl.) of the components to determine the extent of changes during the heat run.

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4.8.13 Water and air leakage.-

4.8.13.1 Specimen.- The specimen shall be a 30-inch section of completed cable with or without end seal. Armored specimens shall be tested with armor in place.

4.8.13.2 Apparatus.- The apparatus shall consist of a testing tank, of 15-inch length and suitable diameter to accommodate the test specimen, fitted with Navy standard watertight stuffing tubes with approved packing, connected to the pressure system, a pressure control and leakage detection device permitting test by either air or water under pressure, and necessary connections, gages and valves between testing tank and pressure control system.

4.8.13.3 Procedure.- The specimen shall be sealed in place into or through the testing tank. Air pressure shall be applied to detect leakage through the cable sheath or end seal, and to enable location and correction of leaks through stuffing tubes and packing. The cable sheath or end seal and the gland shall not leak under this air pressure. After the assembly has withstood the air pressure for a period of 30 minutes without leakage, the specimen shall be subjected to water pressure at room temperature for a period of 4 hours. The amount of water leakage, if any, shall be determined to a precision of 1 cubic inch.

4.8.13.4 Pressures.- Except where otherwise specified in section 3, the air and water pressures applied shall be 75 p.s.i. for cable sheath tests and 25 p.s.i. for cable end seal tests.

NOTE: The water and air leakage test is intended primarily for armored cables and end seals (see 3.12.20), but is required in section 3 for certain unarmored cables as a means for checking the integrity of the sheath in instances where this feature appears doubtful. This test shall not be made on unarmored cables unless visual examination discloses a patch, thin spot, cut, gouge, pock-mark, blister, porosity or other defect in the sheath.

4.8.14 Flexing endurance.-

4.8.14.1 Specimens.-

4.8.14.1.1 Bending.- There shall be two bending test specimens, each a 30-inch section of completed cable.

4.8.14.1.2 Twisting.- There shall be two twisting test specimens of completed cable, 24 inches in length for cables of 5/8-inch diameter or less, and 10 times the cable diameter plus 18 inches in length for cables of larger than 5/8-inch diameter.

4.8.14.2 Apparatus.- The bending apparatus and the twisting apparatus may be built into a portable, compact assembly, powered by a single drive and equipped with means for counting the number of cycles impressed upon a specimen. A suitable electric circuit should be provided to stop the apparatus in the event the series conductor circuit of a specimen is broken.

4.8.14.3 Procedure.-

4.8.14.3.1 Conditioning.- Specimens shall be conditioned and tested in a controlled temperature room held at the specified temperature. Specimens for the bending test and for the twisting test shall be exposed to the specified temperature until thermal equilibrium is established before testing is started. All conductors in each specimen shall be connected in a single series circuit for the purpose of detecting breakage of the conductors.

4.8.14.3.2 Bending.- A specimen shall be loosely inserted between a pair of rollers and subjected to 90-degree bending in each direction at the rate of 12 to 14 complete cycles, 360 degrees total travel per minute. The bends shall be made approximately midway along the length of the specimen, and the portion of the specimen not coming in contact with the rollers shall be loosely restricted to prevent appreciable bending at other points. The upper portion of the specimen shall be anchored to the bending apparatus. A weight which stresses the specimen in tension to approximately 40 pounds p.s.i. of cross section shall be attached to the free lower end. The grips or clamps shall be designed to apply uniform radial pressure to the core of the specimen. The diameter of the rollers shall be approximately but not less than 2-1/2 times the maximum overall diameter specified for the particular type and size undergoing test, except that when the maximum

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overall diameter specified is greater than 0.800 inch, the approximate roller diameter shall be selected from the following:

Cable type and size	Roller diameter Inches
MHOF-7, MHOF-10	2-1/2
MHOF-14, TTOP-15, DHOF-23, TTRS-6, MHOF-19	3
MHOF-24, DHOF-30, THOF-23, TTRS-8	3-1/2
MHOF-30, TTRS-10, TTRS-12, CVSF, MHOF-37	4
MHOF-44, THOF-42, TTRS-16	4-1/2
DHOF-83	7
(No test required for DHOF-250 and 400, SHOF all sizes, THOF-150, 250 and 400, FHOF-133.)	

4.8.14.3.3 Twisting.- Specimens shall be clamped in cable grips and subjected to 180 degrees of twist in each direction (360 degrees total twist) at the rate of 12 to 14 complete cycles per minute. The distance between grips shall be 6 inches plus 10 times the maximum specified diameter of the specimen. The upper grip shall be oscillated by the twisting apparatus. The lower grip shall be free to move vertically but restrained from oscillation, and have a weight attached which stresses the specimen in tension to approximately 40 p.s.i. of cable section.

4.8.14.4 Duration of tests.-

4.8.14.4.1 Qualification.- The suitability of cables for repeated flexing service (see 1.2.2.1) shall be investigated by bending and twisting tests conducted for 6000 cycles at 75°C. and for 2000 cycles at -30°C., unless the specimen fails in some manner prior thereto. During all tests, frequent visual examination of the sheath shall be made for the purpose of detecting rupture. Upon completion of a test, the specimen shall be removed from the apparatus and checked for open or short circuits between conductors. The specimen shall then be opened, its internal condition examined, and record made of any evidence of damage caused by the test. If failure occurs within 2 inches of either grip, the results shall be disregarded and the test repeated.

4.8.14.4.2 Quality conformance.- Cables for repeated flexing service (see 1.2.2) having an overall diameter not greater than 1.500 inches shall be tested for the duration of twisting and bending cycles as required in the detail requirements. Flexing endurance tests will not be required in the examination of cables larger than 1.500 inches in diameter. The specimens then shall be examined for weakening or failure of sheath and conductors and the nature of damage, if any, reported with the results. If failure occurs within 2 inches of either grip, the results shall be disregarded and the test repeated.

4.8.15 Impact resistance.-

4.8.15.1 Specimen.- The specimen shall be a 3-foot section of completed cable.

4.8.15.2 Procedure.- The specimen shall be securely fastened to a steel block and shall be subjected to the blows of a 6-pound steel hammer having sides tapered at a 45-degree angle to a 1/16-inch radius striking edge. The steel block and hammer shall be connected to one side of an alternating current high potential test set and grounded, and all conductors in the specimen connected to the other side of this circuit. With 5000 volts impressed, the hammer shall be dropped from a height of 24 times the overall diameter of the specimen to one spot on the specimen at a rate of 12 blows per minute until potential breakdown occurs. For specimens having an overall diameter of 1.500 inches or less, the point of hammer impact shall then be moved at least 6 inches and the test repeated with 2500 volts applied with the hammer's being dropped from a height of 48 times the overall diameter of the specimen. The tests shall be made at room temperature and the number of hammer falls to cause potential breakdown at each height determined and recorded. Upon completion of the hammer tests, the specimen shall be checked for short circuits between conductors and the cable then opened and examined for any damage caused by the tests.

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4.8.16 Abrasion resistance.-

4.8.16.1 Specimen.- The specimen for cables of 1.250-inch diameter and smaller shall be a 2-foot length of completed cable. The specimen for cables larger than 1.250 inch diameter shall be a 2-foot strip of cable sheath wall of 1-inch width.

4.8.16.2 Procedure.- The specimen shall be hung with a 90-degree arc of contact over a 7-inch diameter abrasion wheel made up of twelve 3/8-inch diameter fine grade tungsten carbide rods and two 7-inch diameter phenolic disks arranged in the form of a squirrel cage. A 15-pound weight shall be attached to the free end of the specimen. Specimens of the sheath strip from the larger cable sizes shall be provided with a backing of soft sheet metal. The tests shall be made at room temperature with the abrasion wheel rotating at 60 revolutions per minute (r.p.m.) in the direction which tends to lift the 15-pound weight. The number of wheel revolutions required to wear through a measured thickness of the sheath wall shall be determined and recorded.

4.8.17 Hot oil test.-

4.8.17.1 Specimens.- Specimens shall be duplicates of those prepared for the tensile tests (see 4.8.1).

4.8.17.2 Procedure.- Specimens shall be immersed in lubricating oil, conforming to Navy Symbol No. 3100 of Pamphlet N.B.S. 431, or equal, at a temperature of 120°C. After immersion for 18 hours, the specimen shall be removed from the hot oil, blotted lightly to remove excess oil, and suspended in air at room temperature for not less than 3-1/2 hours nor more than 4-1/2 hours. After this conditioning, determinations shall be made of tensile strength based on the original cross section before immersion, and of the ultimate elongation based upon the length between bench marks as measured before immersion.

4.8.18 Oil, water and chemical resistance.-

4.8.18.1 Specimens.- Specimens shall be strips of cable sheath suitable for subsequent measurement of tensile strength and elongation, and shall be buffed on the inner surface to a uniform thickness of 0.040 inch.

4.8.18.2 Procedure.- Specimens shall be weighed and surface area measured before immersion. Five specimens shall then be immersed at room temperature for a period of 10 days in each of the following liquids: Distilled water, storage battery electrolyte, fuel oil, and hydraulic fluid (phosphate ester type). After removal from the liquids, each specimen shall be wiped clean and again weighed. The change in weight of the specimens shall be expressed in terms of weight per unit initial area of surface exposed to the liquids. The specimens shall then be tested to determine change in tensile strength and elongation.

4.8.19 Dielectric tests.-

4.8.19.1 Unless otherwise specified herein, the dielectric tests shall be made on full coil or reel lengths of completed wire, cord or cable to be shipped. Where tests are to be made on short lengths, as required in section 3 for the particular type and size of cable under consideration, these tests shall be made only as part of the regular comparison inspection (see 4.6.5).

4.8.19.2 Apparatus.-

4.8.19.2.1 High potential test.- Except for spark testing, the dielectric tests shall be made with alternating potential from a source of ample capacity, and in no case less than 5 kilovolt-ampere, having a frequency not greater than 100 cycles per second and a wave shape approximately as closely as practicable to a sine curve under all test conditions. The testing voltage may be measured by any method satisfactory to the bureau or agency concerned which gives a root-mean-square value, preferably by means of voltmeter connected to voltmeter coil in the high-tension winding of the testing transformer, or to a separate instrument transformer. A voltmeter on the low-tension side of the transformer is satisfactory if the ratio of transformation does not change under any test condition.

4.8.19.2.2 Spark test.- The power capacity of the sparking equipment shall be as high as practicable, and sufficient to maintain the required testing voltage under all normal load conditions. Mear shall be provided for regulating the testing voltage to within 5 percent of the specified value at all times. The testing voltage may be measured by a kilovoltmeter or electrostatic voltmeter having a burden that will not affect the value or wave shape of the testing voltage. An indicating voltmeter connected to the low-tension side of the

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transformer may be used, provided its readings are calibrated to give a true indication of the voltage across the dielectric of the wire being tested. The equipment shall include a fault-signal system that will automatically stop the machine, and give a visible signal until manually reset. The electrode shall be of a suitable bead chain or fine link mesh construction, and the finished cable or insulated conductor shall be passed through the electrode at a speed such that the insulation will be subjected to the specified test voltage for a period of time not less than 0.2 second, and which shall locate all faults.

4.8.19.3 Rate.- Except for spark testing, the initially applied voltage shall be not greater than 600 volts, and the rate of increase shall be approximately uniform and not over 100 percent in 10 seconds, nor less than 100 percent in 60 seconds.

4.8.19.4 Temperature.- Dielectric tests shall be made at normal room temperature.

4.8.19.5 Time of application.-

4.8.19.5.1 Shipping lengths.- Except for spark-tested lengths, the time of voltage application measured from the time the specified voltage is attained shall be not less than the following:

Each conductor against all adjacent conductors and ground:
5 minutes for cables having 4 conductors or less (see 1.2.1 and 1.2.2).
1 minute for cables having 5 conductors or more (see 1.2.1 and 1.2.2).
1 minute for miscellaneous wire (see 1.2.4).

4.8.19.5.2 Spark tested lengths.- Where a spark test is required in section 3 for miscellaneous wire (see 1.2.3), the completed wire shall be passed through the spark electrode at a speed such that the insulation will be subjected to the specified test voltage for a period of time not less than 0.2 second, and which will locate all faults.

4.8.19.5.3 Short lengths.- The time of voltage application, measured from the time the specified voltage is attained, shall be approximately but not less than 1 minute for each conductor against all adjacent conductors connected to ground, and all conductors connected together against ground.

4.8.19.6 Ground electrodes.-

4.8.19.6.1 The following methods for grounding shall be employed while testing electrically. The method of grounding is determined by the cable construction and the nature of the requirements specified in section 3.

4.8.19.6.2 Cable coverings.- In the case of cables with braided metal armor, the armor shall constitute the ground electrode.

4.8.19.6.3 Water bath.- All unarmored cables shall be immersed in a water bath for not less than 1 hour and then tested while still immersed. The water bath shall constitute the ground electrode. When testing types MCOS, DBSP, TBSP, FBSP, TTRS, DSS, TSS, FSS and 7SS, the shield shall be connected to ground while the insulated conductors are being tested, then the insulated conductors shall be connected together to the shield and potential applied from shield to ground or water bath.

4.8.19.6.4 Spark electrodes.- The copper conductor shall be grounded and potential applied to the outer surface of the completed wire by a suitable bead chain or fine link mesh device that will insure intimate metallic contact over practically the entire surface of the wire under test. A pipe, helical coil spring, and similar electrodes are not considered to be of an intimate type.

4.8.19.6.5 Electrodes for short lengths.- Where dielectric tests are specified in the detail requirements on a short length specimen in the "as received" condition after cold bend (see 4.8.9), or after the flame test (see 4.8.10), the tests shall be made with the external surface of the specimen grounded. Specimens having no braided metal armor shall have the external surface grounded by immersion in water or mercury, by wrapping with metal oil, or by a coating of sprayed metal.

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4.8.20 Cable tension test.-

4.8.20.1 Specimens.- Specimens shall consist of straight sections of completed cable approximately 6 feet in length.

4.8.20.2 Procedure.- The specimen shall be secured at each end with a woven cable grip similar to type HD of MIL-G-160. A tension load shall be applied for 1 hour after which the load shall be released and the specimen examined for damage.

4.8.21 Tensile strength and elongation, conductors.- While physical tests need not be made on each lot of copper conductors submitted for delivery, the inspector shall require tensile tests at such frequent intervals as may be necessary to ascertain that the physical properties of the conductors are in conformance with 3.11 and 3.12.1. All conductors shall, however, be given a careful visual and dimensional examination as specified in 4.7.

4.8.22 Coating, metallic.-

4.8.22.1 Specimens.- Specimens shall be coated wire of approximately 6 inches, but not less than 4-1/2 inches, in length.

4.8.22.2 Preparation.- Specimens shall be thoroughly cleaned by immersion in benzine, ether, alcohol, or carbon tetrachloride for at least 3 minutes, and shall then be removed and wiped dry with a clean, soft cloth.

4.8.22.3 Solutions.-

4.8.22.3.1 Hydrochloric acid.- The hydrochloric acid solution shall have a specific gravity of 1.088 at 15.5°C. During the test, the solution shall be maintained at a temperature between 15.5°C. and 21°C. A portion of the hydrochloric acid solution having a volume of 180 milliliters shall be considered to be exhausted when 2 cycles have been completed on the numbers of specimens given in table XLIV.

Table XLIV - Number of specimens to be tested for 2 cycles in 180 ml HCL.

Diameter of wire	Maximum number of specimens
Inch	
0.0850 to 0.0501	6
.0500 to .0381	10
.0380 to .0301	12
.0300 to .0030	14

4.8.22.3.2 Sodium polysulphide.- The sodium polysulphide solution shall have a specific gravity of 1.142 at a temperature of 15.5°C. During the test, the solution shall be maintained at a temperature between 15.5°C. and 21°C. The sodium polysulphide solution shall contain an excess of sulphur and have sufficient strength to blacken thoroughly a piece of clean bare copper wire in 5 seconds.

4.8.22.4 Procedure.- The cleaned specimen shall be immersed in the hydrochloric acid solution for 1 minute. It shall then be rinsed in clear water, wiped and dried and immersed in the sodium polysulphide solution for 30 seconds, and again rinsed and dried. These cycles of operations shall be repeated, and if the specimen does not become blackened after the second cycle of immersion, the coating shall be regarded as satisfactory.

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4.8.23 Tensile strength, elongation, set and aging, insulation and sheath.- The preparation of specimens and methods of conducting tests shall be as specified in 4.8.1.

4.8.24 Conductor resistance.- The resistance of the insulated conductor shall be measured in all completed wires and cables to determine conformance with the detail requirements. For stranded conductors, the resistance of individual strand wires need not be measured unless there is doubt as to the conductivity.

4.8.25 Insulation resistance.- The apparent insulation resistance shall be measured as soon as practicable after the dielectric test, and shall be not less than the values specified. The leakage current shall be measured after a 1-minute electrification with a direct current potential of not less than 200 nor more than 500 volts. The conductor shall be maintained negative to the sheath, armor, or water. For multiple conductor cables, the insulation resistance of each conductor against all other conductors connected to ground shall be not less than the specified value. Insulation resistance shall be referred to a temperature of 15.5°C. by multiplying the measured value of insulation resistance by a conversion factor corresponding to the temperature of the measurement and the type of cable. These conversion factors are shown in table XLV. For temperature values between those tabulated, the conversion factor shall be obtained by interpolation. Where the values in table XLV are not applicable to approved cable constructions, other conversion factors may be used when specifically authorized.

Table XLV - Temperature conversion factors.

Temperatures		Temperature coefficients	
Degrees F.	Degrees C.	Nonflexing cables (see 1.2.1) Repeated flexing cables (see 1.2.2) Miscellaneous wire (see 1.2.4)	Varnished cambric insulated cables (see 1.2.1)
32	0.0	0.032	0.085
35	1.7	.047	.11
38	3.3	.071	.15
41	5.0	.098	.18
44	6.7	.14	.26
47	8.3	.25	.32
50	10.0	.29	.42
53	11.7	.45	.54
56	13.3	.78	.71
59	15.0	.91	.89
60	15.5	1.00	1.00
62	16.7	1.26	1.20
65	18.3	1.80	1.50
68	20.0	2.50	2.00
71	21.7	3.50	2.45
74	23.3	5.00	3.10
77	25.0	6.60	4.00
80	26.7	8.30	5.15
83	28.3	11.0	6.20
86	30.0	17.5	8.00
89	31.7	24.5	10.0
92	33.3	34.0	12.0
95	35.0	48.0	16.0

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4.8.26 Flame retardant test.- Various component materials employed in shipboard cables are specified in section 3 to require moisture and flame retardant characteristics. The materials used shall be in conformance with those employed in the approved sample, or otherwise specifically approved by the bureau or agency concerned.

4.8.27 Capacitance and characteristic impedance (shielded pairs).-

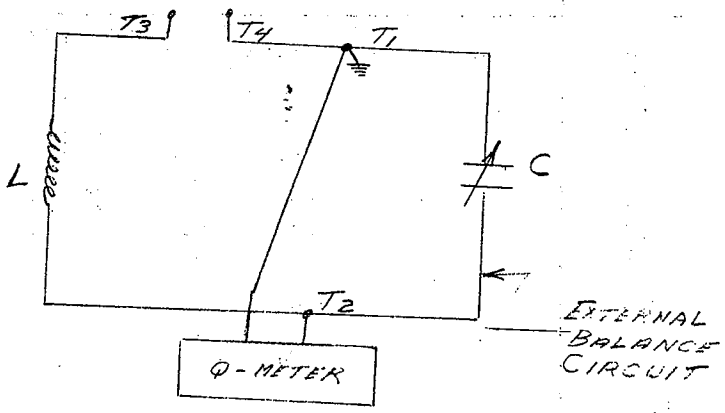
4.8.27.1 Specimens of pair or cables for test.-

4.8.27.1.1 Length.- Measurements of capacitance and characteristic impedance shall be made upon a specimen 5 feet in length. Measurement shall be checked on a specimen 10 feet in length.

4.8.27.1.2 Preparation of specimen.- The shield shall be removed for a distance of 1 inch from both ends of the pair being tested. Each conductor shall be bared for 1/2 inch from each end. The length of the specimen is considered to be the shielded length.

4.8.27.1.3 Conditioning.- Measurements shall be made at 25° ± 2.5°C. and prevailing humidity. If a refrigerator, oven, or other small chamber is used for conditioning, leads from the specimen to the terminals of the measuring instrument shall be 4000 circular mils or larger wire of the shortest practicable length. Due consideration should be given to the effect of internal capacitances in the instrument, and of interlead capacitance and inductance upon the accuracy of the determinations.

4.8.27.2 Test equipment.- The following procedure is specified for a Q-meter and suitable oscillator equipped with the external balance circuit schematically shown hereinafter. The general procedure can be adapted to an alternate method of measurement, where it can be demonstrated that such an alternate will yield results of at least equal accuracy and precision.



L: Coil, of inductance suitable for the magnitude of the reactances to be measured.
C: Variable calibrated precision condenser.
T_1, T_2: Terminals of Q-meter, T_1 grounded.
T_3, T_4: Terminals in the inductive branch of the balance circuit (T_4 can be identical with the terminal T_1).
S: Means for shorting T_3 and T_4.

The leads from L and C to the Q-meter terminals shall be short as practicable. The test specimen, or leads from the test specimen, shall be terminated at, or as near as possible to, the terminals of L.

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4.8.27.3 Determination of capacitance.- The shield shall be grounded at both ends for these measurements. The capacitance shall be measured between the inner conductors of each pair. Designate one conductor as No. 1, the other as No. 2 and the shield as No. 3.

(a) Short terminals T_3 and T_4.
(b) Balance the circuit to resonance by adjusting C to a value C_o, at which point balance (resonance) will be indicated by a maximum reading on the Q-meter.
(c) Connect No. 1 to No. 3 terminate 1-3 at T_1, and terminate 2 at T_2. Balance the circuit by adjusting C to a new value C_1. Designate the capacitance $(C_o - C_1)$ between No. 2 and No. 1-3 by the symbol C_a. Disconnect No. 1 from No. 3.
(d) Connect No. 2 to No. 3, terminate No. 2-3 at T_1, and terminate No. 1 at T_2. Balance the circuitry by adjusting C to a new value C_2. Designate the capacitance $(C_o - C_2)$ between No. 1 and No. 2-3 by the symbol C_b. Disconnect No. 2 from No. 3.
(e) Connect No. 1 to No. 2, terminate No. 3 at T_1 and terminate No. 1-2 at T_2. Balance the circuit by adjusting C to a new value C_3. Designate the capacitance $(C_o - C_3)$ between No. 1-2 and No. 3 by the symbol C_c. Disconnect No. 1 from No. 2.
(f) The capacitance (C_p) between the two inner conductors is equal to $\frac{2(C_a + C_b) - C_c}{4}$.
(g) The capacitance per foot is the above value of capacitance divided by the length of the sample in feet (see 4.8.27.1.1).

4.8.27.4 Determination of characteristic impedance (surge impedance).- These measurements shall be made upon the same pair or specimen upon which the capacitance measurements were made.

4.8.27.4.1 Measurement of inductance (short circuit impedance Z_{sc}).-

(a) Short T_3 and T_4, and balance the circuit by adjusting C to a value C_o.
(b) Open the circuit at T_3 and T_4, connect the inner conductors of the specimen at the far end, and terminate the conductors at T_3 and T_4. Ground the shield to T_4.
(c) Balance the circuit by adjusting C to value C_1.
(d) Since the inductance of the inductive branch will be increased by an amount dL due to connecting the specimen in series with L, there will be a corresponding change dC (equal to $C_o - C_1$) in the adjustment of the capacitance C necessary to regain balance. The inductance of the specimen (L_p) can be calculated from the formula:

$$L_p = dL = dC/W^2C_o(C_o - dC) \text{ where } W = 2\pi \text{ times the frequency.}$$

If dC is 1 percent or less of C_o, the formula can be reduced to:

$$L_p = dL = \frac{dC}{W^2C_o^2}$$

4.8.27.4.2 Measurement of capacitance (open circuit impedance - Z_{oc}).- Capacitance, C_p shall be measured by the method specified in 4.8.27.3.

4.8.27.4.3 The characteristic impedance, Z_o, can be obtained from either of the formulas:

$$Z_o = \sqrt{\frac{L_p}{C_p}}$$

or

$$Z_o = \sqrt{Z_{oc}\, Z_{sc}}$$

Where: $Z_{oc} = \frac{1}{WC_p}$

$Z_{sc} = WL_p$

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4.8.28 Breaking strength.-

4.8.28.1 Completed cable.-

4.8.28.1.1 Specimens.- Specimens shall consist of straight sections of completed cable approximately 6 feet in length.

4.8.28.1.2 Apparatus.- The testing machine shall be power-driven. The applied tension shall be indicated on a dial or scale having an accuracy within ±1 percent, and the indicator shall remain at the point of maximum load after rupture of the test specimen. A knurled metal mandrel, not less than 3 inches in diameter, shall be fitted into each jaw of the testing machine.

4.8.28.1.3 Procedure.- One end of the specimen shall be wrapped two full turns around the upper mandrel and secured in place so as to provide sufficient snubbing action to prevent slipping, and the other end secured in a similar manner to the lower mandrel. The distance between the jaws with the specimen in place shall be not less than 6 inches. The jaws shall be separated at a uniform rate of approximately 1 inch per minute until the specimen breaks.

4.8.28.1.4 Number of tests.- If the first specimen representing the lot is rejected, samples from two other reel lengths shall be taken to represent the lot and, if either of these two samples fails to meet the specified breaking strength, the entire lot shall be rejected. Breaks which occur within 1 inch of each mandrel shall be considered not indicative of the true breaking strength of the specimen.

4.8.28.2 Tie cord.- The apparatus and procedure shall be as specified in 4.8.28.1 except that specimens shall consist of lengths of tie cord removed from the completed cable and the rate of separation between the jaws of the testing machine shall be approximately 12 inches per minute.

4.8.29 Induced voltage, completed cable (MCOS-5 and MCS-6 only).-

4.8.29.1 Specimen.- The specimen shall be 10 feet in length.

4.8.29.2 Preparation of specimen.- The insulation shall be removed from 1/2 inch of both ends of all conductors. The braided copper shield shall be removed or pushed back to expose 1-1/2 inch of the twisted pair at both ends. The length of braided copper shield is considered to be the length of the specimen. The braided copper shield or shields shall be grouped at one end only. The conductors of one end of each shielded pair shall be shortened.

4.8.29.3 Procedure.-

4.8.29.3.1 Type MCOS-5.- An approximate 1000 c.p.s. current shall be circulated, first in the red and green conductors, then in the red and blue conductors, then in the green and blue conductors. The induced voltage appearing across the open end of the shielded pair shall be measured to an accuracy of 1 microvolt while the current is flowing.

4.8.29.3.2 Type MCOS-6.- An approximate 1000 c.p.s. current shall be circulated in the yellow and green conductors. The induced voltage appearing across the open end of each shielded pair shall be measured to an accuracy of 1 microvolt while the current is flowing.

4.29.3.3 Calculations.- Mutual inductance in henries is:

$$\frac{\text{Volts}}{2 \times \text{frequency (c.p.s.)} \times \text{current (amps)}}$$

4.8.30 Heat shock.- A specimen of the completed cable shall be wound for six adjacent turns about a mandrel not greater than three times the diameter of the specimen. The wound specimen shall be placed in an oven and subjected to air at a temperature of 121°C. for 1 hour and, after cooling, shall withstand while immersed in water at room temperature the dielectric strength tests shown in the applicable table of the detail requirements.

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4.8.31 Hydrostatic test.-

4.8.31.1 Open end.- Samples shall be selected and tested as specified in the Appendix of this specification, except that a double-ended stuffing tube of appropriate size (each end) conforming to Drawing 815-F-1197101 shall be used and water pressure shall be applied for 2 hours at the pressure specified for the type of cable being tested.

4.8.31.2 Pressure cycling.- A 15-foot specimen shall be installed in the water pressure chamber with both ends brought out through a double-ended stuffing tube of appropriate size (each end) conforming to Drawings 815-F-1197101 for electrical testing. Five water pressure cycles at the pressure specified for the type of cable being tested, shall be applied: first, 30 minutes, then release; second, third and fourth, build up pressure and hold for approximately 1 minute or long enough to measure capacitance and insulation resistance, then release; and fifth, 30-minute electrical tests (see test requirements for specific cable types) shall be made before, during, and after this pressure cycling.

4.8.32 Capacitance unbalance.- The coefficient of asymmetry of a shielded twine-conductor cable expressed in percent shall be determined by the following formula:

$$\text{Capacitance asymmetry} = \frac{400\,(Ca - Cb)}{2\,(Ca + Cb) - Cc}$$

where:

Ca = capacitance between the No. 1 conductor and the No. 2 conductor connected to the shield.
Cb = capacitance between the No. 2 conductor and the No. 1 conductor connected to the shield.
Cc = capacitance between the No. 1 and No. 2 conductors connected together and the shield.

The unbalance shall be determined at any frequency between 1 kc. and 1 mc. per second on a specimen of cable not exceeding 10 feet in length.

4.8.33 Endurance.- A 7-foot long specimen shall be subjected to 500 cycles. A cycle shall consist of drawing the specimen under alternate loads of 490 and 180 pounds tension back and forth once through 90 degrees around a 3-inch diameter pulley with a total travel of 18 inches in one direction.

4.9 Test reports.- Laboratory reports covering the comparison inspection (see 4.4.6.4; 4.6.5) shall be forwarded directly to the cognizant Government inspector and two copies of each test report shall be addressed to the bureau or agency concerned.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 Cable shall be wound in coils or on reels as follows:

5.1.1.1 Coils.- Coils shall each contain one continuous length of cable (see 6.2.2.1). The following cable types, up to and including the sizes shown, shall be furnished in coils:

MRI-T-2-1/2	DBSP	DHOF-4	DSS
DCOP-2	TBSP	THOF-4	TSS
TCOP-2	FBSP	FHOF-3	FSS
MMOP-5	SHOF-3	DSWS-4	7SS

Both ends of each coil, except for miscellaneous wires as specified in 1.2.4, shall be hermetically sealed as specified in 3.12.20 to prevent the entrance of moisture. Individual coils shall be securely tied, wrapped spirally with one layer of 30-30 waterproof creped paper, a second layer or waterproof reinforced paper or cloth-backed paper, and securely tied with a minimum of four individual ties. As an alternate wrapping, an inner layer of waterproof-creped paper and a single outer wrap of 30 to 45 water-resistant, neutral, synthetic resin impregnated and laminated creped paper may be used.

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5.1.1.2 Reels.- Reels shall each contain one continuous length of cable (see 6.2). Sizes of cable larger than these specified in 5.1.1.1 shall be wound on reels. Where the gross weight of the reel and cable is more than 1,000 pounds, the reels shall have suitable metal hubs. Reels shall be of suitable design strongly constructed with flanges of sufficient diameter to protect the cable from injury in handling, shipment, and storage. The diameter of the drum shall be not less than 1/4 times the diameter of the cable for non-flexing service, and not less than 8 times the diameter of the cable for repeated flexing service. The cable on a reel shall have both ends brought out so that the cable can be readily tested without unreeling. These ends the inner through a hole in the flange near the drum and the outer end at the top of the reel, shall be hermetically sealed in accordance with 3.12.20, to prevent the entrance of moisture and the inner or drum ends shall be fully protected from damage during the shipment by a suitable metal shield. Both ends of the cable shall be fastened securely to the reel flange with 1/4-inch rope, applied with two half-hitches, one 6 inches and the other 4 inches approximately back of the end seal, then twisting the loose rope ends together, knotting them, pulling the cable up tight, and stapling the rope to the flange with at least two 1-1/2 inch staples driven into the flange.

5.1.1.2.1 Lagging.- The cable wound on reels shall be covered with flexible waterproof barrier-material secured in a suitable manner to the flange rims. The material used shall conform to UU-P-271 or 0.004-inch thick polyethylene film conforming to Publication CS227-59. Reels shall be completely enclosed with wooden lagging with boards touching each other. Lagging boards shall be the same thickness as the outside wooden reel flanges but not thicker than 2-inch commercial lumber, and shall be securely nailed to the outside wooden flanges of the reel with cement-coated nails. On reels 36 inches in diameter and larger, lagging shall extend within 1/4 inch of the outside edges of the flanges. On reels smaller than 36 inches in diameter, lagging shall be flush with the outside edges of the flanges. Lagging of reels of cable weighing 1000 pounds or more shall be strapped with two 0.1620 ± 0.003-inch (8-gage) high-tensile round wires or two 3/4 by 0.035-inch steel straps. Lagging of reels of cable weighing less than 1000 pounds shall be strapped with two 0.0915 ± 0.003-inch (13-gage) high-tensile round wires or two 5/8 by 0.020-inch steel straps. All straps shall be stapled at intervals of approximately 15 inches. Lagging on metal flanges shall be secured by notching lagging of suitable thickness over the flanges and by steel strapping. A minimum clearance of 2 inches, but not less than 1 cable diameter, shall be provided between the inner face of the lagging and the outer layer of cable.

5.1.1.2.2 Plywood casing.- Plywood casing may be used in lieu of lagging, but shall be limited to reels smaller than 42 inches in diameter and which contain less than 1000 pounds of cable. Plywood casing shall be applied as follows: A cushioning layer of asphalt-impregnated, corrugated fiberboard or its equivalent shall be applied directly over the cable. Two layers of three-ply water-resistant toxic-treated plywood shall be secured tightly in place between the reel flanges directly over the cushioning material with not less than two 3/8 by 0.020-inch steel straps.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Reels.- Cable wound on reels will require no further packing for domestic shipment and storage or overseas shipments (all levels).

5.2.2 Coils.-

5.2.2.1 Cable in coils weighing over 50 pounds will require no further packing (all levels).

5.2.2.2 Cable in coils weighing 50-pounds and under shall be packed in boxes for the level of packing as specified (see 6.2).

5.2.2.2.1 Level A.- Boxes shall conform to any one of the following specifications at the option of the contractor:

Specification	Box	Type or class
PPP-B-585	Wood, wirebound	Class 3 use
PPP-B-591	Fiberboard, wood-cleated	Overseas type
PPP-B-601	Wood, cleated-plywood	Overseas type
PPP-B-621	Wood, nailed and lockcorner	Class 2
PPP-B-636	Fiberboard	Class 2

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Specification	Box	Type or class
PPP-B-640	Fiberboard, corrugated, triplewall	Class 2
PPP-B-576	Wood-cleated, veneer, paper overlaid	Class 2

Fiberboard boxes conforming to PPP-B-636 shall have all corner and edge seams and manufacturer's joint sealed with tape in accordance with the appendix to PPP-B-636, or minimum 2-inch wide tape conforming to PPP-T-76. Boxes shall be closed, strapped, or banded in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds.

5.2.2.2.2. Level B. - Boxes shall conform to any one of the following specifications at the option of the contractor:

Specification	Box	Type or class
PPP-B-585	Wood, wirebound	Class 1 or 2 use
PPP-B-591	Fiberboard, wood-cleated	Domestic type
PPP-B-601	Wood cleated-plywood	Domestic type
PPP-B-621	Wood, nailed and lock corner	Class 1
PPP-B-636	Fiberboard	Class 1
PPP-B-576	Wood-cleated, veneer, paper overlaid	Class 1

Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.3 Level C. - Cable wound on reels or in coil form shall be packed in a manner that will insure carrier acceptance and will afford protection against damage during direct shipment from the supply source to the first receiving activity for early use. Shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required herein or in the contract or order, coils, reels, and exterior shipping containers, shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. -

5.3.1.1 Sheath material. - All coils, reels, or shipping containers for cables for repeated flexing service (see 1.2.2) shall be marked with the phrase "Thermosetting sheath" or "Thermoplastic sheath", as applicable.

5.3.1.2 Yearly marking. - In addition to the marking specified in 5.3.1.2 and 5.3.1.3, all coils, reels and shipping containers shall be marked with a keyed series of colors to indicate the year of manufacture. This marking shall consist of a stripe approximately 2 inches wide and colored as follows for the particular year of manufacture. This cycle of colors shall be repeated every fifth year:

Year of manufacture			Identifying color
1960	1965	1970	White
1961	1966	1971	Red
1962	1967	1972	Green
1963	1968	1973	Orange
1964	1969	1974	Blue

5.3.1.2.1 Coils. - On coils wrapped for shipment the colored stripe shall be applied in at least 2 locations spaced equidistant around the coil. At each location the stripe shall be applied around the cross-section of the coil and each marking shall cover not less than the outer half of the periphery of the cross-section.

5.3.1.2.1.1 Shipping containers. - When the color-identified wrapped coils are packed more than one in a container, such containers also shall be marked with the colored stripe on one side and one end.

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5.3.1.2.2 Reels. - On reels the colored stripe shall be applied circumferentially over the lagging and midway between the flanges. The stripe shall consist of one coat, commercial quality, outside paint of appropriate color. In addition to the stripe, both flanges of all reels shall be stenciled with 4-inch high figures to show the year of manufacture.

5.3.1.3 Coils. - Two shipping tags shall be securely attached to each coil, both inside and outside the wrapping, and marked with the following information completed:

(a) Type.
(b) Size.
(c) Contract or order number.
(d) Contractor's name.
(e) Manufacturer's name (if other than contractor).
(f) Gross weight.

5.3.1.4 Reels. - Each reel shall be plainly marked on both flanges with the following information completed:

(a) Reel No.
(b) Type.
(c) Size.
(d) Contract or order number.
(e) Contractor's name.
(f) Manufacturer's name (if other than contractor).

Metal tags, stencils, or paper labels containing the information requested shall be securely attached to reels. Where paper labels are used, they shall be securely attached and protected by a transparent compound to prevent deterioration of the markings.

6. NOTES

6.1 Intended use. - Ratings, characteristics, and applications for cables covered by this specification are specified in Section 9620-2 of General Specifications for Ships.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and size of cable required (see 1.2 and 1.3)
(c) Quantity (see 6.2.1).
(d) Level of packing required (see 5.2).

6.2.1 Quantity. - The quantity of each type and size of wire or cable on a contract or order should be specified as an integral multiple of the unit ordering length given in table XLVI for reels; the nominal length or unit ordering length for wire or cable in coils (see 5.1.1.1) is 500 feet, unless otherwise specified in the detail requirements.

Preparing activity:
Navy - SH
(Project 6145-N037(NAVY)

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Table XLVI - Unit ordering lengths - reels

Designation		Feet	Designation		Feet	Designation		Feet
Type	Size	Nominal	Type	Size	Nominal	Type	Size	Nominal
CVSF	6	1000	MHOF	7	1500	THOF	6	2000
DHFR	4	2000		10	1500		9	1500
DHOF	6	2000		14	1000		14	1500
	9	1500		19	1000		23	1000
	14	1500		24	1000		42	1000
	23	1000		30	1000		150	1000
	30	1000		37	1000		250	500
	83	1000		44	1000		400	500
	250	500	PBTM	5	2500		600	500
	400	500		15	1500	TRF	105	1000
DLT	4	1/ 615		30	1000	TRXF	133	1000
FCSF	66	1620	SDU	500	1000		168	1000
FHOF	4	2000		800	700	TSP	11	2500
	9	1500	SHFR	4	2500		31	1000
	133	500	SHOF	23	2000	TTHFWA	1-1/2	2500
JAS	250	500		60	1000	TTOP	3	2500
MCSF	4	1620		150	1000		5	2500
MCOS	2	2000		200	1000		10	2500
	4	2000		250	1000		15	1500
	5	2000		500	500		20	1500
	6	2000		650	500		30	1000
	7	2000		800	500		40	1000
	6	1000	SSF	300	2/ 1100		50	1000
MDU	14	1000	THFR	4	2000		60	1000
MDY	23	1000						
	40	1000						
	60	1000						

See footnotes at end of table.

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Table XLVI - Unit ordering lengths - reels (cont'd.)

Designation		Feet	Designation		Feet
Type	Size	Nominal	Type	Size	Nominal
TTRS	2	1500	T3P	42	1000
TTRSA	4	1500		400	500
	6	1500			
	8	1500			
	10	1500			
	12	1500			
	16	1500			

1/ See 3.38.

2/ See 3.26.

6.2.2 In order to compensate for handling and probable loss on issue and for reasons specified in 4.7, remnant lengths are subject to the price reductions given in table XLVII. Scrap lengths are not acceptable.

6.2.2.1 Tabulation. - The range of standard, random, remnant and scrap lengths for each nominal length is shown in table XLVII.

Table XLVII - Cable lengths.

Type of package	Nominal length	Standard lengths	Random lengths	Remnant lengths		Scrap lengths
				Price reduction		
		No price reduction	No price reduction	5 percent	10 percent	Not acceptable
	Feet	Feet	Feet	Feet	Feet	Feet
Coil	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Coil	600	660 to 540	539 to 180	179 to 120	119 to 60	59 to 0
Reel	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Reel	800	880 to 720	719 to 240	239 to 160	159 to 80	79 to 0
Reel	1000	1100 to 900	899 to 300	299 to 200	199 to 100	99 to 0
Reel	1500	1650 to 1350	1349 to 450	449 to 300	299 to 150	149 to 0
Reel	2000	2200 to 1800	1799 to 600	599 to 400	399 to 200	199 to 0
Reel	2500	2750 to 2250	2249 to 750	749 to 500	499 to 250	249 to 0
Reel	3000	3300 to 2700	2699 to 900	899 to 600	599 to 300	299 to 0

6.3 Time required. - The length of time normally required for conducting quality conformance inspection on completed cable will be approximately 21 days from the time the necessary samples are forwarded to the laboratory, to the date the results will be reported to the inspector. All bidders should take this time interval into account and due allowance should be made for it in the computation of the bidder's delivery dates. If a longer time than 21 days is actually required for conducting such laboratory tests, the time taken in excess of 21 days will not be charged against the contractor. It should be noted, however, that the foregoing estimate is based upon the assumption that only one series of tests will be required for a given set of samples (that is, that the results of the first series of tests are satisfactory); where rejections are found necessary, the Government shall be considered as responsible for only that time, in excess of 21 days, required for the first series of tests.

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6.4 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-915, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.4.1).

6.4.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

6.5 Preproduction. - Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.6 Supersession of types. - The following types of cable listed in MIL-C-915A(SHIPS) have been replaced as shown in table XLVIII:

Table XLVIII - Superseded cable types.

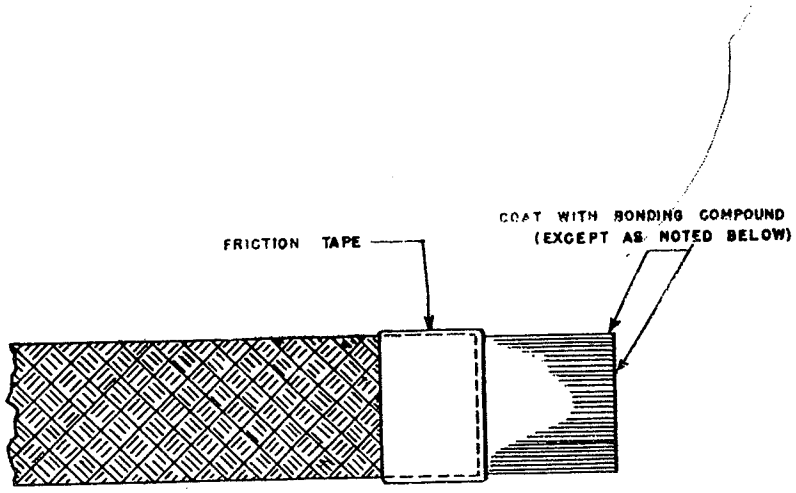
Type of MIL-C-915A(SHIPS)	Replaced by type of MIL-C-915B(NAVY)	Replaced by type of MIL-C-2194	Replaced by type of MIL-W-16878
MDGA	MDU		
MDGL	MDY		
MSS-6	~~7SS~~		
MHFF	MHOF		
PBJX		TCJA	
PBTX		TCTA	
MDGB		MSCA	
SHFS			Type D
Types of MIL-C-2681			
MDGT	MDY		
MDGY	MDY		
MDGD		MSCA	
DDGT		DSGA	

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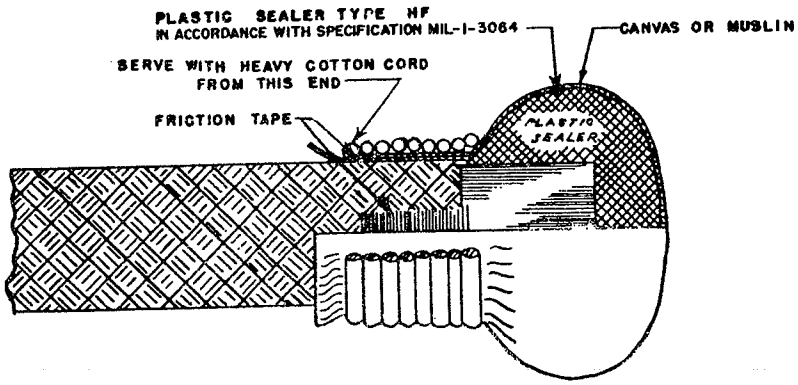
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(1) REMOVE ABOUT ONE INCH OF CABLE ARMOR
(2) SEIZE ARMOR WITH ONE LAYER OF FRICTION TAPE AS SHOWN
(3) APPLY BONDING AGENT IF NECESSARY SUCH AS A SOLUTION OF BATTERY SEALING COMPOUND AND CARBON TETRACHLORIDE TO CABLE END AND SHEATH



(4) PREPARE A PAD OF PLASTIC SEALER ABOUT 1/2 INCH THICK, KNEAD AND SHAPE AS SHOWN.
(5) COVER WITH CANVAS OR MUSLIN.
(6) SERVE WITH HEAVY COTTON CORD SO AS TO FORCE PLASTIC SEALER TIGHTLY AGAINST CABLE SIDE AND END.

Figure 1 - Method of treating cable ends on reels to prevent entrance of water.

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APPENDIX I

WATERTIGHTNESS

10. Purpose. - The manufacturer shall endeavor to fill all types of watertight cable (see 3.7) so that no water can be forced into or through the cable core, and so that the electrical characteristics of the cable are not affected by exposure of the open ends to water under pressure. The quality of the watertight feature shall be judged by the total amount of water which leaks from the open free end of the specimen, when tested as specified herein.

20. Sampling. - One 5-foot specimen from each sample of watertight cable shall be tested and the leakage found shall represent the watertight quality of the entire lot from which the specimen is cut. If there is any question as to the satisfactory processing of the completed wire or cable up to the starting and finishing ends, the contractor shall cut off and discard such ends before the lot is submitted for inspection to insure that the samples are truly representative of the completed wire or cable offered for delivery.

30. Procedure. - One end of the 5-foot specimen shall be subjected to a continuous period of 6 hours to to tap water under 25 pounds per square inch pressure. The method employed for applying the water pressure shall not constrict nor permit expansion of the specimen within the gland. Since different packing methods will influence the results and because all plants must be judged on the same basis, all tests shall be made with a low-melting point (not exceeding 190°F.) metallic alloy packing material. No other method shall be employed without prior approval in writing by the Bureau of Ships.

40. Permissible leakage. - The following tabulation shows several degrees of water leakage for each type and size of watertight cable. The values are based on a long-time analysis of test data from all suppliers. The objective is that eventually, zero leakage will be specified.

50. Limits. - The following tabulation, and the explanatory notes thereto, constitute a part of this specification.

All cable types with conductor sizes	Total water leakage during 6 hour test at 25 p. s. i.	
	Column 1	Column 2
	Accept	Reject
	Cubic inches	Cubic inches
9 and smaller	1.0 or less	1.1 or over
14 to 40 incl.	5.0 or less	5.1 or over
50 to 100 incl.	10.0 or less	10.1 or over
125 to 300 incl.	20.0 or less	20.1 or over
350 and larger	30.0 or less	30.1 or over

NOTE 1. - The water leakage reported for any specimen will determine whether the cable footage represented by that specimen comes within the limits of column 1 or column 2.

NOTE 2. - Cable footage represented by specimens which do not exceed the leakage permitted in column 1 is acceptable at the contract price.

NOTE 3. - Cable footage represented by specimens which exceed the leakage permitted in column 1 will be considered a failure and shall be cause for rejection.

60. Reports. - The results of all tests and retests made at the plant under quality conformance inspection shall be reported, together with the footage represented by each specimen, to the Bureau of Ships by the Government inspector. The results of all tests made at a laboratory shall be reported to the cognizant Government inspector with a copy to the Bureau of Ships.

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APPENDIX II

QUALITY CONTROL SYSTEM REQUIREMENTS FOR

CABLE, CORD, AND WIRE, ELECTRICAL (SHIPBOARD USE)

10. SCOPE

10.1 This appendix requires the establishment of a quality control system by the contractor to assure that supplies meet the quality standards established by the contract. This system, including procedures, is subject to surveillance by the Government representative. The procedures shall be designed by the contractor. The contractor's procedures used to implement the requirements of this appendix shall be subject to the disapproval of the Government representative.

10.2 Applicability. - This appendix shall apply to all supplies on which Government inspection is required.

10.3 Significance. - This appendix and any procedure or document executed in implementation thereof, shall be in addition to and not in derogation of other contract requirements.

20. REQUIREMENTS

20.1 Outline. - The contractor shall maintain an effective and economical quality control system planned and developed in conjunction with other planning functions. The system, including procedures, shall be adjusted to suit production procurement. The system shall be based upon consideration of the complexity of product design, quantity under procurement, interchangeability and reliability requirements, and manufacturing techniques. The system shall assure that adequate control of quality is maintained throughout all areas of contract performance, including, as applicable, the receipt, identification, stocking and issue of material, and the entire process of manufacture, packaging, shipping, storage, and maintenance. All supplies under the contract, whether manufactured or performed within the contractor's plant or at any other source, shall be subject to control at such points as necessary to assure conformance to contractual requirements. The system shall provide for the prevention and ready detection of discrepancies and for timely and positive corrective action. The contractor shall make objective evidence of quality conformance readily available to the Government representative. It is the intent of this appendix to provide a quality control procedure suitable for production of magnet wire for military use utilizing as much as possible normal control procedures for commercial production. Where differences exist between items for Naval use and commercial use, those items for Naval use shall be identified during production process. Where items for Naval use are identical to those produced for commercial use, the magnet wire supplier shall verify that these items meet the requirements specified herein. The commercial product should be utilized to satisfy contract requirements insofar as practicable.

20.2 Description of procedures. - The contractor shall provide and maintain a description of procedures for control of quality. To the extent necessary, written examination and test procedures shall be prepared to supplement the applicable drawings and specifications, and shall make clear the manner in which such examination and test procedures are to be used. This description may be a compilation of existing shop travelers, routing cards, inspection method sheets, test procedures, route sheets, or other documents normally used by the contractor to define inspection operations. The description of the quality control system and all applicable examination and test procedures shall be available to the Government representative.

20.3 Drawing and change control. - A procedure shall be maintained by the contractor to assure that the latest applicable drawing, technical requirement, and contract change information will be available at the time and place of contractor's inspection. Concurrently with the effectivity of revised drawings or changes, the contractor's drawing and change control shall assure that obsolete information is removed from all points of issue and use. All changes shall be processed in a manner which will assure accomplishment on the affected supplies at the specified effective points. The contractor shall maintain a record of the point of effectivity of changes. This record shall be available for ready reference by the Government representative.

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20.4 Measuring and testing equipment. - Unless otherwise specified in the contract, the contractor shall provide and maintain gages and other measuring and testing devices necessary to assure that supplies conform to contract requirements. These devices shall be calibrated against measurement standards or designed measuring equipment at established periods to assure continued accuracy. The contractor shall prepare and maintain a written schedule for the maintenance and calibration of such equipment based on type, purpose, and degree of usage.

20.4.1 Production tooling used as media of inspection. - When production jigs, fixtures, tool masters and other such devices are used as media of inspection they shall be initially inspected or, by other suitable means, proved for accuracy prior to release for production use. The devices shall be reinspected or proved at established intervals.

20.4.2 Use of contractor's inspection equipment. - The contractor's gages, measuring, and testing devices shall be made available for reasonable use by the Government, when required, to determine conformance with contract requirements. If conditions warrant, contractor's personnel shall be made available for operation of such devices and for verification of their accuracy and condition.

20.5 Inspection during manufacture. - The contractor shall establish and maintain inspection at appropriately located points in the manufacturing process to assure continuous control of quality of parts, components, and assemblies.

20.6 Special processes. - When Government approval or certification of processes, equipment, or personnel is required under the contract, the contractor shall assure that he is fully qualified prior to requesting Government approval.

20.7 Inspection of completed supplies. - The contractor shall inspect completed supplies as necessary to assure that contract requirements have been met.

20.8 Sampling inspection. - Any sampling procedures, in addition to those required by contract, used by the contractor to determine the acceptability of supplies, shall afford reliable assurance of the maintenance of acceptable quality levels.

20.9 Indication of inspection status. - The contractor shall maintain a system for identifying the inspection status of supplies. Identification may be accomplished by means of stamps, tags, routing cards, move tickets, tote box cards or other normal control devices. Such controls shall be of a design distinctly different from Government inspection identification.

20.10 Nonconforming supplies. - Procedures shall be provided for control of nonconforming supplies, including procedures for the identification, presentation and disposition of reworked, repaired or waived supplies. The acceptance of nonconforming supplies is a prerogative of and shall be as prescribed by the Government. All nonconforming supplies shall, when practicable, be diverted from normal material movement channels. The nonconforming supplies shall be positively identified to prevent use until disposition is made. Holding areas mutually agreeable to the contractor and Government representative shall be provided.

20.11 Evidence of approvals. - When engineering inspection, tests or Government approvals are contractually required on supplies, such as engineering models, qualification test articles, preproduction test articles and "first articles", these supplies shall be subject to the requirements of this appendix. The contractor shall maintain current records of such approvals.

20.12 Storage. - The contractor shall provide adequate procedures for control of supplies stored for the Government or to be applied to Government contracts to insure preservation and treatment in accordance with applicable requirements. Procedures shall define inspections to be conducted at scheduled intervals.

20.13 Transportation. - The contractor shall provide procedures for protecting the quality of supplies during transit in accordance with contract requirements.

20.14 Quality control records. - The contractor shall maintain adequate records throughout all stages of contract performance of examination and tests, including checks made to assure accuracy of inspection,

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testing equipment and other control media. All quality control records shall be available for review by the Government representative, and copies of individual records shall be furnished him upon request.

20.15 Corrective action. - The contractor shall take prompt action to correct conditions which might result in defective supplies or services. Use shall be made of feedback data generated and furnished by using activities as well as that generated in the contractor's facility."

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